Rockchip Linux Software Developer Guide

ID: RK-KF-YF-902

Release Version: V3.0.0

Release Date: 2026-06-20

Security Level: □Top-Secret □Secret □Internal ■Public

DISCLAIMER

THIS DOCUMENT IS PROVIDED "AS IS". ROCKCHIP ELECTRONICS CO., LTD. (“ROCKCHIP”) DOES NOT PROVIDE ANY WARRANTY OF ANY KIND, EXPRESSED, IMPLIED OR OTHERWISE, WITH RESPECT TO THE ACCURACY, RELIABILITY, COMPLETENESS, MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE OR NON-INFRINGEMENT OF ANY REPRESENTATION, INFORMATION AND CONTENT IN THIS DOCUMENT. THIS DOCUMENT IS FOR REFERENCE ONLY. THIS DOCUMENT MAY BE UPDATED OR CHANGED WITHOUT ANY NOTICE AT ANY TIME DUE TO THE UPGRADES OF THE PRODUCT OR ANY OTHER REASONS.

Trademark Statement

"Rockchip", "瑞芯微", "瑞芯" are Rockchip’s registered trademarks and owned by Rockchip. All the other trademarks or registered trademarks mentioned in this document are owned by their respective owners.

All rights reserved. ©2026. Rockchip Electronics Co., Ltd.

Beyond the scope of fair use, neither any entity nor individual shall extract, copy, or distribute this document in any form in whole or in part without the written approval of Rockchip.

Rockchip Electronics Co., Ltd.

No.18 Building, A District, Software Park, Tongpan Road, Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service Email: fae@rock-chips.com

Preface

Overview

This document serves as a software development guide for Rockchip Buildroot/Debian/Yocto Linux systems, aimed at helping software development engineers and technical support engineers get started more quickly with development and debugging on the Rockchip Linux platform.

Target Readers

This document (guide) is primarily intended for the following engineers:

- Technical support engineers
- Software development engineers

Revision History

Date	Author	Version	Change Description
2021-04-10	Caesar Wang	V1.0.0	Initial version
2021-05-20	Caesar Wang	V1.1.0	Added support for RK3399, rk3288, rk3326/px30
2021-09-30	Caesar Wang	V1.2.0	Updated support for Linux 4.4 and Linux4.19
2022-01-15	Caesar Wang	V1.3.0	Added support for RK3588 Linux 5.10 Added support for RK3358 Linux4.19 Updated SDK version information
2022-04-14	Caesar Wang	V1.4.0	Added support for RK3326S Updated RK3588 Updated FAQ
2022-05-20	Caesar Wang	V1.4.1	Updated chip support list and 2022 roadmap
2022-06-20	Caesar Wang	V1.4.2	Updated SDK version and support
2022-09-20	Caesar Wang	V1.5.0	Updated support for Linux 5.10
2022-11-20	Caesar Wang	V1.6.0	Updated the support status and roadmap of each chip system Secure boot update instructions Update FAQ
2023-04-20	Caesar Wang	V1.7.0	Updated the support status and roadmap of each chip system Update the latest SDK directory structure Add support for RK3562 The document is split into multiple chapters
2023-05-20	Caesar Wang	V1.8.0	Corrected some mistakes Version update Added SDK version description Updated the SDK development environment setup chapter
2023-06-20	Caesar Wang	V1.9.0	Updated documentation Updated development environment setup chapter
2023-07-20	Caesar Wang	V1.9.1	Updated RK3566/RK3568/RK3399 Linux 4.19 SDK version
2023-09-20	Caesar Wang	V2.0.0	Updated the contents of each chapter
2023-12-05	Ruby Zhang	V2.0.1	Fixed some descriptions

Date	Author	Version	Change Description
2023-12-20	Caesar Wang	V2.1.0	Updated chip system support status Added Roadmap sections Updated documentation sections Updated SDK compilation instructions Updated SDK development sections Updated FAQ section
2024-06-20	Caesar Wang	V2.2.0	Added support for the RK3576 chip Updated the status of chip support, and remove Linux 4.4 and Linux 4.19 Update more module development instructions, such as post-processing for rootfs, Overlay handling, etc. Update FAQ
2024-09-20	Caesar Wang	V2.3.0	Updated Camera Development Introduction Updated Security Development Introduction Added Detailed Explanation of Recovery Function Updated SDK Launch Method Added Development Introduction for Boot Logo, Animation, and Other Boot Screen Images Add Section on Shared Remote Desktop Development Update Frequently Asked Questions
2024-12-20	Caesar Wang	V2.4.0	Updated the system support status for various chipsets Added SDK version support and recommendation section Added SDK partition management section Updated the FAQ section Updated the documentation section Added RK3506 support
2025-06-20	Caesar Wang	V2.5.0	Added support for RK3308 Linux 6.1 Added support for RV1126B Added AMP support status Updated SDK security update section Updated FAQ section Updated SDK permissions
2025-12-20	Caesar Wang	V2.6.0	Added support for RK3328 Linux 6.1 Updated SDK Roadmap Updated SDK documentation section

Date	Author	Version	Change Description
2026-06-20	Caesar Wang	V3.0.0	Added support for Linux 6.12 SDK (RK3572/RK3588/RK3576) Added support for Buildroot2026.02/Debian13/Yocto6.0 Updated SDK toolchain (GCC 14.3) Updated contents of each chapter Fixed spelling errors and terminology consistency (U-Boot/FUEL GAUGE/Trusted Application, etc.) Fixed EOL date consistency Cleaned up residual HTML tags Added Preface section and corrected company address

RK Linux SDK Supported System List

Linux6.12 SDK

Chipset	Buildroot Version	Debian Version	Yocto Version	AMP/RTT Version	Kernel Version	SDK Version	TAG Version
RK3588	2026.02	13	6.0	N/A	6.12	V1.0.0_20260620	linux-6.12-stan-rkr1
RK3576	2026.02	13	6.0	N/A	6.12	V1.0.0_20260620	linux-6.12-stan-rkr1
RK3572	2026.02	13	6.0	N/A	6.12	V1.0.0_20260620	linux-6.12-stan-rkr1

Linux6.1 SDK

Chipset	Buildroot Version	Debian Version	Yocto Version	AMP/RTT Version	Kernel Version	SDK Version	TAG Version
PX30	2024.02	12	5.0	N/A	6.1	V1.2.0_20251220	linux-6.1-stan-rkr7
RK3588	2024.02	12	5.0	4.1	6.1	V1.4.0_20251220	linux-6.1-stan-rkr7
RK3576	2024.02	12	5.0	4.1	6.1	V1.3.0_20251220	linux-6.1-stan-rkr7
RK3568	2024.02	12	5.0	3.1	6.1	V1.3.0_20251220	linux-6.1-stan-rkr7
RK3566	2024.02	12	5.0	3.1	6.1	V1.3.0_20251220	linux-6.1-stan-rkr7
RK3562	2024.02	12	5.0	4.1	6.1	V1.3.0_20251220	linux-6.1-stan-rkr7
RK3506	2024.02	N/A	5.0	4.1	6.1	V2.0.0_20260418	linux-6.1-stan-rkr4.4
RK3399	2024.02	12	5.0	N/A	6.1	V1.2.0_20251220	linux-6.1-stan-rkr7
RK3358	2024.02	N/A	N/A	N/A	6.1	V1.2.0_20251220	linux-6.1-stan-rkr7
RK3328	2024.02	N/A	5.0	N/A	6.1	V1.0.0_20251220	linux-6.1-stan-rkr7
RK3326	2024.02	N/A	N/A	N/A	6.1	V1.2.0_20251220	linux-6.1-stan-rkr7
RK3308	2024.02	N/A	N/A	N/A	6.1	V1.1.0_20251220	linux-6.1-stan-rkr7
RV1126B	2024.02	12	5.0	4.1	6.1	V1.1.0_20251220	linux-6.1-stan-rkr7

Linux5.10 SDK

Chipset	Buildroot Version	Debian Version	Yocto Version	AMP/RTT Version	Kernel Version	SDK Version	TAG Version
RK3588	2021.11	11	4.0	4.1	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
RK3566	2021.11	11	4.0	3.1	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
RK3568	2021.11	11	4.0	3.1	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
RK3562	2021.11	11	4.0	4.1	5.10	V1.3.0_20241220	linux-5.10-stan-rkr4
RK3399	2021.11	11	4.0	N/A	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
RK3358	2021.11	N/A	N/A	N/A	5.10	V1.5.0_20241220	linux-5.10-gen-rkr9
RK3328	2021.11	N/A	4.0	N/A	5.10	V1.2.0_20241220	linux-5.10-gen-rkr9
RK3326	2021.11	N/A	N/A	N/A	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
PX30	2021.11	11	4.0	N/A	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
RK3308	2021.11	N/A	N/A	4.1	5.10	V1.6.0_20241220	linux-5.10-gen-rkr9
RK3288	2021.11	11	4.0	N/A	5.10	V1.3.0_20241220	linux-5.10-gen-rkr9
RK312X	2021.11	N/A	N/A	N/A	5.10	V1.5.0_20241220	linux-5.10-gen-rkr9
RK3036	2021.11	N/A	N/A	N/A	5.10	V1.5.0_20241220	linux-5.10-gen-rkr9

2026-2027 Linux SDK Upgrade Schedule

Chipset	Buildroot Version	Debian Version	Yocto Version	AMP/RTT Version	Kernel Version	Release Schedule
RK3568	2026.02	13	6.0	5.2	6.12	2026 Q4
RK3566	2026.02	13	6.0	5.2	6.12	2026 Q4
RK3562	2026.02	13	6.0	5.2	6.12	2026 Q4

Contents

Rockchip Linux Software Developer Guide

1. Chapter-1 SDK Version Support and Recommendations
    - 1.1 SDK Version Maintenance Plan
        - 1.1.1 Kernel Roadmap
        - 1.1.2 Buildroot Roadmap
        - 1.1.3 Yocto Roadmap
        - 1.1.4 Debian Roadmap
    - 1.2 SDK Recommendations
2. Chapter-2 SDK
    - 2.1 Overview
    - 2.2 Obtain the General SDK
        - 2.2.1 Get Source Code from Rockchip Code Server
            - 2.2.1.1 Rockchip Linux 6.12 SDK Downloading
            - 2.2.1.2 Rockchip Linux 6.1 SDK Downloading
            - 2.2.1.3 Rockchip Linux 5.10 SDK Downloading
        - 2.2.2 Get Source Code from Local Compression Package
    - 2.3 Summary of SDK Build Commands
3. Chapter-3 Documentation Directory Overview
    - 3.1 Chinese Documentation Directory (cn)
    - 3.2 English Documentation Directory (en)
    - 3.3 Software Bill of Materials (SBOM)
        - 3.3.1 SBOM Descriptions by System
    - 3.4 Patches
    - 3.5 Release Notes
    - 3.6 General Development Guidance Document (Common)
        - 3.6.1 Asymmetric Multi-Processing System Development Guide (AMP)
        - 3.6.2 Audio Module Document (AUDIO)
        - 3.6.3 Peripheral Components Support List (AVL)
            - 3.6.3.1 DDR Support List
            - 3.6.3.2 eMMC Support List
            - 3.6.3.3 Flash Support List
            - 3.6.3.4 UFS Support List
            - 3.6.3.5 WIFI/BT Support List
            - 3.6.3.6 Camera Support List
        - 3.6.4 CAN Module Document (CAN)
        - 3.6.5 Clock Module Document (CLK)
        - 3.6.6 CRYPTO Module Document (CRYPTO)
        - 3.6.7 DDR Module Document (DDR)
        - 3.6.8 Debug Module Document (DEBUG)
        - 3.6.9 Display Module Document (DISPLAY)
        - 3.6.10 DSMC Module Documentation (DSMC)
        - 3.6.11 Dynamic Frequency and Voltage Adjustment Module Documentation (DVFS)
        - 3.6.12 FLEXBUS Module Documentation (FLEXBUS)
        - 3.6.13 Ethernet Module Document (GMAC)
        - 3.6.14 HDMI-IN Module Document (HDMI-IN)
        - 3.6.15 I2C Module Document (I2C)
        - 3.6.16 IO Power Domain Module Document (IO-DOMAIN)
        - 3.6.17 IOMMU Module Document (IOMMU)
        - 3.6.18 Image Module Document (ISP)
        - 3.6.19 MCU Module Document (MCU)

3.6.20 MMC Module Document (MMC)
3.6.21 Memory Module Document (MEMORY)
3.6.22 MPP Module Document (MPP)
3.6.23 NPU Module Document (NPU)
3.6.23.1 RKNN-TOOLKIT2
3.6.23.2 RKNN Runtime
3.6.23.3 RKNN Driver
3.6.23.4 RKNN LLM
3.6.24 NVM Module Document (NVM)
3.6.25 PCIe Module Document (PCIe)
3.6.26 Performance Module Document (PERF)
3.6.27 PINCTRL Module Document (PINCTRL)
3.6.28 PMIC Module Document (PMIC)
3.6.29 Power Module Document (POWER)
3.6.30 PWM Module Document (PWM)
3.6.31 RGA Module Document (RGA)
3.6.32 SARADC Module Document (SARADC)
3.6.33 SPI Module Document (SPI)
3.6.34 Thermal Module Document (THERMAL)
3.6.35 Tools Module Document (TOOL)
3.6.36 Security Module Document (TRUST)
3.6.37 UART Module Document (UART)
3.6.38 UBOOT Module Document (UBOOT)
3.6.39 USB Module Document (USB)
3.6.40 Watchdog Module Document (WATCHDOG)
3.6.41 Algorithm Module (ALGORITHM)
3.6.42 GPIO Module (GPIO)
3.6.43 IMU Module (IMU)
3.6.44 RT-Thread Module (RTT)
3.7 Linux System Development Documents (Linux)
3.7.1 ApplicationNote
3.7.2 Audio Development Documents (Audio)
3.7.3 Camera Development Documents (Camera)
3.7.4 Docker Development Documents (Docker)
3.7.5 DPDK Module (DPDK)
3.7.6 Graphics Development Documents (Graphics)
3.7.7 Multimedia
3.7.8 SDK Profile introduction (Profile)
3.7.9 OTA Upgrade (Recovery)
3.7.10 Security Solution (Security)
3.7.11 System Development (System)
3.7.12 UEFI Booting (UEFI)
3.7.13 Network Module (RKWIFIBT)
3.7.14 AI Framework (RKAI)
3.7.15 Multimedia Development (Multimedia)
3.8 Chip Platform Related Documents (Socs)
3.8.1 Release Note
3.8.2 Quick Start
3.8.3 Software Development Guide
3.9 Datasheet
3.9.1 Hardware Development Guide
3.10 Other Documents (Others)
3.11 Documents List (docs_list_en.txt)

4. Chapter-4 Tools Introduction
    4.1 Driver Installation Tool
    4.2 Burning Tools
    4.3 Packaging Tools
    4.4 Tools used to Make SD Card Upgrading and Booting
    4.5 Device Information Writing Tool
    4.6 Programmer Upgrade Tool
    4.7 PCBA Testing Tools
    4.8 DDR Soldering Test Tool
    4.9 eFuse Programming Tool
    4.10 Mass Production Upgrade Tool
    4.11 Partition Modification Tool
5. Chapter-5 SDK Software Framework
    5.1 Introduction to SDK project Directory
        5.1.1 SDK Overview
        5.1.2 Buildroot
        5.1.3 Yocto
        5.1.4 Debian
    5.2 SDK Software Architecture
    5.3 SDK Development Process
6. Chapter-6 SDK Development Environment Setup
    6.1 Overview
    6.2 Preparation Work before SDK Development
        6.2.1 Install and Configure Git
        6.2.2 Install and Configure repo
        6.2.3 SDK Obtaining
            6.2.3.1 SDK Download Command
            6.2.3.2 Compressed Package of SDK Code
            6.2.3.3 Software Update History
            6.2.3.4 SDK Update
            6.2.3.5 SDK Issue Feedback
    6.3 Setting Up a Linux Development Environment
        6.3.1 Preparing the Development Environment
        6.3.2 Installing Libraries and Toolkits
            6.3.2.1 Checking and Upgrading the Host's Python Version
            6.3.2.2 Checking and Upgrading the Host's `make` Version
            6.3.2.3 Checking and Upgrading the Host's LZ4 Version
    6.4 Window PC Development Environment Setup
        6.4.1 Development Tool Installation
        6.4.2 Rockchip USB Driver Installation
        6.4.3 Windows Burning Tool Usage
        6.4.4 Target Hardware Board Preparation
    6.5 Docker Environment Setup
    6.6 Introduction to Cross-compilation Tool Chain
        6.6.1 U-Boot and Kernel Compilation Tool Chain
            6.6.1.1 Packaging Toolchain
    6.7 SDK Board Level Configuration
    6.8 SDK Customization Configuration
    6.9 SDK Environment Variable Configuration
    6.10 Fully Automatic Compilation
    6.11 Build Modules
        6.11.1 Build U-Boot
        6.11.2 Build Kernel

6.11.3 Build Recovery
6.11.4 Build Buildroot
6.11.5 Build Debian
6.11.6 Build Yocto
6.11.7 Cross-Compilation
6.11.7.1 SDK Directory Built-in Cross-Compilation
6.11.7.2 Built-in Cross Compilation in Buildroot
7. Chapter-9 Introduction to SDK Partition Configuration
7.1 SDK Partition Introduction
7.2 Preview of Parameter File
7.3 Explanation of Parameter File Content
7.3.1 FIRMWARE_VER:1.0
7.3.2 MACHINE_MODEL:RK3576
7.3.3 MACHINE_ID:007
7.3.4 MANUFACTURER:RK3576
7.3.5 MAGIC:0x5041524B
7.3.6 ATAG:0x60000800
7.3.7 MACHINE:0xffffffff
7.3.8 CHECK_MASK:0x80
7.3.9 TYPE:GPT
7.3.10 GROW_ALIGN
7.3.11 CMDLINE
7.4 Parameter Partition Definition Specification
7.5 SDK Partition Configuration Modification Guide
7.5.1 Command Overview
7.5.2 Partition Information Printing
7.5.3 Partition Information Modification
7.5.4 Adding a New Partition
8. Chapter-10 SDK Firmware Upgrade
8.1 Burning Mode Introduction
8.1.1 Windows Flashing Instructions
8.1.2 Linux Flashing Instructions
8.1.3 System Partition Introduction
9. Chapter-11 SDK Development
9.1 Bootloader Development
9.1.1 U-Boot Development
9.1.1.1 Introduction to U-Boot
9.1.1.2 Version
9.1.1.3 Preliminary Preparation
9.1.1.4 Booting Process
9.1.1.5 Shortcut Keys
9.1.2 UEFI Development
9.2 Kernel Development
9.2.1 Introduction to DTS
9.2.1.1 DTS Overview
9.2.1.2 Adding a New Product DTS
9.2.2 Module Development Documentation
9.2.3 Common Module Commands
9.2.3.1 CPU-Related Commands
9.2.3.1.1 CPU Frequency Locking Operation
9.2.3.1.2 View Current CPU Frequency
9.2.3.1.3 Checking Current CPU Voltage
9.2.3.1.4 CPU Independent Frequency and Voltage Adjustment

9.2.3.1.5 How to View Frequency and Voltage Table
9.2.3.1.6 How to Check CPU Temperature
9.2.3.2 GPU-Related Commands
9.2.3.2.1 GPU Clock Setting Operation
9.2.3.2.2 View Current GPU Frequency
9.2.3.2.3 Inspect Current GPU Voltage
9.2.3.2.4 GPU Frequency and Voltage Adjustment
9.2.3.2.5 Inspect GPU Utilization
9.2.3.2.6 Inspect GPU Temperature
9.2.3.3 DDR Related Commands
9.2.3.3.1 Inspect DDR Frequency
9.2.3.3.2 DDR Frequency Setting Operation
9.2.3.3.3 View DDR Bandwidth Utilization
9.2.3.3.4 DDR Bandwidth Statistics
9.2.3.3.5 Memory Debugging
9.2.3.3.6 Memory Stress Test
9.2.3.4 NPU-related Commands
9.2.3.4.1 Viewing NPU Frequency
9.2.3.4.2 NPU Frequency Setting Operation
9.2.3.4.3 NPU Support Query Settings
9.2.3.4.4 Related Materials
9.2.3.5 RGA-related Commands
9.2.3.5.1 Query RGA Frequency
9.2.3.5.2 Modifying RGA Frequency
9.2.3.5.3 RGA Driver Version Query
9.2.3.5.4 Query the version number of the librga library
9.2.3.5.5 Enable librga Logging
9.2.3.5.6 RGA Debug Node
9.2.3.5.7 RGA Load Query
9.2.3.5.8 RGA Memory Manager Inquiry
9.2.3.5.9 RGA Task Request Inquiry
9.2.3.5.10 RGA Hardware Information Inquiry
9.2.3.5.11 Dump Runtime Data
9.2.3.6 VPU-related Commands
9.2.3.6.1 Querying VPU Driver Version
9.2.3.6.2 Checking VPU Frame Rate
9.2.3.6.3 Viewing Frequency
9.2.3.6.4 Modify Frequency
9.2.3.6.5 Enabling VPU Debug Print
9.2.4 Pin Configuration Control
9.2.4.1 General-Purpose Input/Output (GPIO)
9.2.4.1.1 I/O Multiplexing (IOMUX)
9.2.4.2 PULL (Port Pull-Up and Pull-Down)
9.2.4.3 DRIVE-STRENGTH (Port Drive Strength)
9.2.4.4 SMT (Synchronous Metastability Trigger)
9.2.5 Thermal Control Development
9.2.6 DDR Development Guide
9.2.7 SD Card Configuration
9.3 Recovery Development
9.3.1 Introduction
9.3.2 Debugging
9.3.3 Recovery Feature Summary
9.4 Buildroot Development

9.5 Debian Development
9.6 Yocto Development
9.7 Audio Development
9.7.1 Kernel Audio Driver Development
9.7.2 Audio Pulseaudio Path Adaptation
9.8 Multimedia Development
9.9 Camera Development
9.9.1 Image Processing Module (ISP)
9.9.2 USB CAMERA
9.10 Graphics Development
9.10.1 Boot Screen Development
9.10.1.1 Boot Logo Development
9.10.1.2 Boot Animation Development
9.10.2 Desktop Application Development
9.10.2.1 Weston Desktop Development
9.10.2.2 Enlightenment Desktop Development
9.10.2.3 LVGL Desktop Development
9.10.2.4 Flutter Linux Desktop Development
9.11 Secure Development
9.11.1 Current Security Features Supported by Each Chip
9.11.2 Secure Boot Initialization
9.11.3 Compiling Secureboot
9.11.4 Rockchip Anti-Cloning Feature
9.12 WIFI/BT Development
9.13 Post-Processing Development for ROOTFS
9.14 Overlays Development
9.15 SDK Launch Method
9.15.1 Device Boot Process
9.15.2 BOOTROM Process
9.15.3 System Boot Methods
9.16 Shared Remote Desktop Development
9.16.1 Controlling Remote Desktop with wlfreerdp
9.16.2 Weston VNC Remote Desktop Access
9.17 SDK Testing
9.17.1 Integrating Rockchip Stress Test Scripts
9.17.2 Benchmark Testing
9.17.3 Rockchip Modules and Stress Testing
10. Chapter-12 SDK System Debugging Tools Introduction
10.1 ADB Tool
10.1.1 Overview
10.1.2 USB ADB Usage Instructions
10.2 Busybox Tool
10.3 GDB Tool
10.4 IO Tool
10.5 kmsgrab Tool
10.6 modetest Tool
10.7 Perf Tool
10.8 Pmap Tool
10.9 ProcrankTool
10.10 PS Tool
10.11 Slabtop Tool
10.12 Strace Tool
10.13 Top Tools

- 10.14 Update Tool
- 10.15 Vendor_storage Tool
- 10.16 Vmstat Tool
- 10.17 Watch Tool
- 10.18 weston Debugging Method
- 10.19 Obtain System Log Information Automatically

11. Chapter-13 SDK Software License Instructions
    - 11.1 Copyright Detection Tool
        - 11.1.1 Buildroot
        - 11.1.2 Debian
        - 11.1.3 Yocto
    - 11.2 License List
12. Chapter-14 SDK Security Update
    - 12.1 Kernel Security Updates
    - 12.2 Buildroot Security Updates
        - 12.2.1 `pkg-stats` Command
    - 12.3 Yocto Security Updates
    - 12.4 Debian Security Updates
13. Chapter-15 Rockchip open source information
    - 13.1 Github
    - 13.2 Wiki
    - 13.3 Upstream
14. Chapter-16 SDK FAQ
    - 14.1 How to confirm the current SDK version and system/kernel version
    - 14.2 DMA-BUF Related Issues
        - 14.2.1 Introduction to dma_heaps related nodes
    - 14.3 System Issues
        - 14.3.1 Dual Systems
        - 14.3.2 Year 2038 Problem
        - 14.3.3 How to call hardware codec resources and NPU modules on Debian system
    - 14.4 Partition Size Settings
    - 14.5 Issues about SDK Building
        - 14.5.1 Sync Issues Caused by Repo
        - 14.5.2 Abnormal Issues Caused by ./build.sh Building
        - 14.5.3 Compilation issue in the Docker environment with ./build.sh
        - 14.5.4 Buildroot Building Issues
    - 14.6 Recovery FAQ
        - 14.6.1 How to prevent recovery from being compiled into the update.img
    - 14.7 Buildroot FAQ
        - 14.7.1 Dual Screen Asynchronous Clock Times
        - 14.7.2 How to Set Buildroot System to Boot into Command Line Mode
    - 14.8 Debian FAQ
        - 14.8.1 "noexec or nodev" Issue
        - 14.8.2 Failed to Download "Base Debian"
        - 14.8.3 Abnormal Operation Causes an error to Mount /dev
        - 14.8.4 Multiple Mounts lead to /dev error
        - 14.8.5 How to Check System Related Information
            - 14.8.5.1 How to Check the Debian Version of Your System
            - 14.8.5.2 How to Check Whether the Debian Display Uses X11 or Wayland
            - 14.8.5.3 How to Check System Partition Status
            - 14.8.5.4 The ssh.service Is Abnormal in System
        - 14.8.6 Debian11 Base Package Fails to Build
        - 14.8.7 How to Decompress, Modify and Repackage Debian deb Package

14.8.8 How to Add the Swap Partition in Debian
14.8.9 Update Debian System for the First Time Will Restart the Display Service
14.8.10 Errors Occurring in Debian When Calling libGL related dri.so
14.8.11 How to Confirm that the Hardware Mouse Layer is Useful in Debian
14.8.12 The log is Too Large in Debian
14.8.13 Debian Setting Multi User Mode Issue
14.8.14 Debian Username and Password
14.8.15 Debian XFCE Desktop icon Double-click Abnormal
14.8.16 Chromium browsers will have a command line flag: --no-sandbox
14.8.17 Setting Up DRI2 Extension in Debian System's X11
14.8.18 Installing GCC Toolchain on Debian
14.8.19 Auto-Completion for Installing Packages on Debian
14.8.20 Supporting DRI3 Extension in Debian X11 System
14.8.21 Setting System to Boot into Command Line Mode
14.8.22 Configuring Screen Rotation
14.8.23 Implementing No Black Screen Function in Debian
14.8.24 Removing Desktop Mouse Pointer Display in Debian System
14.8.25 Steps for Compiling and Porting the rkaiq/rkisp Repository in Debian
14.8.25.1 Overview of Steps
14.8.26 How to Download Offline Deb Packages in Debian
14.8.27 How to Check glibc Version in Debian
14.8.28 Support for Screen Splitting in Debian Systems
14.8.29 Debian XFCE4 Desktop Lock Screen Not Working
14.8.30 Debian Partition Repair Issue
14.8.31 Debian Old Version Mirrors Unavailable
14.9 Linux Video Related Issues
14.9.1 When playing videos, there is stuttering, and the logs show frame dropping errors
14.9.2 gst-launch-1.0 Camera Video Preview Command
14.9.3 The Playback Screen Jitters after Turning on AFBC
14.9.4 How does GStreamer implement zero-copy
14.9.5 How to Test the Highest Decoding Performance of gst-launch-1.0
14.9.6 The Screen Jitters or Ripples Appear during Playback
14.9.7 How to Quickly Connect GStreamer to Opengles
14.10 Third-party OS Porting Issues
14.10.1 Is There Any Introduction to Kylin System Porting, Which is to Download the Standard iso Image and Extract rootfs.squashfs for Porting
14.10.2 Which Domestic OS Have Been Adapted
14.10.3 Whether to Support UEFI Booting
14.11 Display Related Issues
14.11.1 Mouse Blinking Issue
14.11.2 How to Send Video to the Video Layer
14.11.3 How to Configure the Position of Each Screen in Wayland Multi-screen with Differential Display Mode, Such As Left and Right or Up and Down Positions
14.11.4 What is the Debian Xserver Version
14.12 Browser Related Issues
14.12.1 Chromium R114 Scaling Issue
15. Chapter-17 SSH Public Key Operation Introduction
15.1 Key Authority Management
15.2 Reference Documents

1. Chapter-1 SDK Version Support and Recommendations

1.1 SDK Version Maintenance Plan

The following table lists different SDK versions along with their maintenance periods and release dates.

Version	SDK Released	Projected EOL
Linux4.4	2018-06-20	Jan, 2024
Linux4.19	2020-12-11	Dec, 2024
Linux5.10	2022-01-15	Dec, 2026
Linux6.1	2023-12-20	Dec, 2027
Linux6.12	2026-06-20	Dec, 2028

> **Note:** The estimated EOL time in the table above is based on the current Linux kernel release and lifecycle plan, and may change depending on adjustments to the official kernel EOL schedule.
> For the latest information, please refer to the official Linux kernel release page:
> https://www.kernel.org/category/releases.html

1.1.1 Kernel Roadmap

Version	Kernel Released	SDK Released	Projected EOL
4.4	2017-11-12	2018-06-20	Jan, 2024
4.19	2018-10-22	2020-12-11	Dec, 2024
5.10	2020-12-13	2022-01-15	Dec, 2026
6.1	2022-12-11	2023-12-20	Dec, 2027
6.12	2024-11-17	2026-06-20	Dec, 2028

The Linux Kernel's last stable release each year is designated as the LTS version. However, the exact number of versions released annually varies, usually around 5 to 6, making it difficult to predict the specific version number of the next LTS. The maintenance duration for LTS versions also varies, based on their usage extent. Typically, it starts with 2 years and may extend to 4 or 6 years (not guaranteed).

For specific Kernel EOL, refer to the official release:

Kernel Release

The currently released SDKs for Kernel versions 5.10, 6.1, and 6.12 are long-term supported versions by Rockchip.

> Minor versions are updated quarterly (e.g., 6.1.75 to 6.1.99).
> Major versions are updated every 2-3 years (e.g., 4.4 to 4.19 to 5.10 to 6.1 to 6.12).

Regarding the next kernel LTS version: Rockchip's kernel update strategy involves releasing an LTS (Long-Term Support) version every two years, which is offset by one year compared to the mainline kernel's strategy of releasing an LTS version every year. For example, if the mainline releases one LTS version annually, Rockchip releases one every two years. Therefore, the subsequent LTS version for Rockchip was released in 2024 (Linux 6.1), followed by another in 2026 (Linux 6.12), and so on.

1.1.2 Buildroot Roadmap

The current officially supported Buildroot versions by RK are `2021.11`, `2024.02`, and `2026.02`.

Version	Buildroot Released	SDK Released	Projected EOL
2018.02	2018-03-04	2018-06-20	Jan, 2024
2021.11	2021-12-05	2022-05-20	Dec, 2026
2024.02	2024-03	2024-06-20	Dec, 2027
2026.02	2026-03	2026-06-20	Dec, 2028

The Buildroot community releases a new version every three months, and one stable version is released annually.

For more details, refer to the official Buildroot release information:

Buildroot Release

1.1.3 Yocto Roadmap

The current officially supported Yocto versions by RK range from Kirkstone (4.0) to Wrynose (6.0), with the main maintenance versions being Kirkstone (4.0), Scarthgap (5.0), and Wrynose (6.0).

Version	Yocto Released	SDK Released	Support Level
Kirkstone(4.0)	April 2022	2022-12-20	Long Term Support (until April 2026)
Scarthgap(5.0)	April 2024	2024-12-20	Future - Long Term Support (until April 2028)
Wrynose(6.0)	April 2026	2026-06-20	Future - Long Term Support (until April 2030)

The Yocto community typically releases two major versions each year, usually in the spring and autumn seasons.

For more details, refer to the official Yocto release information:

Yocto Release

1.1.4 Debian Roadmap

The current officially supported Debian versions by RK range from Stretch (9) to Trixie (13), with the main maintenance version being Bookworm (12) and Trixie (13).

Version	Debian Released	SDK Released	EOL LTS
Stretch(9)	2017-06-17	2018-06-20	2022-07-01
Buster(10)	2019-07-06	2020-12-11	2024-06-30
Bullseye(11)	2021-08-14	2022-01-15	~ 2026
Bookworm(12)	2023-06-10	2024-06-20	~ 2028
Trixie(13)	2025-08-09	2026-06-20	~ 2030

Debian LTS versions are updated biennially.

For more details, refer to the official Debian release information:

Debian Release

1.2 SDK Recommendations

To help customers choose the right chip and SDK version based on their needs, the following table provides recommendations.

Product Category	Recommended Chips	Recommended SDK Version	Reasons & Key Features
Automotive Dashboard	RK3358, PX30	Linux4.19	Deep optimization for automotive scenarios, supporting fast boot, FUSA secure upgrade, boot animation and dedicated UI, meeting automotive-grade stability and safety requirements.
Industrial Control	RK3588, RK3576, RK3572, RK3568, RK3562, RK3506, RV1126B	Linux6.12 or 6.1	Enhanced real-time performance (Preempt-RT/Xenomai), heterogeneous multi-core (AMP) architecture, EtherCAT bus support, designed for industrial automation and power systems. Linux6.12 supports RK3572/RK3588/RK3576.
Commercial Cleaning & Robotics	RK3576, RK3572, RK3566, RK3562, RK3506, RK3326, RK3308, RV1126B	Linux6.12 or 6.1	Highly optimized lightweight system with low memory/storage footprint; integrated AI-ISP for improved image processing, suitable for cleaning robots and commercial AIoT devices. RK3572 is the next-generation dedicated chip for cleaning robots/commercial robots.
Smart Learning & Office	RK3588, RK3572, RK3399, RK3566, RK3288, RK3326, RK3126C, RK3128	Linux6.12 or 6.1	Widely deployed in dictionary pens, ARM PCs, conference cameras, gaming consoles, audio pickup devices, etc. Mature ecosystem and stable solutions.
Vision & Image Processing	RK3576, RK3572, RK3588, RV1126B	Linux6.12 or 6.1	Deep integration of Rockit media framework with full compatibility to RKIPC, greatly reducing development complexity and cycle for vision applications such as smart NVR and AI cameras.
Open Source & Deep Customization	RK3588, RK3576, RK3572, RK3288, RK3399, RK3568, PX30, RK3328	Linux6.12 or 6.1	Supports Buildroot, Debian, Yocto and other distributions, providing complete open-source support from core board to system, flexible for SBC/SOM products. Linux6.12 provides the latest kernel feature support.

2. Chapter-2 SDK

2.1 Overview

Rockchip Linux SDK supports Buildroot, Yocto and Debian systems. The kernel is based on Linux 5.10, 6.1, or 6.12, and boot is based on U-Boot next-dev. It is suitable for all Linux products developed or secondarily developed based on Rockchip EVB development board.

The development kit is suitable for but not limited to industrial applications, smart homes, consumer electronics, office, conference, and other AIoT (Artificial Intelligence of Things) products, and provides flexible data path combination interfaces to meet the customized requirements of customers for flexible combination. For specific function debugging and interface description, please read the documents under the project directory docs/.

2.2 Obtain the General SDK

2.2.1 Get Source Code from Rockchip Code Server

To get Rockchip Linux SDK, customers need an account to access the source code repository provided by Rockchip. In order to get code synchronization permission, please provide SSH public key for server authentication and authorization when applying for the SDK from the Rockchip technical window (FAE). About Rockchip server SSH public key authorization, please refer to SSH Public Key Operation Introduction.

2.2.1.1 Rockchip Linux 6.12 SDK Downloading

Chipset	Version	Download Command
RK3572/RK3588/RK3576	Linux 6.12	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b linux6.12 -m \ rockchip_linux6.12_release.xml

2.2.1.2 Rockchip Linux 6.1 SDK Downloading

Chipset	Version	Download Command
PX30	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b px30 -m \ px30_linux6.1_release.xml
RK3588	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3588 -m \ rk3588_linux6.1_release.xml
RK3576	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3576 -m \ rk3576_linux6.1_release.xml
RK3566_RK3568	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3566_rk3568 -m \ rk3566_rk3568_linux6.1_release.xml
RK3562	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3562 -m \ rk3562_linux6.1_release.xml
RK3506	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3506 -m \ rk3506_linux6.1_release.xml
RK3399	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3399 -m \ rk3399_linux6.1_release.xml

Chipset	Version	Download Command
RK3358	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3358 -m \ rk3358_linux6.1_release.xml
RK3328	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3328 -m \ rk3328_linux6.1_release.xml
RK3326	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3326 -m \ rk3326_linux6.1_release.xml
RK3308	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3308 -m \ rk3308_linux6.1_release.xml
RV1126B	Linux 6.1	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rv1126b -m \ rv1126b_linux6.1_release.xml

2.2.1.3 Rockchip Linux 5.10 SDK Downloading

Chipset	Version	Download Command
RK3562	Linux 5.10	repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -u \ https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b rk3562 -m \ rk3562_linux_release.xml
RK3588	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3588_linux_release.xml
RK3566, RK3568	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk356x_linux5.10_release.xml
RK3399	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3399_linux5.10_release.xml
RK3328	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3328_linux5.10_release.xml
RK3326	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3326_linux5.10_release.xml
RK3358	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3358_linux5.10_release.xml

Chipset	Version	Download Command
PX30	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ px30_linux5.10_release.xml
RK3308	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3308_linux5.10_release.xml
RK312X	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk312x_linux5.10_release.xml
RK3036	Linux 5.10	repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo -u \ ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \ rk3036_linux5.10_release.xml

Repo, a tool written in Python by Google to help manage git repositories, is mainly used to download and manage software repository of projects. The download address is as follows:

```
git clone ssh://git@www.rockchip.com.cn/repo/rk/tools/repo
```

2.2.2 Get Source Code from Local Compression Package

For quick access to SDK source code, Rockchip technical window usually provides the corresponding version of SDK initial compression package. In this way, developers can get SDK source code through decompressing the initial compression package, which is the same as the one downloaded by repo.
Take ROCKCHIP_LINUX6.12_SDK_RELEASE_V1.0.0_20260620.tgz for example, After getting an initialization package, you can get source code by running the following command:

```
mkdir rockchip_linux6.12
tar xvf ROCKCHIP_LINUX6.12_SDK_RELEASE_V1.0.0_20260620.tgz -C rockchip_linux6.12
cd rockchip_linux6.12
.repo/repo/repo sync -l
.repo/repo/repo sync -c
```

Developers can update via `.repo/repo/repo sync -c` command according to update introductions that are regularly released by technical window (FAE).

> Note:
> The software release version can be checked through the project xml file. The detailed way is as follows:

```
.repo/manifests$ realpath rockchip_linux6.12_release.xml
```

In the following example: the printed version number is v1.0.0 and the update time is 20260620.

```
<SDK>/.repo/manifests/release/rockchip_linux6.12_release_v1.0.0_20260620.xml
```

The initial compressed SDK packages currently released for Linux 6.12 are as follows:

Chipset	Compressed Package	Version
RK3572/RK3588/RK3576	ROCKCHIP_LINUX6.12_SDK_RELEASE_V1.0.0_20260620.tgz	v1.0.0

The initial compressed SDK packages currently released for Linux 6.1 are as follows:

Chipset	Compressed Package	Version
RK3588	RK3588_LINUX6.1_SDK_RELEASE_V1.0.0_20231220.tgz	v1.0.0
RK3576	RK3576_LINUX6.1_SDK_RELEASE_V1.0.0_20240620.tgz	v1.0.0
RK3566, RK3568	RK3566_RK3568_LINUX6.1_SDK_RELEASE_V1.0.0_20240920.tgz	v1.0.0
RK3562	RK3562_LINUX6.1_SDK_RELEASE_V1.0.0_20240920.tgz	v1.0.0
RK3506	RK3506_LINUX6.1_SDK_RELEASE_V1.0.0_20241020.tgz	v1.0.0
PX30	PX30_LINUX6.1_SDK_RELEASE_V1.0.0_20241220.tgz	v1.0.0
RK3399	RK3399_LINUX6.1_SDK_RELEASE_V1.0.0_20241220.tgz	v1.0.0
RK3358	RK3358_LINUX6.1_SDK_RELEASE_V1.0.0_20241220.tgz	v1.0.0
RK3326	RK3326_LINUX6.1_SDK_RELEASE_V1.0.0_20241220.tgz	v1.0.0
RK3308	RK3308_LINUX6.1_SDK_RELEASE_V1.0.0_20250620.tgz	v1.0.0
RV1126B	RV1126B_LINUX6.1_SDK_RELEASE_V1.0.0_20250620.tgz	v1.0.0
RK3328	RK3328_LINUX6.1_SDK_RELEASE_V1.0.0_20251220.tgz	v1.0.0

The initial compressed SDK packages currently released for Linux 5.10 are as follows:

Chipset	Compressed Package	Version
RK312X	RK312X_LINUX5.10_SDK_RELEASE_V1.0.0_20220420.tgz	v1.0.0
RK3036	RK3036_LINUX5.10_SDK_RELEASE_V1.0.0_20220420.tgz	v1.0.0
RK3588	RK3588_LINUX5.10_SDK_RELEASE_V1.0.0_20220520.tgz	v1.0.0
RK3566, RK3568	RK356X_LINUX5.10_SDK_RELEASE_V1.0.0_20220920.tgz	v1.0.0
RK3399	RK3399_LINUX5.10_SDK_RELEASE_V1.0.0_20220920.tgz	v1.0.0
RK3326, RK3326S	RK3326_LINUX5.10_SDK_RELEASE_V1.0.0_20220920.tgz	v1.0.0
PX30, PX30S	PX30_LINUX5.10_SDK_RELEASE_V1.0.0_20220920.tgz	v1.0.0
RK3308	RK3308_LINUX5.10_SDK_RELEASE_V1.0.0_20220920.tgz	v1.0.0
RK3358	RK3358_LINUX5.10_SDK_RELEASE_V1.0.0_20230420.tgz	v1.0.0
RK3562	RK3562_LINUX5.10_SDK_RELEASE_V1.0.0_20230620.tgz	v1.0.0
RK3328	RK3328_LINUX5.10_SDK_RELEASE_V1.0.0_20231220.tgz	v1.0.0

> Note:
> The initial compressed package may be replaced and updated with a new version!

2.3 Summary of SDK Build Commands

Chipset	Type	Board / Device Name	Build Kernel & U-Boot Command	Build Rootfs	Build Image Command	Pack / Flash Command
PX30	Dev Board	PX30 EVB	./build.sh lunch:rockchip_px30_evb_ddr3_v11_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 px30_linux_defconfig; make ARCH=arm64 px30-evb-ddr3-v11-linux.img -j24	./make.sh px30
RK3588S	Dev Board	RK3588S EVB1	./build.sh lunch:rockchip_rk3588s_evb1_lp4x_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3588.config rk3588_linux.config; make ARCH=arm64 rk3588s-evb1-lp4x-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3588 --spl-new
RK3588	Dev Board	RK3588 EVB1	./build.sh lunch:rockchip_rk3588_evb1_lp4_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3588.config rk3588_linux.config; make ARCH=arm64 rk3588-evb1-lp4-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3588 --spl-new
RK3588	Dev Board	RK3588 EVB7 (Linux 6.12)	./build.sh chip:rk3588:rockchip_rk3588_evb7_v11_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3588.config rk3588_linux.config; make ARCH=arm64 rk3588-evb7-v11-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3588 --spl-new
RK3576	Dev Board	RK3576 EVB1 (Linux 6.12)	./build.sh chip:rk3576:rockchip_rk3576_evb1_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3576.config; make ARCH=arm64 rk3576-evb1-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3576 --spl-new
RK3572	Dev Board	RK3572 EVB1	./build.sh chip:rk3572:rockchip_rk3572_evb1_v10_defconfig && ./build.sh all-release	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3572.config; make ARCH=arm64 rk3572-evb1-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3572 --spl-new
RK3572	Dev Board	RK3572 EVB2	./build.sh chip:rk3572:rockchip_rk3572_evb2_v10_defconfig && ./build.sh all-release	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3572.config; make ARCH=arm64 rk3572-evb2-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3572 --spl-new
RK3572	Dev Board	RK3572 EVB3	./build.sh chip:rk3572:rockchip_rk3572_evb3_v10_defconfig && ./build.sh all-release	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3572.config; make ARCH=arm64 rk3572-evb3-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3572 --spl-new
RK3572	Robot	RK3572 EVB1 Robot	./build.sh chip:rk3572:rockchip_rk3572_evb1_robot_defconfig && ./build.sh all-release	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3572.config rk3572_robot.config; make ARCH=arm64 rk3572-evb1-robot-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3572 --spl-new
RK3568	Dev Board	RK3568 EVB1	./build.sh lunch:rockchip_rk3568_evb1_ddr4_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3568.config; make ARCH=arm64 rk3568-evb1-ddr4-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3568 --spl-new
RK3566	Dev Board	RK3566 EVB2	./build.sh lunch:rockchip_rk3566_evb2_lp4x_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3566.config; make ARCH=arm64 rk3566-evb2-lp4x-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3566 --spl-new
RK3562	Sample	Dictionary Pen	./build.sh lunch:rockchip_rk3562_dictpen_test3_v20_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rk3562_linux_dictpen_defconfig rk3562.config; make ARCH=arm64 rk3562-dictpen-test3-v20.img BOOT_ITS=boot.its -j24	./make.sh rk3562 --spl-new
RK3562	Robot	RK3562 EVB1	./build.sh lunch:rockchip_rk3562_evb1_lp4x_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3562.config rk3562_robot.config; make ARCH=arm64 rk3562-evb1-lp4x-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3562 --spl-new
RK3562	Robot	RK3562 EVB2	./build.sh lunch:rockchip_rk3562_evb2_ddr4_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rk3562.config rk3562_robot.config; make ARCH=arm64 rk3562-evb2-ddr4-v10-linux.img BOOT_ITS=boot.its -j24	./make.sh rk3562 --spl-new
RK3399	Industry	RK3399 EVB IND	./build.sh lunch:rockchip_rk3399_evb_ind_lpddr4_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig; make ARCH=arm64 rk3399-evb-ind-lpddr4-linux.img -j24	./make.sh rk3399
RK3399	Excavator	RK3399 SAPPHIRE EXCAVATOR	./build.sh lunch:rockchip_rk3399_sapphire_excavator_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig; make ARCH=arm64 rk3399-sapphire-excavator-linux.img -j24	./make.sh rk3399
RK3358	Dev Board	RK3358 EVB	./build.sh lunch:rockchip_rk3358m_automotive_ddr3_v11_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 px30_linux_defconfig rk3358_linux.config; make ARCH=arm64 rk3358m-automotive-ddr3-v11-linux.img -j24	./make.sh evb-rk3358
RK3328	Dev Board	RK3328 EVB	./build.sh lunch:rockchip_rk3328_evb_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig; make ARCH=arm64 rk3328-evb.img -j24	./make.sh rk3328
RK3326	Dev Board	RK3326 EVB	./build.sh lunch:rockchip_rk3326_evb_lp3_v12_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 px30_linux_defconfig rk3326_linux.config; make ARCH=arm64 rk3326-evb-lp3-v12-linux.img -j24	./make.sh rk3326

Chipset	Type	Board / Device Name	Build Kernel & U-Boot Command	Build Rootfs	Build Image Command	Pack / Flash Command
RK3308	Dev Board	RK3308 EVB	./build.sh lunch:rockchip_rk3308_evb_audio_v11_64bit_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rk3308_linux_defconfig rk3308bs_mipi_display.config; make ARCH=arm64 rk3308-evb-audio-v11-display-rgb.img BOOT_ITS=boot.its -j24	./make.sh rk3308
RK3288	Dev Board	RK3288 EVB RK808	./build.sh lunch:rockchip_rk3288w_evb_rk808_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm rockchip_linux_defconfig; make ARCH=arm rk3288-evb-rk808-linux.img -j24	./make.sh rk3288
RV1126B	Industry	RV1126B EVB1	./build.sh lunch:rockchip_rv1126b_evb1_v10_defconfig && ./build.sh	./build.sh rootfs	make ARCH=arm64 rockchip_linux_defconfig rv1126b.config; make ARCH=arm64 rv1126b-evb1-v10.img BOOT_ITS=boot.its -j24	./make.sh rv1126b --spl-new

> Note:
>
> - **Rootfs Building**:
>   The SDK builds the Buildroot system by default. If you need other systems, you can set environment variables. For example:
>   Build Buildroot system: RK_ROOTFS_SYSTEM=buildroot ./build.sh
>   Build Debian system: RK_ROOTFS_SYSTEM=debian ./build.sh
>   Build Yocto system: RK_ROOTFS_SYSTEM=yocto ./build.sh
> - **Build Kernel, U-Boot:** a tool chain needs to be specified.
> - SDK has a built-in 64-bit tool chain, for example:
>
> export CROSS_COMPILE=../prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu/bin/aarch64-none-linux-gnu-
>
> - SDK has a built-in 32-bit tool chain, for example:
>   export CROSS_COMPILE=../prebuilts/gcc/linux-x86/arm/gcc-arm-10.3-2021.07-x86_64-arm-none-linux-gnueabihf/bin/arm-none-linux-gnueabihf-

3. Chapter-3 Documentation Directory Overview

In the Rockchip Linux SDK, the `docs` directory is divided into Chinese documentation (`cn`), English documentation (`en`), Software Bill of Materials (`sbom`), and Patches (`Patches`) directories.

```
docs/
├── cn/                 # Chinese documentation
├── en/                 # English documentation
├── sbom/               # Software Bill of Materials and security reports
├── Patches/            # Patch files
├── LICENSE             # Document license statement
├── readme_cn.md        # Chinese documentation overview
└── readme_en.md        # English documentation overview
```

3.1 Chinese Documentation Directory (cn)

The Chinese documentation directory contains common development guides, Linux system development documents, release notes, chip platform related documents, and other reference documents.

```
docs/cn/
├── Common/                                 # Common development guides
│   ├── ALGORITHM/                          # Algorithm modules (EIS/IDC)
│   ├── AMP/                                # Asymmetric Multi-Processing
│   ├── AUDIO/                              # Audio module
│   ├── AVL/                                # Approved Vendor List
│   ├── CAMERA/                             # Camera module
│   ├── CAN/                                # CAN bus module
│   ├── CLK/                                # Clock module
│   ├── CRYPTO/                             # Crypto module
│   ├── DDR/                                # DDR module
│   ├── DEBUG/                              # Debug module
│   ├── DISPLAY/                            # Display module (DP/HDMI/LVDS/MIPI etc.)
│   ├── DSMC/                               # Double Data Rate Serial Memory Controller
│   ├── DVFS/                               # Dynamic Voltage and Frequency Scaling
│   ├── FLEXBUS/                            # FLEXBUS module
│   ├── GMAC/                               # Ethernet module
│   ├── GPIO/                               # GPIO module
│   ├── HDMI-IN/                            # HDMI-IN module
│   ├── I2C/                                # I2C module
│   ├── IMU/                                # Inertial Measurement Unit
│   ├── IO-DOMAIN/                          # IO Power Domain module
│   ├── IOMMU/                              # IOMMU module
│   ├── ISP/                                # Image Signal Processor
│   ├── MCU/                                # Microcontroller Unit
│   ├── MEMORY/                             # Memory module (CMA/DMABUF etc.)
│   ├── MMC/                                # MMC/SD/eMMC module
```

```
│   ├── MPP/                                          # Multimedia Processing Platform
│   ├── NPU/                                          # Neural Network Processing Unit
│   ├── NVM/                                          # Non-Volatile Memory module
│   ├── PCIe/                                         # PCIe module
│   ├── PERF/                                         # Performance analysis module
│   ├── PINCTRL/                                      # Pin Control module
│   ├── PMIC/                                         # Power Management IC
│   ├── POWER/                                        # Power module
│   ├── PWM/                                          # Pulse Width Modulation module
│   ├── RGA/                                          # 2D Graphics Acceleration
│   ├── RTT/                                          # RT-Thread RTOS
│   ├── SARADC/                                       # SARADC module
│   ├── SECURITY/                                     # Security module
│   ├── SPI/                                          # SPI module
│   ├── THERMAL/                                      # Thermal module
│   ├── TOOL/                                         # Tool module
│   ├── TRUST/                                        # Trust security module
│   ├── UART/                                         # UART module
│   ├── UBOOT/                                        # U-Boot module
│   ├── USB/                                          # USB module
│   └── WATCHDOG/                                     # Watchdog module
├── Linux/                                            # Linux system development documents
│   ├── ApplicationNote/                              # Application notes
│   ├── Audio/                                        # Audio development
│   ├── Camera/                                       # Camera development
│   ├── DPDK/                                         # DPDK development
│   ├── Docker/                                       # Docker container development
│   ├── Graphics/                                     # Graphics development
│   ├── Multimedia/                                   # Multimedia development
│   ├── Profile/                                      # Software test/Benchmark
│   ├── RKAI/                                         # AI framework development
│   ├── Recovery/                                     # OTA upgrade/Recovery
│   ├── Security/                                     # Secure boot solution
│   ├── System/                                       # System porting development
│   ├── Uefi/                                         # UEFI boot solution
│   └── Wifibt/                                       # WIFI/BT development
├── Others/                                           # Other reference documents
├── Release/                                          # Release notes and quick start
├── Rockchip_Developer_Guide_Linux_Software_CN.pdf  # Linux software development guide
└── docs_list_cn.txt                                  # Document list
```

3.2 English Documentation Directory (en)

The English documentation directory has the same structure as the Chinese documentation directory, containing the same modules and subdirectories with English versions of the development documents.

```
docs/en/
├── Common/                                    # Common development guides
├── Linux/                                     # Linux system development documents
├── Others/                                    # Other reference documents
├── Release/                                   # Release notes and quick start
├── Rockchip_Developer_Guide_Linux_Software_EN.pdf  # Linux software development guide
└── docs_list_en.txt                           # Document list
```

3.3 Software Bill of Materials (SBOM)

`sbom` stands for Software Bill of Materials, which provides detailed information about the **open-source components, third-party libraries, version information, licenses, dependencies, and vulnerability mapping data** included in the software. The `sbom` directory in Rockchip Linux SDK provides complete SBOM and security vulnerability scan reports for different operating systems (Buildroot, Debian, Yocto), SDK, and U-Boot.

Each operating system and SDK provides SBOM and security analysis reports in 4 formats: SPDX JSON, CycloneDX JSON, CSV, and HTML, to meet the needs of different tools and scenarios.

```
sbom/
├── buildroot/
│   ├── sbom/                              # Buildroot system SBOM files
│   │   ├── sbom.spdx.json                 # SPDX format
│   │   ├── sbom.cyclonedx.json            # CycloneDX format
│   │   ├── sbom.csv                       # CSV format
│   │   └── sbom.html                      # HTML format
│   └── security/                          # Buildroot security scan reports
│       ├── cve-report.spdx.json           # SPDX format
│       ├── cve-report.cyclonedx.json      # CycloneDX format
│       ├── cve-report.csv                 # CSV format
│       └── cve-report.html                # HTML format
├── debian/
│   ├── sbom/                              # Debian system SBOM files
│   │   ├── sbom.spdx.json                 # SPDX format
│   │   ├── sbom.cyclonedx.json            # CycloneDX format
│   │   ├── sbom.csv                       # CSV format
│   │   └── sbom.html                      # HTML format
│   └── security/                          # Debian security scan reports
│       ├── cve-report.spdx.json           # SPDX format
│       ├── cve-report.cyclonedx.json      # CycloneDX format
│       ├── cve-report.csv                 # CSV format
│       └── cve-report.html                # HTML format
├── yocto/
│   ├── sbom/                              # Yocto system SBOM files
│   │   ├── sbom.spdx.json                 # SPDX format
│   │   ├── sbom.cyclonedx.json            # CycloneDX format
│   │   ├── sbom.csv                       # CSV format
│   │   └── sbom.html                      # HTML format
│   └── security/                          # Yocto security scan reports
│       ├── cve-report.spdx.json           # SPDX format
```

```
│       ├── cve-report.cyclonedx.json      # CycloneDX format
│       ├── cve-report.csv                 # CSV format
│       └── cve-report.html                # HTML format
├── sdk/
│   ├── sbom/                               # SDK SBOM files
│   │   ├── sdk-sbom.spdx.json             # SPDX format
│   │   ├── sdk-sbom.cyclonedx.json        # CycloneDX format
│   │   ├── sdk-sbom.csv                   # CSV format
│   │   └── sdk-sbom.html                  # HTML format
│   └── security/                           # SDK security scan reports
│       ├── sdk-security-report.spdx.json   # SPDX format
│       ├── sdk-security-report.cyclonedx.json  # CycloneDX format
│       ├── sdk-security-report.csv         # CSV format
│       └── sdk-security-report.html        # HTML format
├── kernel/
│   ├── Latest-Release-Rockchip-Kernel-CVEs-Link.txt  # Kernel CVE latest link
│   └── cve/                                # Kernel monthly CVE reports
│       ├── 2025-06.md
│       ├── ...
│       └── 2026-05.md
├── u-boot/
│   ├── Latest-Release-Rockchip-U-Boot-CVEs-Link.txt  # U-Boot CVE latest link
│   └── cve/                                # U-Boot CVE reports
│       ├── U-Boot-CVE-2026-06_CN.md
│       └── U-Boot-CVE-2026-06_EN.md
├── SBOM_Security_Scan_Reference_CN.md      # SBOM and security scan reference (Chinese)
└── SBOM_Security_Scan_Reference_EN.md      # SBOM and security scan reference (English)
```

3.3.1 SBOM Descriptions by System

- **Buildroot System**: The `sbom/buildroot/sbom/` directory contains the complete SBOM files for the Buildroot system. The `sbom/buildroot/security/` directory contains security vulnerability scan reports. Open `sbom/buildroot/security/cve-report.html` in a browser to view the complete report.
- **Debian System**: The `sbom/debian/sbom/` directory contains the SBOM (SPDX format) for the Debian system. The `sbom/debian/security/` directory contains security vulnerability scan reports. Open `sbom/debian/security/cve-report.html` in a browser to view the complete report.
- **Yocto System**: The `sbom/yocto/sbom/` directory contains the complete SBOM for the Yocto system. The `sbom/yocto/security/` directory contains CVE vulnerability summary information. Open `sbom/yocto/security/cve-report.html` in a browser to view the complete report.
- **SDK**: The `sbom/sdk/sbom/` directory contains the complete SBOM files for the SDK. The `sbom/sdk/security/` directory contains the SDK's security vulnerability scan reports. Open `sbom/sdk/security/sdk-security-report.html` in a browser to view the complete report.
- **Kernel**: Kernel CVE reports are organized monthly and stored in the `sbom/kernel/cve/` directory. Each report contains the CVE vulnerability list for the kernel in that month, affected kernel versions, vulnerability severity, and fix status. For the latest CVE link, please refer to `sbom/kernel/Latest-Release-Rockchip-Kernel-CVEs-Link.txt`.
- **U-Boot**: U-Boot CVE reports are stored in the `sbom/u-boot/cve/` directory, available in both Chinese

and English. For the latest CVE link, please refer to `sbom/u-boot/Latest-Release-Rockchip-U-Boot-CVEs-Link.txt`.

> **Note**: For detailed usage instructions on SBOM file formats and security vulnerability scanning tools, please refer to `<SDK>/docs/sbom/SBOM_Security_Scan_Reference_EN.md`.

3.4 Patches

The Patches directory contains various patch files for the Rockchip Linux SDK, used for system feature enhancements and bug fixes.

```
Patches/
├── Kernel-6.1/                  # Linux Kernel 6.1 patches
│   ├── rkr1-rkr4/              # RKR1-RKR4 version patches
│   └── rkr5+/                  # RKR5+ version patches
├── ROS2/                        # ROS2 related patches
│   ├── cross-compile/          # Cross-compilation related
│   └── patches/                # ROS2 patch files
└── Real-Time-Performance/       # Real-time performance optimization patches
│   ├── PREEMPT_RT/             # PREEMPT_RT real-time patches
│   └── XENOMAI/                # Xenomai real-time patches
```

> **Note**: For detailed usage instructions on patches, please refer to the readme.txt or related documents in each subdirectory.

The documentation released with the Rockchip Linux SDK is intended to help developers quickly get started with development and debugging. The content covered in the documentation does not encompass all development knowledge and issues. The list of documents will continue to be updated. If you have any questions or needs regarding the documentation, please contact our FAE support at fae@rock-chips.com.

In the Rockchip Linux SDK, the docs directory is divided into Chinese (cn) and English (en). The English directory includes Common (general development guidance documents), Release (release notes), Linux (Linux system development-related documents), Others (other reference documents), and docs_list_en.txt (docs file directory structure). The specific introduction is as follows:

3.5 Release Notes

The Release directory contains Rockchip Linux SDK release notes and quick start guides.

```
en/Release/
├── Rockchip_Linux6.12_SDK_Release_V1.0.0_20260620_EN.pdf  # SDK release notes
├── Rockchip_Quick_Start_Linux6.12_EN.pdf                  # Quick start guide
└── Linux6.12_SDK_Note.md                                  # SDK version notes
```

3.6 General Development Guidance Document (Common)

Please see the documentation in each subdirectory of `<SDK>/docs/en/Common` for details.

3.6.1 Asymmetric Multi-Processing System Development Guide (AMP)

For details, see the `<SDK>/docs/en/Common/AMP` directory. The AMP system is a general-purpose multicore heterogeneous system solution provided by Rockchip, which has been widely used in industrial applications such as power and industrial control, as well as consumer products like vacuum cleaners.

3.6.2 Audio Module Document (AUDIO)

It includes development documents for audio modules. The reference documents are as follows:

```
docs/en/Common/AUDIO/
├── Rockchip_Developer_Guide_Audio_EN.pdf
└── Rockchip_Developer_Guide_Linux_RV_Series_ACodec_EN.pdf
```

3.6.3 Peripheral Components Support List (AVL)

Please refer to `<SDK>/docs/en/Common/AVL` directory for details, which contains the support list for DDR/eMMC/FLASH/UFS/WIFI-BT. The support list is updated in real time on the redmine. The link is as follows:

```
https://redmine.rockchip.com.cn/projects/fae/documents
```

3.6.3.1 DDR Support List

The Rockchip platform DDR chip support list can be found in `<SDK>/docs/en/Common/AVL` under the document **Rockchip_DDR_Approved_Vendor_List_xxx.pdf**. The table below shows the support level for DDR chips. It is recommended to select chips marked with √ or T/A only.

Table 1-1 Rockchip DDR Support Symbol

Symbol	Description
√	Fully Tested and Mass Production
T/A	Fully Tested and Applicable
S/A	Sample Tested and Applicable, but there is a risk of small margins or other issues requiring validation in mass production
N/A	Not Applicable

3.6.3.2 eMMC Support List

The Rockchip platform eMMC chip support list can be found in `<SDK>/docs/en/Common/AVL` under the document **Rockchip_EMMC_Approved_Vendor_List_xxx.pdf**. The table below shows the support level for eMMC chips. It is recommended to select chips marked with √ or T/A only.

Table 1-2 Rockchip eMMC Support Symbol

Symbol	Description
√	Fully Tested, Applicable, and Mass Production
T/A	Fully Tested, Applicable, and Ready for Mass Production
D/A	Datasheet Applicable, requires Sample Testing
N/A	Not Applicable

- **Selection of High-Performance eMMC Chips**

To enhance system performance, it is necessary to choose high-performance eMMC chips. Before selecting an eMMC chip, please refer to the models in the support list provided by Rockchip and pay close attention to the `performance` section in the manufacturer's datasheet.

Please select eMMC chips based on the manufacturer's size specifications and the read/write speeds. It is recommended to choose chips with sequential read speeds >200MB/s and sequential write speeds >40MB/s.

If there are any doubts about selection, you can also directly contact the Rockchip FAE team fae@rock-chips.com.

6.1.5 Performance

[Table 23] Performance

Density	Partition Type	Performance	
		Read(MB/s)	Write (MB/s)
16GB	General	285	40
32GB		310	70
64GB		310	140
128GB		310	140
16GB	Enhanced	295	80
32GB		320	150
64GB		320	245
128GB		320	245

Figure 1-1 eMMC Performance Example

3.6.3.3 Flash Support List

For the support status of SPI Nor, SPI NAND, SLC, and other Flash devices on the Rockchip platform, please refer to `<SDK>/docs/en/Common/AVL` in **Rockchip_Flash_Approved_Vendor_List_xxx.pdf**. This document contains support and validation information for each Flash model and is useful for device selection and design evaluation. The table below shows the Flash support levels; it is recommended to select chips marked

with √ or T/A only.

Table 1-3 Rockchip SPI Nor, SPI NAND, SLC Support Symbol

Symbol	Description
√	Fully Tested, Applicable and Mass Production
T/A	Fully Tested, Applicable and Ready for Mass Production
D/A	Datasheet Applicable, requires Sample Testing
N/A	Not Applicable

3.6.3.4 UFS Support List

The Rockchip platform UFS support list can be found in `<SDK>/docs/en/Common/AVL` under **Rockchip_UFS_Approved_Vendor_List_xxx.pdf**. The document lists UFS models for device selection. The table below shows the UFS support levels; it is recommended to select chips marked with √ or T/A only.

Table 1-5 Rockchip UFS Support Symbol

Symbol	Description
√	Fully Tested, Applicable and Mass Production
T/A	Fully Tested, Applicable and Ready for Mass Production
D/A	Datasheet Applicable, requires Sample Testing
N/A	Not Applicable

3.6.3.5 WIFI/BT Support List

WIFI/BT Support List for Rockchip Platform can be found in the document titled 'Rockchip_Support_List_Linux_WiFi_BT_Ver2.0_20250619.pdf' in the `<SDK>/docs/en/Common/AVL` directory. This document contains a comprehensive list of WIFI/BT chips tested extensively on the Rockchip platform. It is advisable to select models based on the list. For the testing of other WIFI/BT chips, corresponding kernel drivers should be provided from the original manufacturers of the WIFI/BT chips.

For any questions about chip selection, it is recommended to contact the Rockchip FAE team at fae@rock-chips.com.

3.6.3.6 Camera Support List

Camera Support List for Rockchip Platform can be found in the Camera Module Support List. This online list contains a comprehensive collection of Camera Modules extensively tested on the Rockchip platform. It is advisable to select models based on the list.

For any questions about module selection, it is recommended to contact the Rockchip FAE team at sw.fae@rockchips.com.

3.6.4 CAN Module Document (CAN)

CAN bus, also known as Controller Area Network, is an efficient serial communication network used for distributed control or real-time control. The following documents mainly cover CAN driver development, communication testing tools, common command interfaces, and frequently asked questions.

```
docs/en/Common/CAN/
├── Rockchip_Developer_Guide_CAN_FD_EN.pdf
└── Rockchip_Developer_Guide_Can_EN.pdf
```

3.6.5 Clock Module Document (CLK)

This document primarily covers clock development on the Rockchip platform, including Clock, GPIO, PLL spreading, etc.

```
docs/en/Common/CLK/
├── Rockchip_Developer_Guide_Clock_EN.pdf
├── Rockchip_Developer_Guide_Gpio_Output_Clocks_EN.pdf
└── Rockchip_Developer_Guide_Pll_Ssmod_Clock_EN.pdf
```

3.6.6 CRYPTO Module Document (CRYPTO)

The following documents primarily focus on the development of Rockchip Crypto and HWRNG (TRNG), including driver development and upper-layer application development.

```
docs/en/Common/CRYPTO/
└── Rockchip_Developer_Guide_Crypto_HWRNG_EN.pdf
```

3.6.7 DDR Module Document (DDR)

This module document mainly includes DDR development guide, DDR issue troubleshooting, DDR chip validation process, DDR board layout instructions, DDR bandwidth tool usage, and DDR DQ eye diagram tool, etc. for Rockchip platform.

```
docs/en/Common/DDR/
├── Rockchip-Developer-Guide-DDR-EN.pdf
```

3.6.8 Debug Module Document (DEBUG)

This module document mainly includes introduction to the use of debugging tools such as Eclipse_OpenOCD, etc. for Rockchip platform.

```
docs/en/Common/DEBUG/
└── Rockchip_Developer_Guide_GNU_MCU_Eclipse_OpenOCD_CN.pdf
```

3.6.9 Display Module Document (DISPLAY)

This module document mainly includes development documents about DRM, DP, HDMI, LVDS, MIPI, RGB, RK628 and other display modules for Rockchip platform.

```
docs/en/Common/DISPLAY/
├── BT656-BT1120
├── DP
├── DRM
├── HDCP
├── HDMI
├── LVDS
├── MIPI
├── RGB
├── RK628
├── Vsync
└── eDP
```

3.6.10 DSMC Module Documentation (DSMC)

This module documentation mainly contains the development documentation for the Double Data Rate Serial Memory Controller (DSMC) on the Rockchip platform. The Double Data Rate Serial Memory Controller (DSMC) utilizes time-division multiplexing of command, address, and data lines, with data transfer occurring on both the rising and falling edges, featuring fewer pins and higher bandwidth.

```
docs/en/Common/DSMC/
├── Rockchip_Developer_Guide_DSMC_EN.pdf
└── Rockchip_Developer_Guide_SLAVE_DSMC_EN.pdf
```

3.6.11 Dynamic Frequency and Voltage Adjustment Module Documentation (DVFS)

This module document primarily covers CPU/GPU/DDR and other dynamic frequency and voltage adjustment modules for Rockchip platform.

Cpufreq and Devfreq are a set of framework models defined by kernel developers that support dynamic frequency and voltage adjustment based on specified governors. It effectively reduces power consumption while balancing performance.

```
docs/en/Common/DVFS/
├── Rockchip_Developer_Guide_CPUFreq_EN.pdf
└── Rockchip_Developer_Guide_Devfreq_EN.pdf
```

3.6.12 FLEXBUS Module Documentation (FLEXBUS)

This module documentation mainly contains development documentation related to FLEXBUS on the Rockchip platform.

```
docs/en/Common/FLEXBUS/
├── Rockchip_Developer_Guide_Linux_FLEXBUS_ADC_and_DAC_MODE_EN.pdf
├── Rockchip_Developer_Guide_Linux_FLEXBUS_EN.pdf
└── Rockchip_Developer_Guide_Linux_FLEXBUS_FSPI_MODE_EN.pdf
```

3.6.13 Ethernet Module Document (GMAC)

This module document primarily includes the development documentation related to the Ethernet GMAC interface on the Rockchip platform.

```
docs/en/Common/GMAC/
├── Rockchip_Developer_Guide_GMAC_PTP1588_EN.pdf
├── Rockchip_Developer_Guide_Linux_GMAC_EN.pdf
├── Rockchip_Developer_Guide_Linux_GMAC_DPDK_EN.pdf
├── Rockchip_Developer_Guide_Linux_GMAC_Mode_Configuration_EN.pdf
├── Rockchip_Developer_Guide_Linux_GMAC_RGMII_Delayline_EN.pdf
└── Rockchip_Developer_Guide_Linux_MAC_TO_MAC_EN.pdf
```

3.6.14 HDMI-IN Module Document (HDMI-IN)

This module document mainly contains development documents related to the HDMI-IN interface of the Rockchip platform.

```
docs/en/Common/HDMI-IN/
└── Rockchip_Developer_Guide_HDMI_RX_EN.pdf
```

3.6.15 I2C Module Document (I2C)

This module document mainly contains development documents related to I2C interface of the Rockchip platform.

```
docs/en/Common/I2C/
└── Rockchip_Developer_Guide_I2C_EN.pdf
```

3.6.16 IO Power Domain Module Document (IO-DOMAIN)

In the Rockchip platform, IO voltages generally include 1.8V, 3.3V, 2.5V, 5.0V, etc. Some IO interfaces support multiple voltage levels. The io-domain is responsible for configuring the IO power domain registers. It configures the corresponding voltage registers based on the actual hardware voltage range. Without proper configuration, the IO interfaces cannot function correctly.

```
docs/en/Common/IO-DOMAIN/
└── Rockchip_Developer_Guide_Linux_IO_DOMAIN_EN.pdf
```

3.6.17 IOMMU Module Document (IOMMU)

It mainly introduces the Rockchip platform IOMMU for converting 32-bit virtual addresses and physical addresses. It has read and write control bits and can generate page missing exceptions and bus exception interrupts.

```
docs/en/Common/IOMMU/
└── Rockchip_Developer_Guide_Linux_IOMMU_EN.pdf
```

3.6.18 Image Module Document (ISP)

ISP1.X is mainly suitable for RK3399/RK3288/PX30/RK3326/RK1808, etc.
ISP21 is mainly suitable for RK3566_RK3568, etc.
ISP30 is mainly suitable for RK3588, etc.
ISP32-lite is mainly suitable for RK3562, etc.
ISP35 is mainly suitable for RV1126B, etc.
ISP351 is mainly suitable for RK3572, etc.
ISP39 is mainly suitable for RK3576, etc.

It contains ISP development documents, VI driver development documents, IQ Tool development documents, debugging documents and color debugging documents. The reference documents are as follows:

```
docs/en/Common/ISP/
└── The-Latest-Camera-Documents-Link.txt
```

> **Note:**
> Reference documents about RK3288/RK3399/RK3326/RK1808 Linux(kernel-4.4) rkisp1 driver, sensor driver, vcm driver is: "RKISP_Driver_User_Manual_v1.3_20190919";
> RK3288/RK3399/RK3326/RK1808 Linux(kernel-4.4) camera_engine_rkisp (3A repositry) reference documents is: "camera_engine_rkisp_user_manual_v2.0";
> Reference document for IQ effect file parameters of RK3288/RK3399/RK3326/RK1808 Linux(kernel-4.4) camera_engine_rkisp v2.0.0 version v2.0.0 and above is: "RKISP1_IQ_Parameters_User_Guide_v1.0_20190606".

3.6.19 MCU Module Document (MCU)

MCU development guide on the Rockchip platform.

```
docs/en/Common/MCU/
└── Rockchip_RK3399_Developer_Guide_MCU_EN.pdf
```

3.6.20 MMC Module Document (MMC)

Development guide for interfaces such as SDIO, SDMMC, and eMMC on the Rockchip platform.

```
docs/en/Common/MMC/
├── Rockchip_Developer_Guide_SDMMC_SDIO_eMMC_EN.pdf
└── Rockchip_Developer_Guide_SD_Boot_EN.pdf
```

3.6.21 Memory Module Document (MEMORY)

Process memory module mechanisms such as CMA and DMABUF on the Rockchip platform.

```
docs/en/Common/MEMORY/
├── Rockchip_Developer_Guide_Linux_CMA_EN.pdf
├── Rockchip_Developer_Guide_Linux_DMABUF_EN.pdf
├── Rockchip_Developer_Guide_Linux_Meminfo_EN.pdf
└── Rockchip_Developer_Guide_Linux_Memory_Allocator_EN.pdf
```

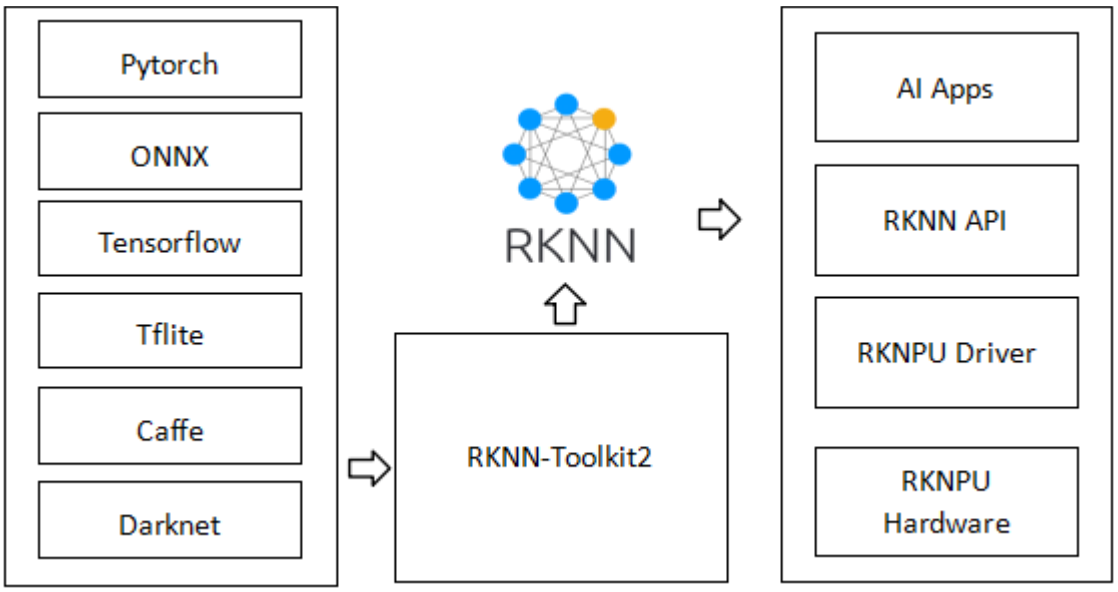
3.6.22 MPP Module Document (MPP)

MPP development instructions on the Rockchip platform.

```
docs/en/Common/MPP/
└── Rockchip_Developer_Guide_MPP_EN.pdf
```

3.6.23 NPU Module Document (NPU)

RKNN software stack can help users to quickly deploy AI models to Rockchip chips. The overall framework is as follows:



In order to use RKNPU, users need to first run the RKNN-Toolkit2 tool on the computer, convert the trained model into an RKNN format model, and then inference on the development board using the RKNN C API or Python API.

3.6.23.1 RKNN-TOOLKIT2

RKNN-Toolkit2 is a software development kit for users to perform model conversion, inference and performance evaluation on PC and Rockchip NPU platforms.

RKNN-Toolkit-Lite2 provides Python programming interfaces for Rockchip NPU platform to help users deploy RKNN models and accelerate the implementation of AI applications.

The development kit is located in the `external/rknn-toolkit2` directory, mainly used to implement a series of functions such as model conversion, optimization, quantization, inference, performance evaluation, and accuracy analysis.

Basic functions are as follows:

Function	Description
Model Conversion	Supports floating-point models of Pytorch / TensorFlow / TFLite / ONNX / Caffe / Darknet Supports quantization-aware models (QAT) of Pytorch / TensorFlow / TFLite Supports dynamic input models (dynamicization/native dynamic) Supports large models
Model Optimization	Constant folding / OP correction / OP Fuse&Convert / Weight sparsification / Model pruning
Model Quantization	Supports quantization types: asymmetric i8/fp16 Supports Layer / Channel quantization methods; Normal / KL/ MMSE quantization algorithms Supports mixed quantization to balance performance and accuracy
Model Inference	Supports model inference through the simulator on the PC Supports model inference on the NPU hardware platform (board-level inference) Supports batch inference, supports multi-input models
Model Evaluation	Supports performance and memory evaluation of the model on the NPU hardware platform
Accuracy Analysis	Supports quantization accuracy analysis function (simulator/NPU)
Additional Features	Supports version/device query functions, etc.

3.6.23.2 RKNN Runtime

The development documentation for RKNN Runtime is located in the `external/rknpu2` directory of the project. It is used to infer models generated by RKNN-Toolkit2 and provides C/C++ programming interfaces for the Rockchip NPU platform, assisting users in deploying RKNN models and accelerating AI application implementation.

For detailed usage instructions, refer to the documentation in the current rknpu2/ directory:

```
├── 01_Rockchip_RKNPU_Quick_Start_RKNN_SDK_V2.3.2_EN.pdf
├── 02_Rockchip_RKNPU_User_Guide_RKNN_SDK_V2.3.2_EN.pdf
├── 03_Rockchip_RKNPU_API_Reference_RKNN_Toolkit2_V2.3.2_EN.pdf
├── 04_Rockchip_RKNPU_API_Reference_RKNNRT_V2.3.2_EN.pdf
├── 05_RKNN_Compiler_Support_Operator_List_V2.3.2.pdf
├── RKNNToolKit2_API_Difference_With_Toolkit1-2.3.2.md
├── RKNNToolKit2_OP_Support-2.3.2.md
...
```

3.6.23.3 RKNN Driver

The RKNN kernel driver handles interactions with NPU hardware and is integrated into the Rockchip kernel code under `drivers/rknpu`.

3.6.23.4 RKNN LLM

RKNN LLM is an inference framework optimized by Rockchip for large language model (LLM) scenarios. It aims to help users efficiently deploy and run large pre-trained language models (e.g., LLaMA, PaLM, ChatGLM) on Rockchip's proprietary NPU hardware platform.

For detailed usage instructions, refer to the documentation in the current rknpu2/ directory:

```
├── Rockchip_RKLLM_SDK_EN_1.2.3.pdf
```

3.6.24 NVM Module Document (NVM)

It mainly introduces the boot process on the Rockchip platform, configuring and debugging storage, OTP OEM area burning and other security interfaces.

```
docs/en/Common/NVM/
├── Rockchip_Application_Notes_Storage_EN.pdf
├── Rockchip_Application_Notes_VendorStorage_EN.pdf
├── Rockchip_Developer_Guide_Linux_Flash_Open_Source_Solution_EN.pdf
└── Rockchip_Developer_Guide_Secure_Boot_Application_Note_EN.pdf
```

3.6.25 PCIe Module Document (PCIe)

It mainly introduces the development instructions of PCIe on Rockchip platform.

```
docs/en/Common/PCIe/
├── Rockchip_Developer_Guide_PCIE_EP_Standard_Card_APIs_EN.pdf
├── Rockchip_Developer_Guide_PCIE_EP_Standard_Card_EN.pdf
├── Rockchip_Developer_Guide_PCIe_EN.pdf
├── Rockchip_Developer_Guide_PCIe_EP_EN.pdf
├── Rockchip_Developer_Guide_PCIe_Performance_EN.pdf
├── Rockchip_PCIe_Virtualization_Developer_Guide_EN.pdf
├── Rockchip_RK3399_Developer_Guide_PCIe_EN.pdf
└── Rockchip_Trouble_Shooting_Linux_PCIe_EN.pdf
```

3.6.26 Performance Module Document (PERF)

Introduction to PERF Performance analysis on Rockchip Platform

```
docs/en/Common/PERF/
├── Rockchip_Developer_Guide_Linux_RealTime_Performance_Test_Report_CN.pdf
└── Rockchip_Quick_Start_Linux_Performance_Analyse_CN.pdf
```

3.6.27 PINCTRL Module Document (PINCTRL)

PIN-CTRL driver and DTS usage method on Rockchip platform.

```
docs/en/Common/PINCTRL/
└── Rockchip_Developer_Guide_Linux_Pinctrl_EN.pdf
```

3.6.28 PMIC Module Document (PMIC)

Developer guide for PMICs such as RK805, RK806, RK808, RK809, and RK817 on the Rockchip platform.

```
docs/en/Common/PMIC/
├── Rockchip_RK801_Developer_Guide_EN.pdf
├── Rockchip_RK805_Developer_Guide_EN.pdf
├── Rockchip_RK806_Developer_Guide_EN.pdf
├── Rockchip_RK808_Developer_Guide_EN.pdf
├── Rockchip_RK809_Developer_Guide_EN.pdf
├── Rockchip_RK816_Developer_Guide_EN.pdf
├── Rockchip_RK817_Developer_Guide_EN.pdf
├── Rockchip_RK818_Developer_Guide_EN.pdf
├── Rockchip_RK818_RK816_Developer_Guide_Fuel_Gauge_EN.pdf
└── Rockchip_RK818_RK816_Introduction_Fuel_Gauge_Log_EN.pdf
```

3.6.29 Power Module Document (POWER)

Basic concepts and optimization methods for chip power consumption on Rockchip platform.

```
docs/en/Common/POWER/
└── Rockchip_Developer_Guide_Power_Analysis_EN.pdf
```

3.6.30 PWM Module Document (PWM)

PWM developer guide on the Rockchip platform.

```
docs/en/Common/PWM
└── Rockchip_Developer_Guide_Linux_PWM_EN.pdf
```

3.6.31 RGA Module Document (RGA)

RGA developer guide on the Rockchip platform.

```
docs/en/Common/RGA/
├── Rockchip_Developer_Guide_RGA_EN.pdf
└── Rockchip_FAQ_RGA_EN.pdf
```

3.6.32 SARADC Module Document (SARADC)

SARADC developer guide on the Rockchip platform.

```
docs/en/Common/SARADC/
└── Rockchip_Developer_Guide_Linux_SARADC_EN.pdf
```

3.6.33 SPI Module Document (SPI)

SPI developer guide on the Rockchip platform.

```
docs/en/Common/SPI/
└── Rockchip_Developer_Guide_Linux_SPI_EN.pdf
```

3.6.34 Thermal Module Document (THERMAL)

Thermal developer guide on the Rockchip platform.

```
docs/en/Common/THERMAL/
└── Rockchip_Developer_Guide_Thermal_EN.pdf
```

3.6.35 Tools Module Document (TOOL)

Instructions for using tools such as partitioning, mass production burning, and factory line burning on the Rockchip platform.

```
docs/en/Common/TOOL/
├── Rockchip_Introduction_Partition_EN.pdf
└── Rockchip_User_Guide_Production_For_Firmware_Download_EN.pdf
```

3.6.36 Security Module Document (TRUST)

Introduction to functions such as TRUST and sleep & wake-up on the Rockchip platform

```
docs/en/Common/TRUST/
├── Rockchip_Developer_Guide_Trust_EN.pdf
├── Rockchip_RK3308_Developer_Guide_System_Suspend_EN.pdf
├── Rockchip_RK3399_Developer_Guide_System_Suspend_EN.pdf
├── Rockchip_RK3506_Developer_Guide_System_Suspend_EN.pdf
├── Rockchip_RK356X_Developer_Guide_System_Suspend_EN.pdf
├── Rockchip_RK3576_Developer_Guide_System_Suspend_EN.pdf
└── Rockchip_RK3588_Developer_Guide_System_Suspend_EN.pdf
```

3.6.37 UART Module Document (UART)

Introduction to serial port functions and debugging on the Rockchip Platform

```
docs/en/Common/UART/
├── Rockchip_Developer_Guide_UART_EN.pdf
└── Rockchip_Developer_Guide_UART_FAQ_EN.pdf
```

3.6.38 UBOOT Module Document (UBOOT)

Introduction to U-Boot related development on the Rockchip platform

```
docs/en/Common/UBOOT/
├── Rockchip_Developer_Guide_U-Boot_V5_EN.pdf
└── Rockchip_Developer_Guide_UBoot_Nextdev_EN.pdf
```

3.6.39 USB Module Document (USB)

Introduction to USB development guide, USB signal testing and debugging tools on the Rockchip platform

```
docs/en/Common/USB/
├── Rockchip_Developer_Guide_Linux_USB_Performance_Analysis_EN.pdf
├── Rockchip_Developer_Guide_USB2_Compliance_Test_EN.pdf
├── Rockchip_Developer_Guide_USB_EN.pdf
├── Rockchip_Developer_Guide_USB_Gadget_UAC_EN.pdf
├── Rockchip_Developer_Guide_USB_SQ_Test_EN.pdf
├── Rockchip_Introduction_USB_SQ_Tool_EN.pdf
├── Rockchip_RK3399_Developer_Guide_USB_EN.pdf
├── Rockchip_RK356x_Developer_Guide_USB_EN.pdf
├── Rockchip_RK3572_Developer_Guide_USB_EN.pdf
├── Rockchip_RK3576_Developer_Guide_USB_EN.pdf
├── Rockchip_RK3588_Developer_Guide_USB_EN.pdf
└── Rockchip_Trouble_Shooting_Linux_USB_Host_UVC_EN.pdf
```

3.6.40 Watchdog Module Document (WATCHDOG)

Development instructions for Watchdog on the Rockchip platform.

```
docs/en/Common/WATCHDOG/
└── Rockchip_Developer_Guide_Linux_WDT_EN.pdf
```

3.6.41 Algorithm Module (ALGORITHM)

This module mainly contains algorithm development documents for Rockchip platform EIS (Electronic Image Stabilization) and IDC (Image Distortion Correction).

```
docs/en/Common/ALGORITHM/
├── EIS/
│   └── Rockchip_Developer_Guide_EIS_EN.pdf
└── IDC/
    ├── Rockchip_IDC_Developer_Guide_EN.pdf
    └── Rockchip_IDC_Developer_Tutorial_EN.pdf
```

3.6.42 GPIO Module (GPIO)

This module mainly contains GPIO related development documents for Rockchip platform.

```
docs/en/Common/GPIO/
└── Rockchip_Developer_Guide_GPIO_EN.pdf
```

3.6.43 IMU Module (IMU)

This module mainly contains IMU (Inertial Measurement Unit) related development documents for Rockchip platform.

```
docs/en/Common/IMU/
└── Rockchip_Developer_Guide_Linux_IMU_EN.pdf
```

3.6.44 RT-Thread Module (RTT)

This module mainly contains RT-Thread real-time operating system related development documents for Rockchip platform, including Display, I2C, PWM, SPI, UART, USB and other peripheral driver development guides.

```
docs/en/Common/RTT/
├── Rockchip_Developer_Guide_RT-Thread_Display_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_I2C_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_PWM_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_SPI2APB_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_SPIFLASH_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_SPI_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_SPI_Screen_EN.pdf
├── Rockchip_Developer_Guide_RT-Thread_UART.pdf
├── Rockchip_Developer_Guide_RT-Thread_USB_EN.pdf
└── Rockchip_Developer_Guide_RT-Thread_rkdemo_EN.pdf
```

3.7 Linux System Development Documents (Linux)

Please refer to documents under the `<SDK>/docs/en/Linux`:

```
├── ApplicationNote
├── Audio
├── Camera
├── DPDK
├── Docker
├── Graphics
├── Multimedia
├── Profile
├── RKAI
├── Recovery
├── Security
├── System
├── Uefi
└── Wifibt
```

3.7.1 ApplicationNote

Development instructions for applications on the Rockchip platform, such as ROS, USB, etc

```
docs/en/Linux/ApplicationNote/
├── Rockchip_Developer_Guide_Linux_RKIPC_EN.pdf
├── Rockchip_Instruction_Linux_ROS2_EN.pdf
├── Rockchip_Quick_Start_Linux_USB_Gadget_EN.pdf
└── Rockchip_Use_Guide_Linux_EtherCAT_IgH_EN.pdf
```

3.7.2 Audio Development Documents (Audio)

Self developed audio algorithm on Rockchip platform.

```
docs/en/Linux/Audio/
├── Algorithms/
│   ├── Rockchip_Developer_Guide_Audio_Algorithm_VQE_Introduction_EN.pdf
│   ├── Rockchip_Developer_Guide_RockAA_Utils_EN.pdf
│   └── Rockchip_Developer_Guide_Sound_Event_Detection_EN.pdf
└── Rockchip_Developer_Guide_PulseAudio_EN.pdf
```

3.7.3 Camera Development Documents (Camera)

MIPI/CSI Camera and Structured Light Development Guide on the Rockchip Platform

```
docs/en/Linux/Camera/
├── Rockchip_Developer_Guide_Linux4.4_Camera_EN.pdf
├── Rockchip_Developer_Guide_Linux_RMSL_EN.pdf
└── Rockchip_Trouble_Shooting_Linux4.4_Camera_EN.pdf
└── Rockchip_Trouble_Shooting_Linux5.10_Camera_EN.pdf
```

3.7.4 Docker Development Documents (Docker)

Docker build and development of third-party systems such as Debian/Buildroot on the Rockchip platform.

```
docs/en/Linux/Docker/
├── Rockchip_Developer_Guide_Debian_Docker_EN.pdf
└── Rockchip_Developer_Guide_Linux_Docker_Deploy_EN.pdf
```

3.7.5 DPDK Module (DPDK)

DPDK development guide on Rockchip Linux platform.

```
docs/en/Linux/DPDK/
└── Rockchip_Developer_Guide_Linux_DPDK_EN.pdf
```

3.7.6 Graphics Development Documents (Graphics)

Linux Graphics related development on Rockchip Platform.

```
docs/en/Linux/Graphics/
├── Rockchip_Developer_Guide_Buildroot_Weston_EN.pdf
└── Rockchip_Developer_Guide_Linux_Graphics_EN.pdf
└── Rockchip_Developer_Guide_Linux_LVGL_EN.pdf
```

3.7.7 Multimedia

The general process of video encoding and decoding on the Rockchip Linux platform:

```
vpu_service  -->  mpp --> gstreamer/rockit --> app
vpu_service: driver
MPP: A video encoding and decoding middleware for the Rockchip platform. Please refer to the
MPP documentation for detailed instructions
Gstreamer/rockit: used to connect components such as apps
```

Currently, gstreamer is used in Debian/Buildroot systems by default to connect apps and codec components.

Currently, the key development documents are as follows:

```
docs/en/Linux/Multimedia/
├── Rockchip_Developer_Guide_Linux_RKADK_EN.pdf
├── Rockchip_User_Guide_Linux_Gstreamer_EN.pdf
└── Rockchip_User_Guide_Linux_Rockit_EN.pdf
```

Encoding and decoding functionalities can also be tested directly through the testing interfaces provided by MPP (such as mpi_dec_test\mpi_enc_test...). MPP source code reference is located in `<SDK>/external/mpp/`. For testing demos, please refer to `<SDK>/external/mpp/test`. Please refer to the SDK document `Rockchip_Developer_Guide_MPP_EN.pdf` for details.

Rockchip chips like RK3588 support powerful multimedia capabilities:

- Supports H.265/H.264/AV1/VP9/AVS2 video decoding, up to 8K60FPS, also supports 1080P multi-format video decoding (H.263, MPEG1/2/4, VP8, JPEG).
- Supports 8K H.264/H.265 video encoding and 1080P VP8, JPEG video encoding.
- Video post-processing: deinterlacing, noise reduction, edge/detail/color optimization.

Below are the reference specifications for common chip encoding and decoding capabilities on each platform.

> **Note:** The maximum testing specifications are related to numerous factors, so the same decoding IP specifications might differ among different chips. Chip support may vary in different systems, leading to differences in supported formats and performance.

- **Decoding Capability Specification Table**

Chip	H264	H265	VP9	JPEG
RV1126B	3840x2160@30fps	3840x2160@30fps	N/A	3840x2160@30fps
RK3588	7680X4320@30f	7680X4320@60f	7680X4320@60f	1920x1088@200f
RK3576	3840x2160@60fps	7680x4320@30fps	7680x4320@30fps	1920x1088@200f
RK3566/RK3568	4096x2304@60f	4096x2304@60f	4096x2304@60f	1920x1080@60f
RK3562	1920x1088@60f	2304x1440@30f	4096x2304@30f	1920x1080@120f
RK3399	4096x2304@30f	4096x2304@60f	4096x2304@60f	1920x1088@30f
RK3328	4096x2304@30f	4096x2304@60f	4096x2304@60f	1920x1088@30f
RK3288	3840x2160@30f	4096x2304@60f	N/A	1920x1080@30f
RK3326	1920x1088@60f	1920x1088@60f	N/A	1920x1080@30f
PX30	1920x1088@60f	1920x1088@60f	N/A	1920x1080@30f
RK312X	1920x1088@30f	1920x1088@60f	N/A	1920x1080@30f

- **Coding Capability Specification Table**

Chip	H264	H265	VP8
RV1126B	3840x2160@30fps	3840x2160@30fps	N/A
RK3588	7680x4320@30f	7680x4320@30f	1920x1088@30f
RK3576	4096x2304@60fps	4096x2304@60fps	N/A
RK3566/RK3568	1920x1088@60f	1920x1088@60f	N/A
RK3562	1920x1088@60f	N/A	N/A
RK3399	1920x1088@30f	N/A	1920x1088@30f
RK3328	1920x1088@30f	1920x1088@30f	1920x1088@30f
RK3288	1920x1088@30f	N/A	1920x1088@30f
RK3326	1920x1088@30f	N/A	1920x1088@30f
PX30	1920x1088@30f	N/A	1920x1088@30f
RK312X	1920x1088@30f	N/A	1920x1088@30f

3.7.8 SDK Profile introduction (Profile)

Including software testing, benchmarks, etc. on Rockchip Linux platform.

```
docs/en/Linux/Profile/
├── Rockchip_Developer_Guide_Linux_PCBA_EN.pdf
├── Rockchip_Introduction_Linux_Benchmark_KPI_EN.pdf
├── Rockchip_Introduction_Linux_PLT_EN.pdf
└── Rockchip_User_Guide_Linux_Software_Test_EN.pdf
```

3.7.9 OTA Upgrade (Recovery)

An introduction to the recovery development process and upgrade during OTA upgrade of Rockchip Linux platform.

```

```

3.7.10 Security Solution (Security)

Introduction to the secure boot solution of Secure Boot and TEE on Rockchip Linux platform

```
docs/en/Linux/Security/
└── Rockchip_Developer_Guide_Linux_Secure_Boot_EN.pdf
```

3.7.11 System Development (System)

Introduction to the porting and development guide for Debian and other third-party systems on the Rockchip Linux platform

```
docs/en/Linux/System/
├── Rockchip_Developer_Guide_Buildroot_EN.pdf
├── Rockchip_Developer_Guide_Debian_EN.pdf
└── Rockchip_Developer_Guide_Third_Party_System_Adaptation_EN.pdf
```

3.7.12 UEFI Booting (UEFI)

Introduction to UEFI boot solution on Rockchip Linux platform.

```
docs/en/Linux/Uefi/
└── Rockchip_Developer_Guide_UEFI_EN.pdf
```

3.7.13 Network Module (RKWIFIBT)

Introduction to the development of WIFI, BT, etc. on Rockchip Linux platform.

```
docs/en/Linux/Wifibt/
├── Rockchip_Developer_Guide_Linux_WIFI_BT_EN.pdf
└── The-Latest-Wifibt-Documents-Link.txt
```

3.7.14 AI Framework (RKAI)

Introduction to RKAI related development on Rockchip Linux platform.

```
docs/en/Linux/RKAI/
└── Rockchip_Developer_Guide_Linux_RKAI_EN.pdf
```

3.7.15 Multimedia Development (Multimedia)

Introduction to multimedia related development on Rockchip Linux platform, including RKADK, Rockit Runtime, Gstreamer, etc.

```
docs/en/Linux/Multimedia/
├── Rockchip_Developer_Guide_Linux_Rkadk_EN.pdf
├── Rockchip_Developer_Guide_Linux_Rockit_Runtime_EN.pdf
├── Rockchip_User_Guide_Linux_Gstreamer_EN.pdf
└── Rockchip_User_Guide_Linux_Rockit_EN.pdf
```

3.8 Chip Platform Related Documents (Socs)

Refer to the documentation in the `<SDK>/docs/en/<chipset_name>` directory. Normally, it will include the release notes, quick start, software development guide, hardware development guide, Datasheet, etc.

3.8.1 Release Note

It contains an overview of the chip, supported main functions, instructions for obtaining the SDK, etc.

Reference document is in the `<SDK>/docs/en/<chipset_name>` directory:
`Rockchip_<chipset_name>_Linux_SDK_Release_<version>_EN.pdf`

3.8.2 Quick Start

Normally, it will include software and hardware development guide, SDK build, SDK pre-build firmware, SDK burning, etc.
Please refer to the documentation in the `<SDK>/docs/en/<chipset_name>/Quick-start` directory.

3.8.3 Software Development Guide

In order to help developers get familiar with the development and debugging of the SDK faster, the "Rockchip_Developer_Guide_Linux_Software_EN.pdf" document can be obtained from the /docs/en/<chip_name>/ directory and will be continuously improved and updated.

3.9 Datasheet

In order to help developers get familiar with chip development and debugging faster, a chip datasheet is released with the SDK.

Please refer to the documentation in the `<SDK>/docs/en/<chipset_name>/Datasheet` directory.

3.9.1 Hardware Development Guide

Rockchip platform will have corresponding hardware reference documents released with the SDK software package. The hardware user guide mainly introduces the basic features, hardware interfaces, and usage methods of the reference hardware board. It aims to assist developers in using the EVB more quickly and accurately, and in developing related products. For more details, please refer to the documents in the `<SDK>/docs/en/<chip_name>/Hardware` directory.

3.10 Other Documents (Others)

For other reference documents, such as Repo mirror environment construction, Rockchip SDK application and synchronization guide, Rockchip Bug system usage guide, etc., please refer to the documents in the `<SDK>/docs/en/Others` directory.

```
docs/en/Others/
├── Rockchip_Developer_Guide_Repo_Mirror_Server_Deploy_EN.pdf
├── Rockchip_User_Guide_Bug_System_EN.pdf
└── Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf
```

3.11 Documents List (docs_list_en.txt)

Please refer to the document /docs/en/docs_list_en.txt.

```
├── Common
├── Linux
├── Others
├── Rockchip_Developer_Guide_Linux_Software_EN.pdf
├──<chipset_name>
└── docs_list_en.txt
```

4. Chapter-4 Tools Introduction

Tools released with the Rockchip Linux SDK are used for development, debugging, and mass production process. The tool versions will be continuously updated along with SDK updates. If you have any questions or requirements about the tools, please contact our FAE team: fae@rock-chips.com.

In the Rockchip Linux SDK, tools are included in the `tools` directory for use in Linux (tools for Linux operating system environment), Mac (tools for macOS operating system environment), and Windows (tools for Windows operating system environment).

- Windows Tools

Tool usage instructions: `<SDK>/tools/windows/ToolsRelease.txt`

Tool name	Description
BoardProofTool	Anti-Piracy Tool
boot_merger	Packing or unpacking loader tool
DDR_UserTool	DDR user testing tool
DriverAssistant	Driver installation tool
EfuseTool	efuse burning tool
FactoryTool	Mass production upgrade tool
ParameterTool	Partition table modification tool
pin_debug_tool	GPIO debugging tool
programmer_image_tool	Programmer upgrade tool
rk_ddrbin_tool	RK ddrbin debugging tool
RKDevInfoWriteTool	Device information writing tool
RKDevTool	Discrete firmware upgrade and entire update firmware tool
RKDevTool_Release	Firmware burning tool
RKPCBATool	PCBA board testing tool
rk_sign_tool	Secureboot signature tool
Rockchip_HdcpKey_Writer	HDCP key burning tool
Rockchip_USB_SQ_Tool	USB PHY signal quality debugging work
SDDiskTool	SD card boot or upgrade image creation
SecureBootTool	Firmware Signature Tool
upgrade_tool	Command line upgrade tool
RKImageMaker	Command line packaging tool

- Linux Tools

Tool usage instructions: `<SDK>/tools/linux/ToolsRelease.txt`

Tool name	Description
boot_merger	Packing or unpacking loader tool
Firmware_Merger	SPI NOR firmware packing tool (the generated firmware can be used in the programmer)
Linux_DDR_Bandwidth_Tool	DDR bandwidth statistics tool
Linux_Diff_Firmware	OTA differential package tool
Linux_Pack_Firmware	Firmware packaging tool (packaged into update.img)
Linux_SecureBoot	Firmware Signature Tool
Linux_SecurityAVB	AVB Signature tool
Linux_SecurityDM	DM Signature Tool
Linux_Upgrade_Tool	Firmware burning tool
pin_debug_tool	GPIO debugging tool
programmer_image_tool	Packaged SPI NOR/SPI NAND/SLC NAND/eMMC programmer firmware
rk_ddrbin_tool	RK ddrbin debugging tool
rk_sign_tool	Secureboot signature tool

- Mac tools

Tool documentation: `<SDK>/tools/mac/ToolsRelease.txt`

Tool name	Description
boot_merger	Packing or unpacking loader tool
upgrade_tool	Command line upgrade tool
rk_sign_tool	Secureboot signature tool

4.1 Driver Installation Tool

Rockchip USB driver installation assistant is stored in `<SDK>/tools/windows/DriverAssistant_v5.13.zip`. xp, win7_32, win7_64, win10_32, win10_64 and other operating systems are supported.

The installation steps are as follows:



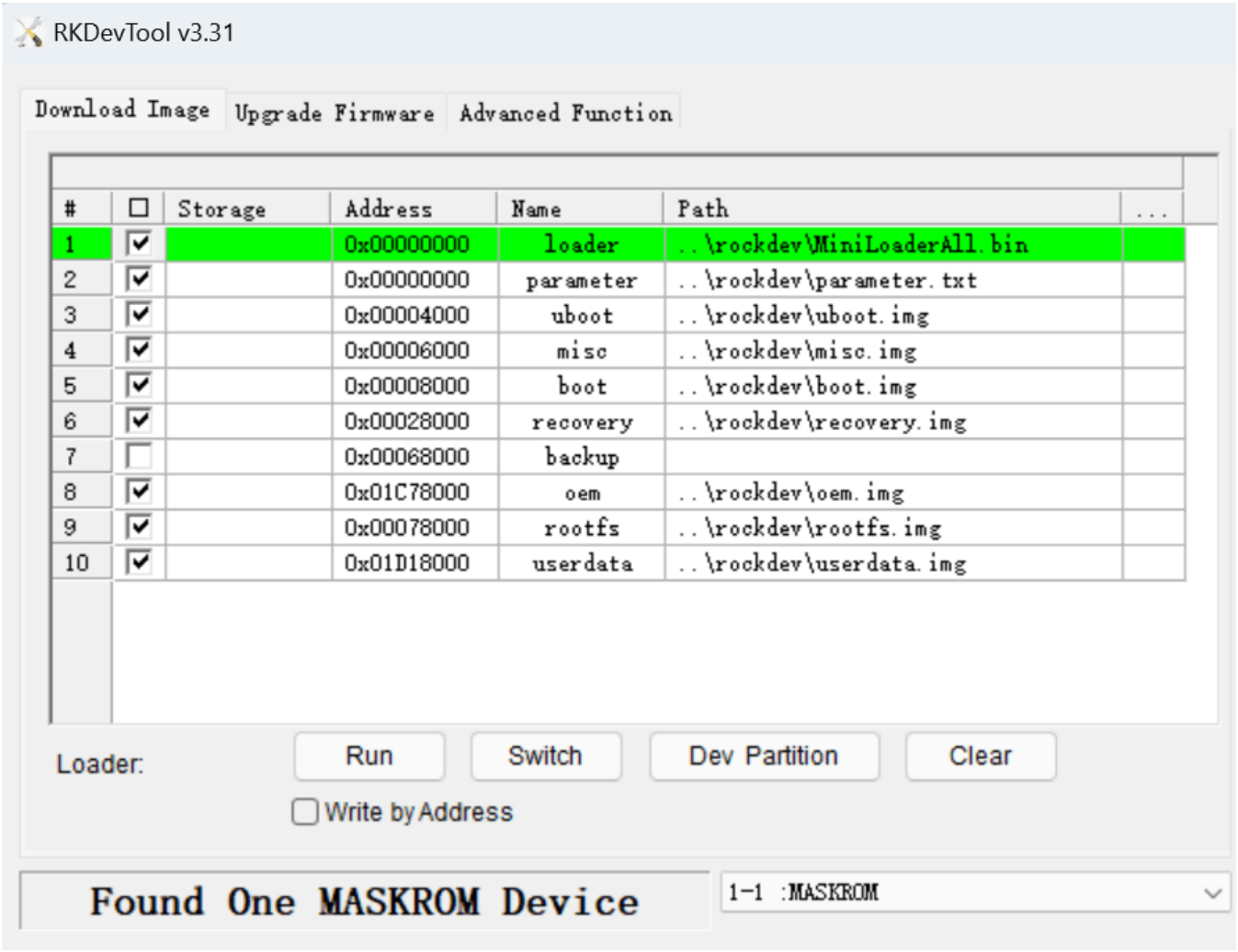




4.2 Burning Tools

- The SDK provides Windows burning tools (the tool version requires V3.31 or above), which is located in the project root directory:

```
<SDK>/tools/windows/RKDevTool/
```



- The SDK provides Linux burning tools (the version of Linux_Upgrade_Tool requires V2.26 or above), the tool is located in the project root directory:

```
<SDK>/tools/linux/Linux_Upgrade_Tool/Linux_Upgrade_Tool
```

```
Linux_Upgrade_Tool$ sudo ./upgrade_tool -h

--------------------Tool Usage --------------------
Help:              H
Version:           V
Log:               LG
-----------------Upgrade Command -----------------
ChooseDevice:        CD
ListDevice:      LD
SwitchDevice:        SD
UpgradeFirmware:     UF <Firmware> [-noreset]
UpgradeLoader:       UL <Loader> [-noreset] [FLASH|EMMC|SPINOR|SPINAND]
DownloadImage:       DI <-p|-b|-k|-s|-r|-m|-u|-t|-re image>
DownloadBoot:        DB <Loader>
EraseFlash:      EF <Loader|firmware>
PartitionList:       PL
WriteSN:         SN <serial number>
ReadSN:          RSN
ReadComLog:      RCL <File>
```

```
CreateGPT:      GPT <Input Parameter> <Output Gpt>
SwitchStorage:      SSD
----------------Professional Command -----------------
TestDevice:     TD
ResetDevice:        RD [subcode]
ResetPipe:      RP [pipe]
ReadCapability:     RCB
ReadFlashID:        RID
ReadFlashInfo:      RFI
ReadChipInfo:       RCI
ReadSecureMode:     RSM
WriteSector:        WS  <BeginSec> <PageSizeK> <PageSpareB> <File>
ReadLBA:        RL  <BeginSec> <SectorLen> [File]
WriteLBA:       WL  <BeginSec> [SizeSec] <File>
EraseLBA:       EL  <BeginSec> <EraseCount>
EraseBlock:     EB <CS> <BeginBlock> <BlockLen> [--Force]
RunSystem:      RUN <uboot_addr> <trust_addr> <boot_addr> <uboot> <trust> <boot>
-------------------------------------------------------
```

4.3 Packaging Tools

Mainly used to package each discrete firmware into a complete update.img firmware for easy upgrade.

- How to package update.img firmware in Windows environment, run the following command to generate update.img

```
<SDK>/tools/windows/RKDevTool/rockdev/mkupdate.bat
```

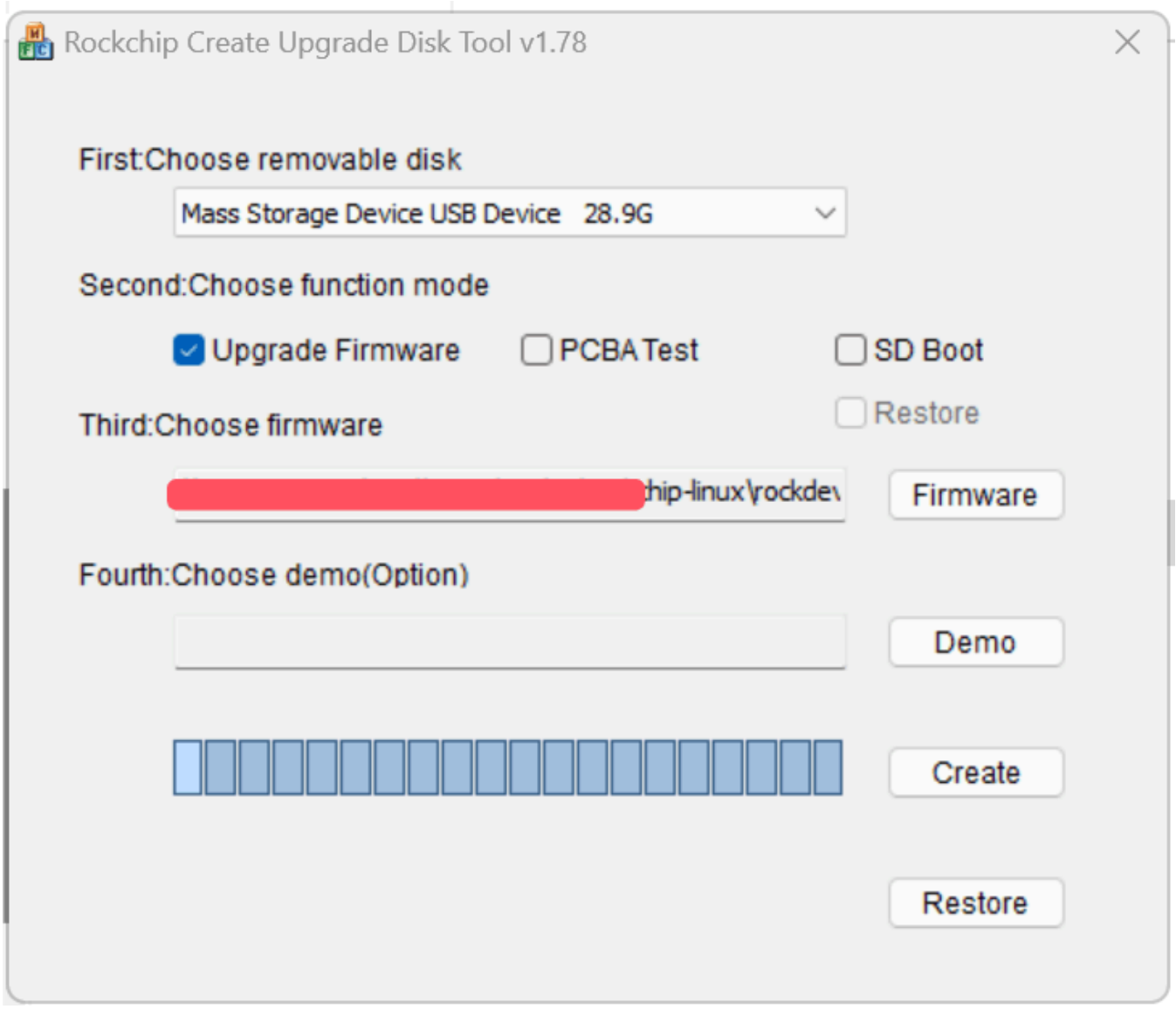
- How to package update.img firmware in Linux environment, run the following command to generate update.img

```
<SDK>/tools/linux/Linux_Pack_Firmware/rockdev/mkupdate.sh
```

4.4 Tools used to Make SD Card Upgrading and Booting

It is used for making SD card upgrades, SD card booting, and SD card PCBA testing.

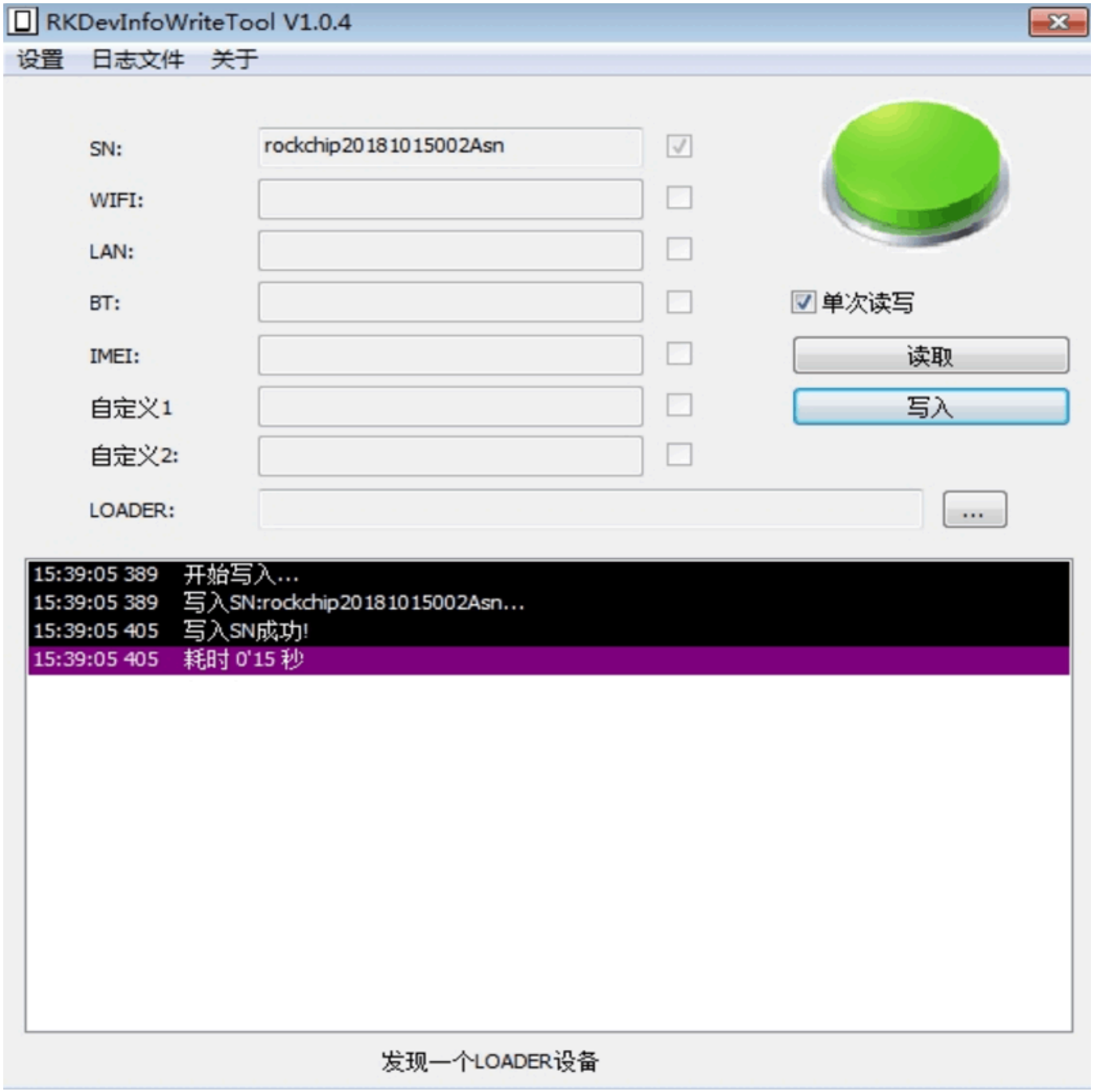
```
<SDK>/tools/windows/SDDiskTool_v1.78.zip
```



4.5 Device Information Writing Tool

`<SDK>/tools/windows/RKDevInfoWriteTool*`

Unzip RKDevInfoWriteTool-v1.3.7.zip and install it. Open the software with administrator rights. Please refer to the `Rockchip_User_Guide_RKDevInfoWriteTool_EN.pdf` in the current directory for tool usage.



4.6 Programmer Upgrade Tool

It is used for mass production of programmer image making tools, this tool is located at:

```
<SDK>/tools/windows/programmer_image_tool or <SDK>/tools/linux/programmer_image_tool
```

```
PS D:\Rockchip\vs_projects\programmer_image_tool> .\programmer_image_tool -h
NAME
        programmer_image_tool - creating image for programming on flash
SYNOPSIS
        programmer_image_tool [-iotbpsvh]
        This tool aims to convert firmware into image for programming
        From now on,it can support slc nand(rk)|spi nand|nor|emmc.
OPTIONS:
          -i    input firmware
          -o    output directory
          -t    storage type,range in[SLC|SPINAND|SPINOR|EMMC]
          -b    block size,unit KB
          -p    page size,unit KB
          -s    oob size,unit B
          -2    2k data in one page
          -l    using page linked list
          -v    show version
```

Programmer image creation steps:

- Burn image to eMMC

```
./programmer_image_tool -i update.img -t emmc
```

- Burn image to SPI Nor Flash

```
./programmer_image_tool -i update.img -t spinor
```

Please refer to the `user_manual.pdf` document in the tool directory for more instructions.

4.7 PCBA Testing Tools

Rockchip-PCBA Tool is a hardware testing tool designed for mass production environments. Its PC-side host software communicates with the board-side service program via ADB (Android Debug Bridge) to automate the testing of preset hardware functions. Focusing on FCT (Functional Test), the tool helps production units quickly identify product functionality, thereby effectively improving production efficiency and inspection consistency. Currently supported test items include screen (LCD), wireless (Wi-Fi), Bluetooth, DDR/EMMC storage, RTC clock, camera, SD card, USB HOST, keys (KEY), speaker/headphone (Codec), etc. Among these, wireless network, DDR/EMMC, RTC clock, and camera are automated test items, while keys, SD card, USB HOST, and Codec are manual test items.

Board-side source code is located in the project directory:

```
<SDK>/external/external/rk_pcba_test
```

PC-side source code can be obtained by contacting our Field Application Engineer (FAE).

Quick Start:

Board-side program deployment: In the /buildroot directory, run `source buildroot/envsetup.sh` to select the corresponding option for `rockchip_xxxx_pcba`. Use `make rockchip-pcba` to generate the base environment. Then return to the directory and execute `./build.sh buildroot` to obtain the rootfs.img firmware with PCBA tools. After flashing the firmware, the service program will be located in the system's `/data/` directory.

PC-side program installation: The PCBA tools are included in the compressed package.

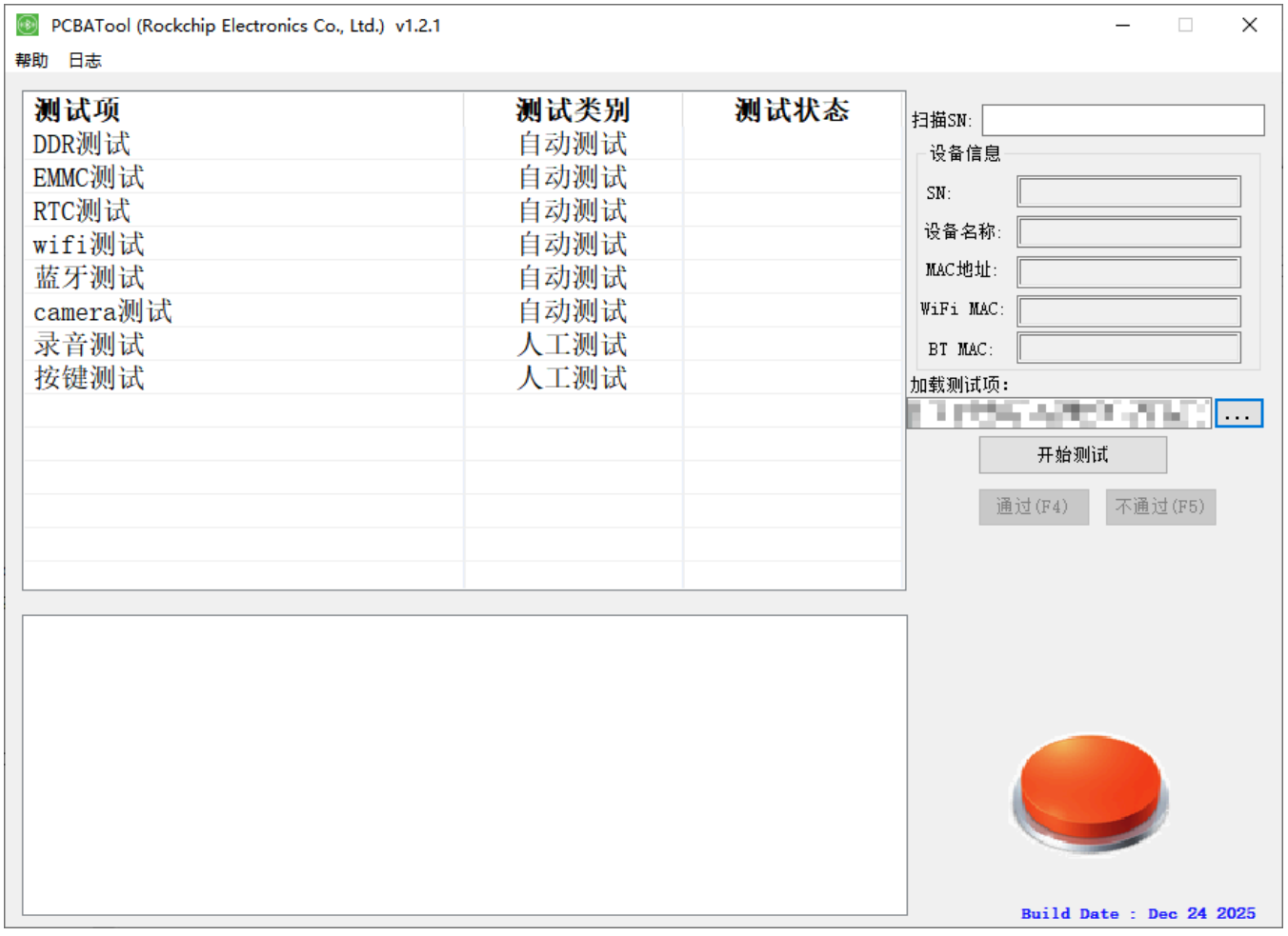
```
<SDK>/tools/windows/RKPCBATool_V1.2.1_20251226.zip
```

Users can install and use the tool with a single click via `RK-PCBATool-setup.exe` in the compressed package.

Secondary Development Notes:

To assist developers in quickly getting started with debugging and developing using the Rockchip-PCBA tool, refer to the documentation:

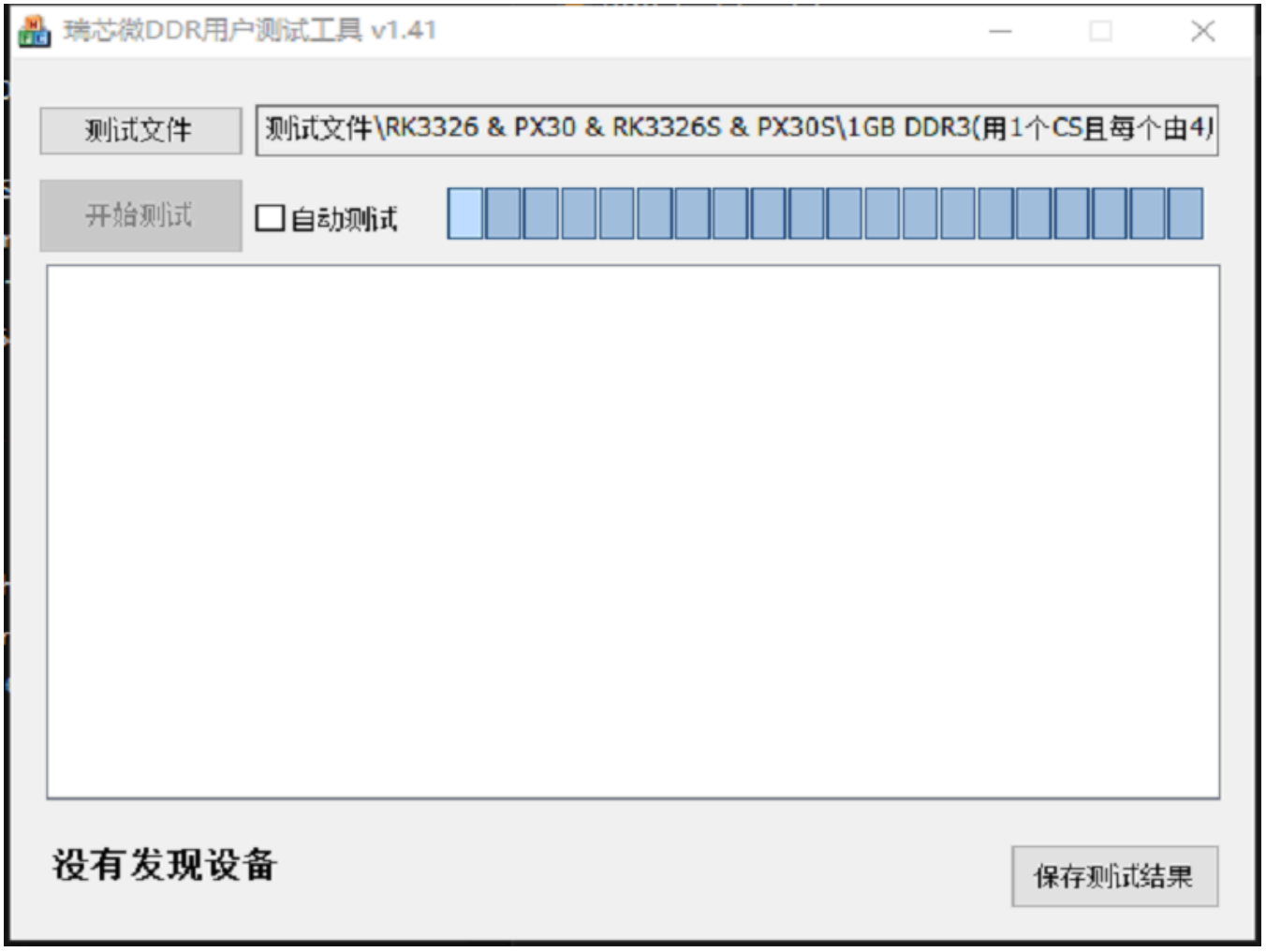
```
<SDK>/tools/windows/RKPCBATool_V1.2.1_20251226/Rockchip_Developer_Guide_Linux_PCBA_CN.pdf
```



4.8 DDR Soldering Test Tool

It is used to test the hardware connection of DDR and troubleshoot hardware problems such as false soldering:

```
<SDK>/tools/windows/DDR_UserTool_v1.41.zip
```

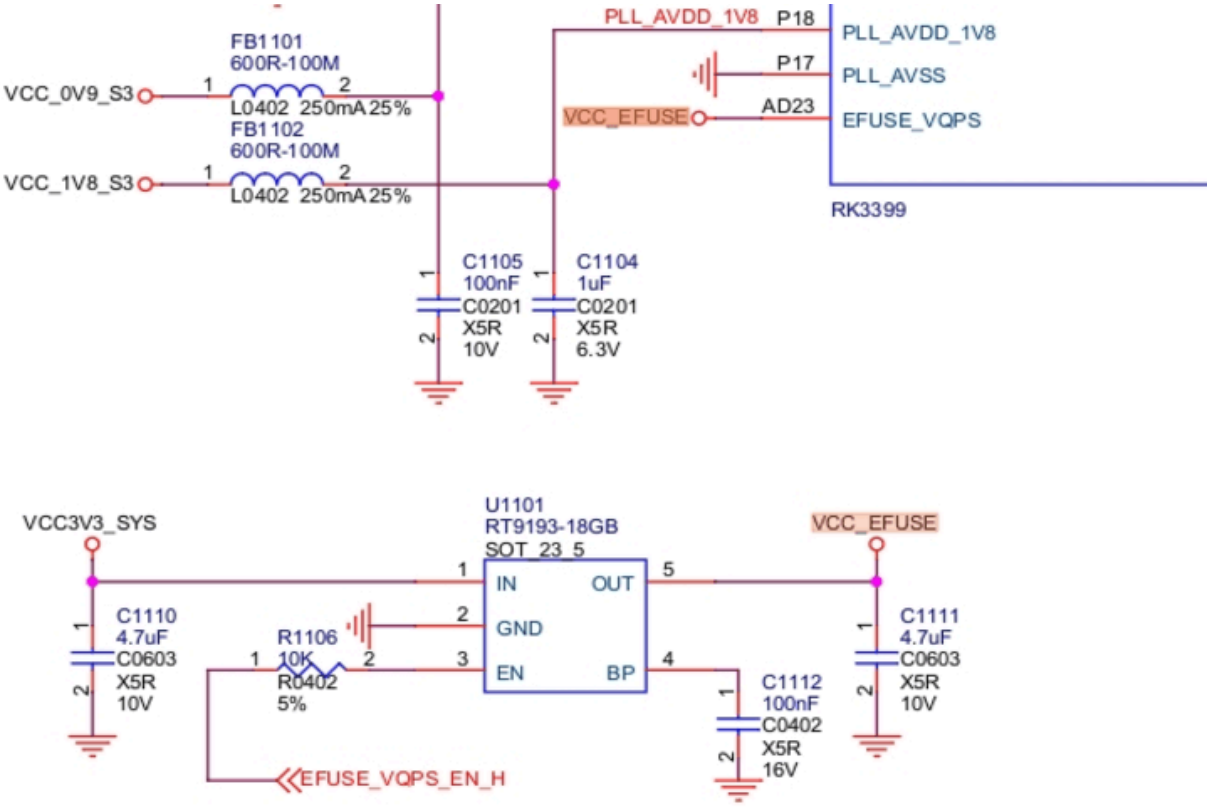


4.9 eFuse Programming Tool

It is used for eFuse programming, suitable for RK3288/RK3368/RK3399/RK3399Pro and other platforms.

```
<SDK>/tools/windows/EfuseTool_v1.42.zip
```

If the chip uses eFuse to enable the SecureBoot function, please ensure that there is no problem with the hardware connection, because when eFuse is programmed, Kernel has not been started yet, so please ensure that VCC_EFUSE has power in MaskRom state before it can be used.



The /Tools/windows/EfuseTool_v1.4.zip is used to let the board enter MaskRom state.
Click "Firmware", select the signed update.img, or Miniloader.bin, click "Start" to start programming eFuse.

4.10 Mass Production Upgrade Tool

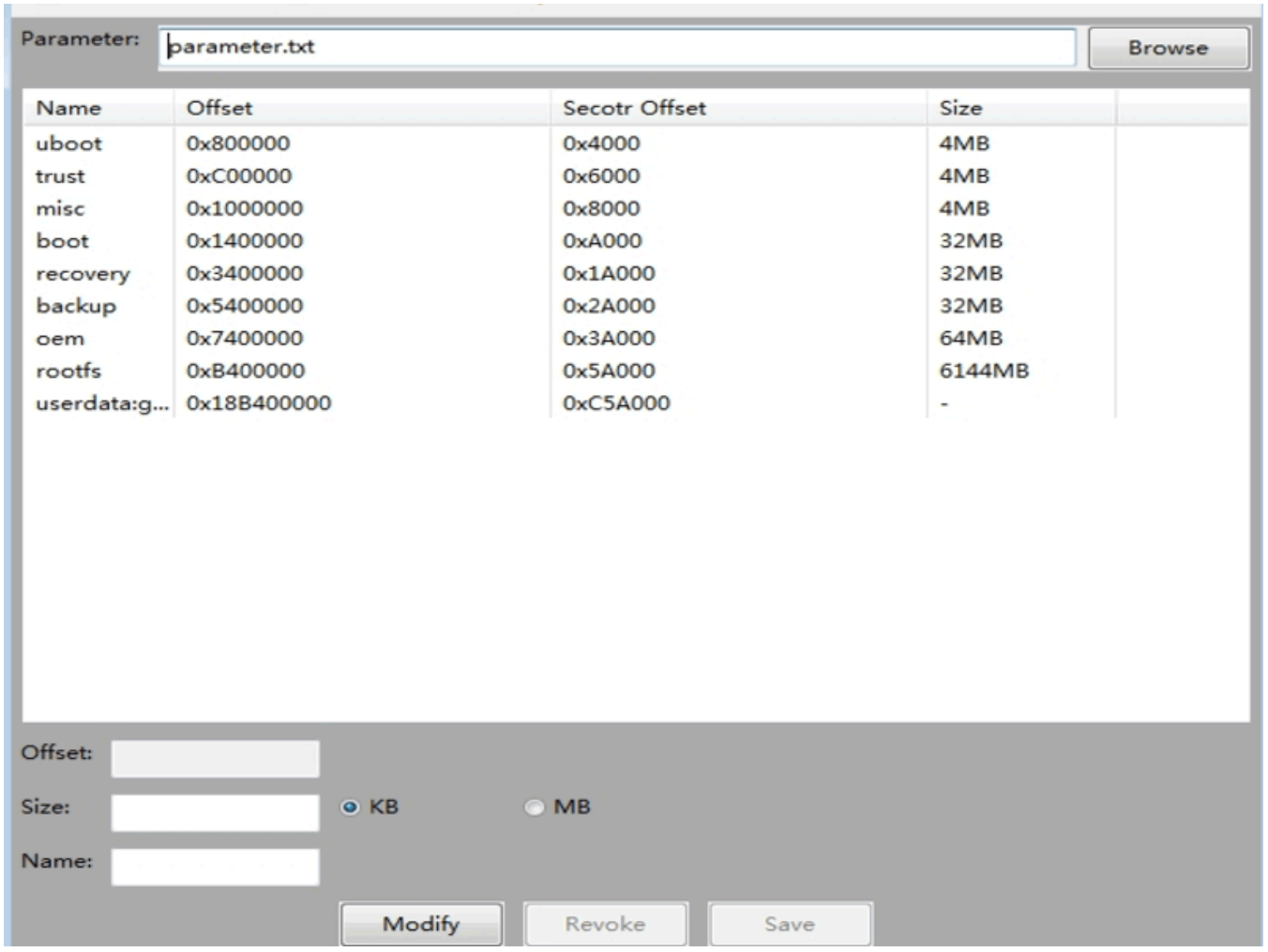
It is used for burning firmware during mass production in factory:

```
<SDK>/tools/windows/FactoryTool_v1.91.zip
```

4.11 Partition Modification Tool

The tool is used for partition modification in Parameter.txt:

```
<SDK>/tools/windows/ParameterTool_v1.2.zip
```


Parameter:
parameter.txt
Browse
Name
Offset
Secotr Offset
Size
uboot
0x800000
0x4000
4MB
trust
0xC00000
0x6000
4MB
misc
0x1000000
0x8000
4MB
boot
0x1400000
0xA000
32MB
recovery
0x3400000
0x1A000
32MB
backup
0x5400000
0x2A000
32MB
oem
0x7400000
0x3A000
64MB
rootfs
0xB400000
0x5A000
6144MB
userdata:g...
0x18B400000
0xC5A000
-
Offset:
Size:
KB
MB
Name:
Modify
Revoke
Save

5. Chapter-5 SDK Software Framework

5.1 Introduction to SDK project Directory

A typical Linux SDK project directory includes buildroot, debian, app, kernel, u-boot, device, docs, external, etc. Repositories are managed by manifests, and the `repo` tool is utilized to manage each directory or its corresponding Git project, including the following subdirectories:

- `app`: Contains upper-level application apps, mainly various application demos.
- `buildroot`: Contains the root file system developed with Buildroot.
- `debian`: Contains the root file system developed with Debian.
- `device/rockchip`: Contains chip board-level configurations and scripts/files for compiling and packaging firmware.
- `docs`: Contains general development guides, Linux system development guides, chip platform-related documents, etc.
- `external`: Contains third-party related repositories, including display, audio/video, camera, network, security, etc.
- `kernel`: Contains Kernel development code. In Linux6.12 SDK, the kernel directory is `kernel-6.12`, and may also contain historical kernel versions such as `kernel-6.1`, `kernel-5.10`, etc.
- `output`: Contains firmware information, compilation details, XML files, host environment, etc., generated during each build.
- `prebuilts`: Contains cross-compilation toolchains.
- `rkbin`: Contains Rockchip-related binaries and tools.
- `rockdev`: Contains compiled output firmware, actually, it soft-linked to xxx `output/firmware`.
- `tools`: Contains commonly used tools for both Linux and Windows operating systems.
- `u-boot`: Contains U-Boot code developed based on version next-dev.
- `yocto`: Contains the root file system developed with Yocto.

5.1.1 SDK Overview

The general Linux SDK currently integrates several mainstream Linux distributions. Distributions refer to the systems we commonly use on Linux hosts, providing users with pre-integrated Linux operating systems and various application software. Linux distributions come in various forms, ranging from fully-featured desktop systems to server versions and lightweight systems for small devices. For example, Debian supports desktop versions, and there are customizable lightweight systems like Buildroot and Yocto.

Customers can choose the system of product based on detailed product requirements. The specific system descriptions are as follows:

5.1.2 Buildroot

The Buildroot system in Rockchip Linux SDK includes various system source code, drivers, tools, and application software packages used for Linux system development. Buildroot is an open-source embedded Linux system automatic build framework on the Linux platform. The entire Buildroot consists of Makefile scripts and Kconfig configuration files. By configuring Buildroot, a complete Linux system software that can be directly flashed and run on the device.

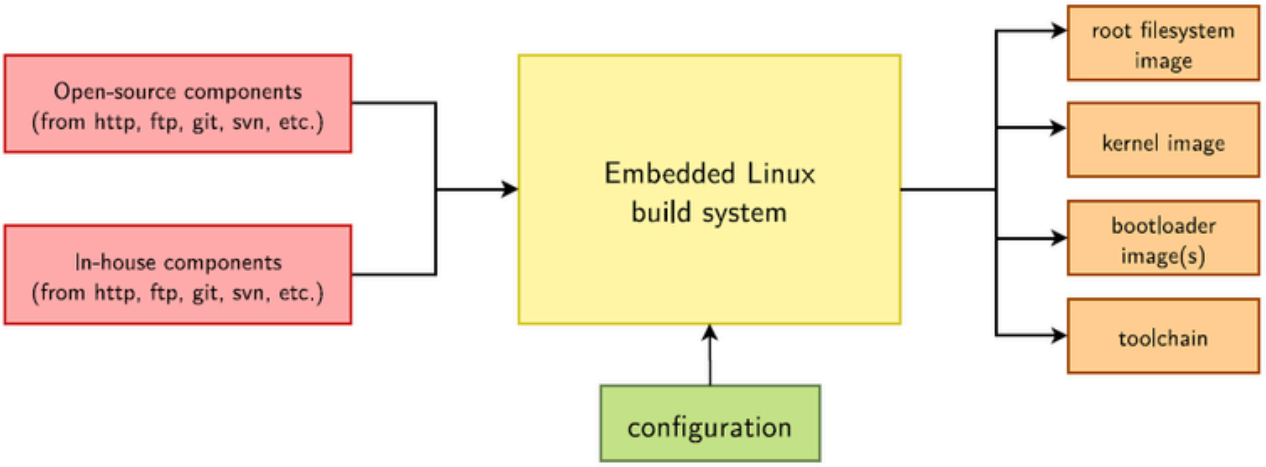


Figure 1-1 Buildroot Compilation Block Diagram

Advantages of Buildroot:

- **Source Code Building:** Buildroot offers great flexibility through source code building.
- **Convenient Cross-Compilation Environment:** It provides an easy-to-use cross-compilation environment for quick and efficient building, making it beginner-friendly.
- **Component Configuration:** Buildroot allows easy customization of various system components, facilitating tailored development.

One of the most famous projects using Buildroot is OpenWrt. Images created with OpenWrt can run on routers with 16 M SPI NOR flash memory, containing only essential components. This simplicity is achieved thanks to Buildroot. The entire Buildroot project is maintained in a single Git repository.

Buildroot uses kconfig and make, a defconfig configuration represents a BSP support.

Buildroot itself does not have the ability to expand, and users need to complete the work through scripts. These listed features are all different from Yocto.

> At present, Buildroot is a critical system that we develop and maintain internally.

5.1.3 Yocto

Yocto, like Buildroot, is a set of tools for building embedded systems, but their styles are completely different. Yocto projects are maintained through individual packages (meta), with some packages responsible for the core and others for peripherals. Some packages are designed for running Rockchip chips, some for installing Weston, and others for running Debian. Similar mechanisms are used for Node.js. The Yocto community is highly active and expansive, allowing everyone to share their achievements on GitHub. This mechanism ensures that we can reuse others' work from the internet, which is very valuable. Compared to the Buildroot

system, Yocto has a stronger compilation mechanism. It automatically handles dependencies and recompilation of third-party packages. However, it is relatively complex and requires a VPN network.

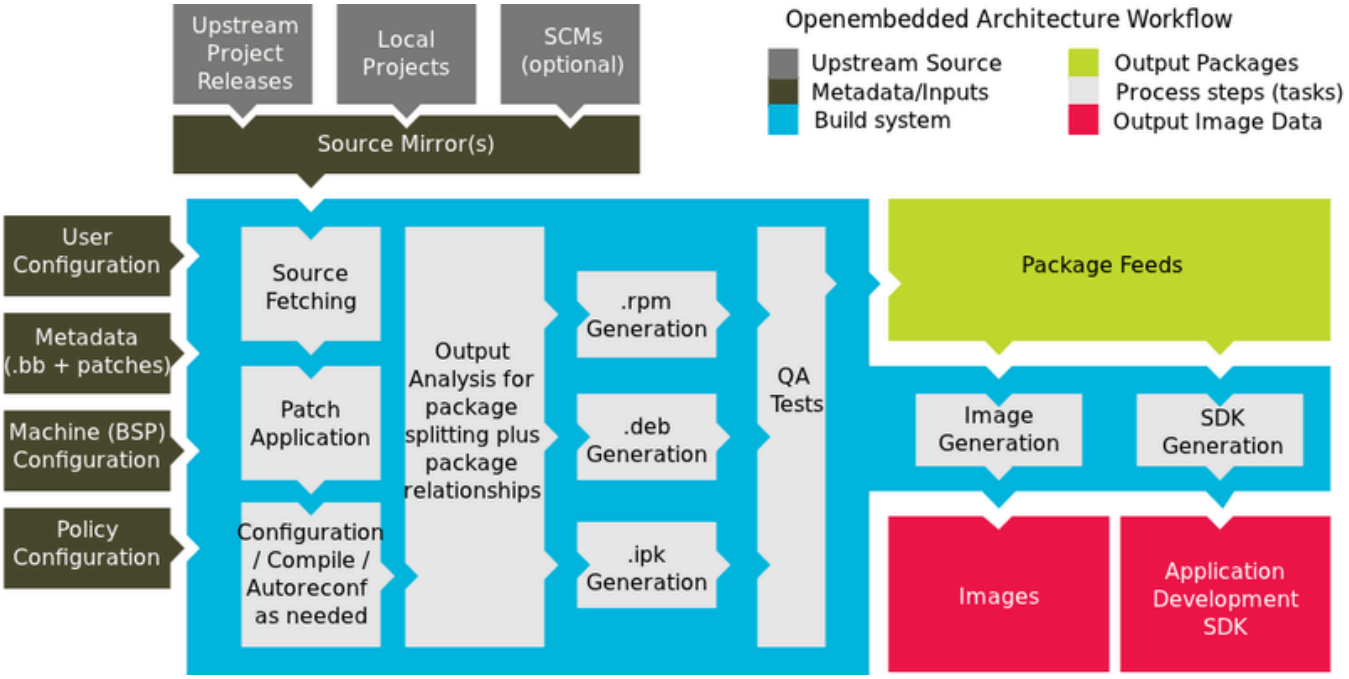


Figure 2-1 Yocto Compilation block diagram

Yocto is a very flexible build system that allows users to use the shell and Python to handle a variety of special cases. Currently, the system is mainly targeted at foreign customers. If you need more information about Yocto, check out the following references:

Yocto

Rockchip Yocto

Yocto stable branch

> At present, Yocto system is mainly used by overseas, enthusiasts or customers with secondary development capabilities.

5.1.4 Debian

Debian is a Linux operating system that is completely free, open-source, and widely used on various devices. The reasons for choosing Debian include:

- **Debian is Free Software:**
  Debian is composed of free and open-source software and will always remain 100% free. Everyone can use, modify, and distribute it freely. Developers can build upon the Debian system created based on Rockchip.
- **Debian is Stable and Secure:**
  Debian is a stable and secure operating system based on Linux, widely used across devices, including laptops, desktops, and servers. Since 1993, its stability and reliability have made it popular among users. Debian provides reasonable default configurations for each software package, and developers strive to provide security updates throughout its lifecycle.
- **Wide Hardware Support:**
  Most hardware is supported by the Linux kernel. When free software cannot provide sufficient support,

dedicated hardware drivers can be used. Currently, chips such as Rockchip RK3588/RK3568/RK3566/RK3399/RK3288 have been adapted and supported.

- **Smooth Updates:**
  Debian is known for its smooth updates within its release cycle, allowing easy transitions to the next major version. Rockchip has upgraded from Debian Stretch (9) to Debian Buster (10) and Bullseye (11).
- **Foundation for Many Other Distributions:**
  Debian serves as the foundation for many popular Linux distributions such as Ubuntu, Knoppix, PureOS, SteamOS, and Tails. Debian provides all the tools so that anyone can extend the software packages in the Debian archive to meet their requirements.
- **A Community Project:**
  Debian is not just a Linux operating system; it is created collaboratively by hundreds of volunteers from around the world. Anyone can be a part of the Debian community, even if they are not a programmer or system administrator.

The customized version of Debian for Rockchip is created through shell scripts, which involve obtaining Debian distribution source code, compiling, and installing the Rockchip hardware acceleration package into the operating system.

> Currently, Debian system is mainly aimed at customer preliminary evaluations, third-party system porting references, and products with complex desktop requirements.

5.2 SDK Software Architecture

As illustrated in Figure 3-1 of the SDK software architecture, it is divided into four layers from bottom to top: the Boot Layer, Kernel Layer, Libraries, and Application Layer. Each layer is described as follows:

- **Boot Layer:** This layer primarily handles system booting processes and provides support for components such as BootROM, U-Boot, ATF, and related functionalities.
- **Kernel Layer:** The Kernel Layer provides the standard implementation of the Linux Kernel. Linux is an open-source operating system. The core of the Rockchip platform's Linux system is based on the standard Linux 4.4/4.19/5.10/6.1/6.12 kernel versions. It offers foundational support for security, memory management, process management, network protocol stack, etc. The Linux kernel manages hardware resources of devices, such as CPU scheduling, caching, memory, I/O, etc.
- **Libraries:** This layer corresponds to general embedded systems and functions as middleware. It includes various system basic libraries and support for third-party open-source libraries. Libraries provide API interfaces to the application layer. System customizers and application developers can develop new applications based on the API provided by the Libraries layer.
- **Application Layer:** The Application Layer implements specific product features and interaction logic. It requires support from system basic libraries and third-party libraries. Developers can create their own applications, leveraging various system capabilities to deliver the final user experience.

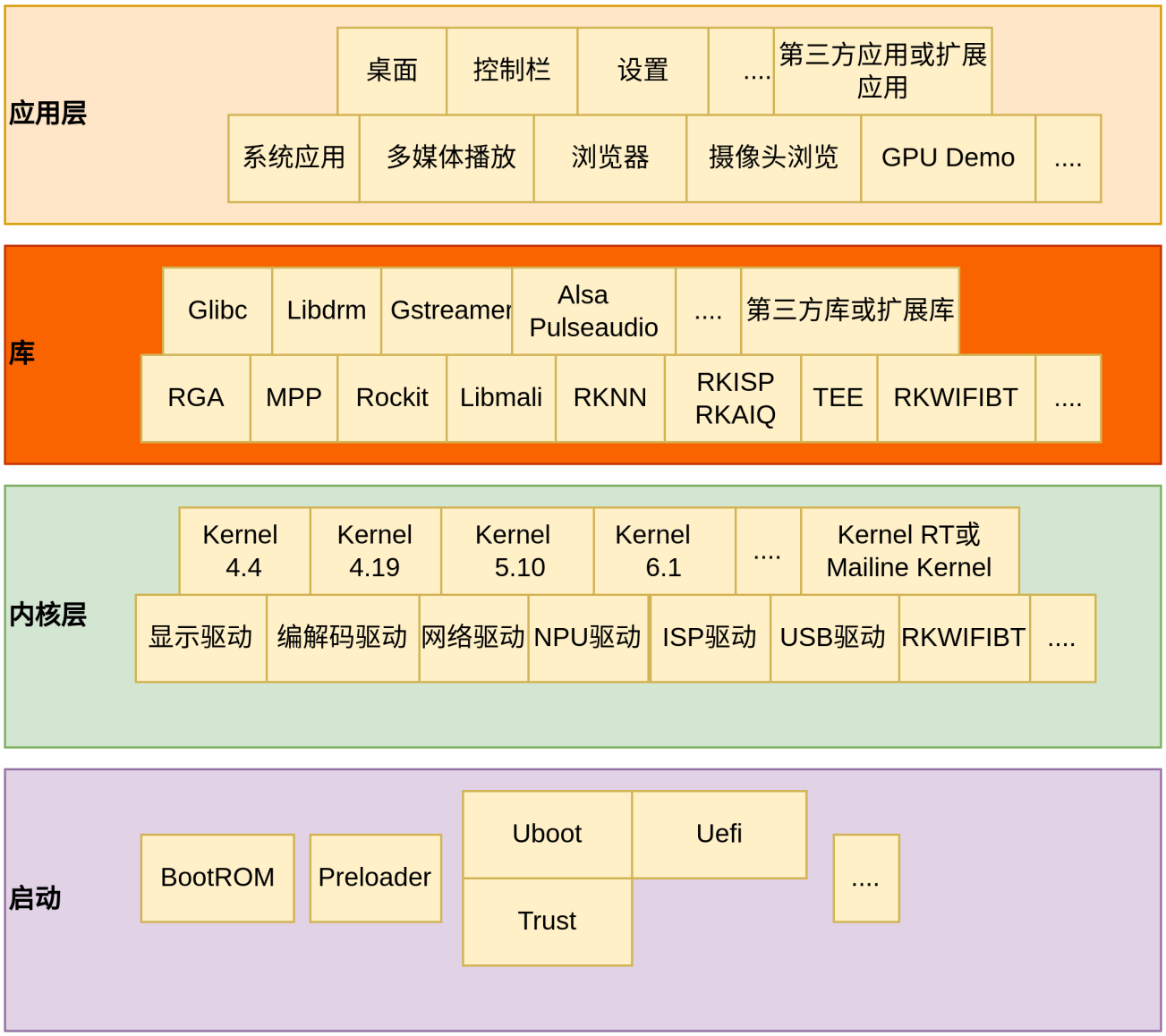
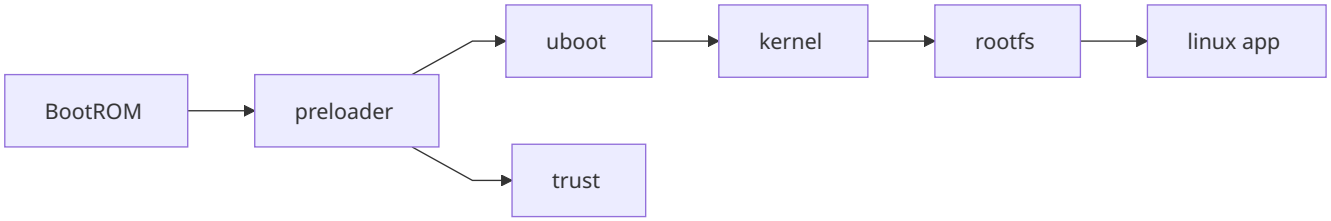


Figure 3-1 SDK Software Block Diagram

The SDK system startup process refers to a software process from system power on to completion of system startup. The following is the Linux system startup process:



> Description:
> Both AP and MCU have a BOOTROM integrated inside. When the system is powered on, it will first run the BOOTROM code, and then the BOOTROM code will detect the peripheral memory and load the Loader code.
> There are currently 3 Pre Loaders: miniloader (non-open source), uboot spl and loader.

5.3 SDK Development Process

The Rockchip Linux system is based on the Buildroot/Yocto/Debian system, and the kernel is developed based on kernel 5.10/6.1/6.12. It is an SDK developed for a variety of different product forms. Based on this SDK, system customization and application porting development can be effectively realized.

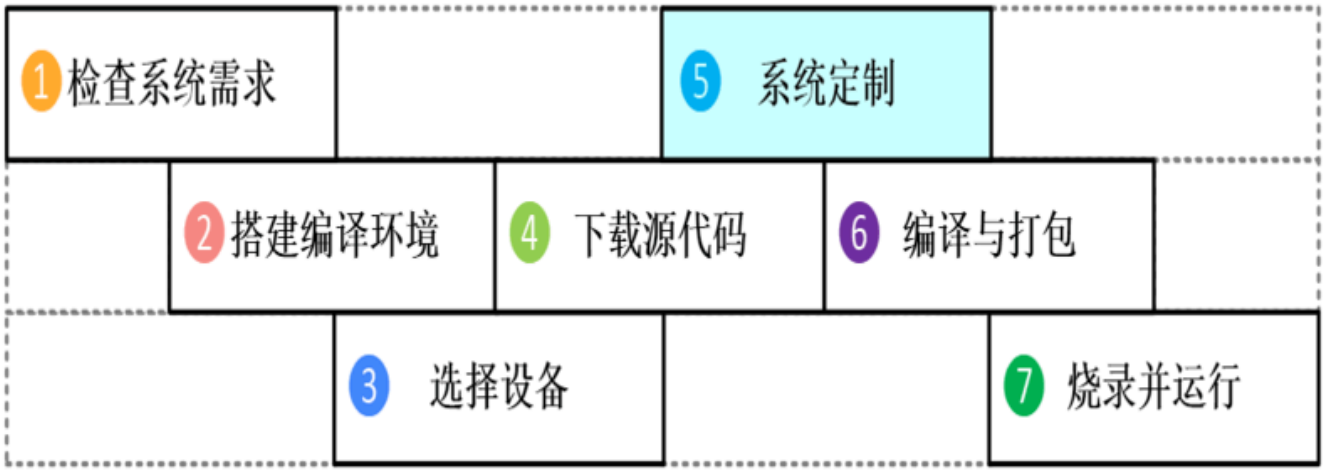


Figure 4-1 SDK Development Process

As shown in Figure 4-1, developers can follow the development process described above to quickly set up the Rockchip Linux system development environment and build code locally. Here is a brief overview of this process:

- **Check System Requirements:** Before downloading code and compiling, ensure that the local development device meets the requirements, including hardware capabilities, software system, toolchain, etc. Currently, the SDK supports compilation under the Linux operating system environment and provides toolchain support only for Linux. Other systems like MacOS and Windows are not supported at the moment.
- **Set Up Compilation Environment:** Provide information about the various software packages and tools that need to be installed on the development device. For details, please refer to SDK Development Environment Setup.
- **Select Device:** During the development process, developers need to choose the appropriate hardware board based on their requirements. Please refer to SDK Adapted Hardware Automatic Compilation Summary.
- **Download Source Code:** After selecting the device type, install the repo tool for batch downloading of source code. For details, please refer to SDK Software Packages.
- **System Customization:** Developers can customize U-Boot, Kernel, and Rootfs based on the hardware board and product definition. Please refer to SDK Development.
- **Compile and Package:** After setting up the source code and selecting the product, initialize the relevant compilation environment. Then execute compilation commands, including overall or module compilation, and cleanup work. For further details, please refer to SDK Compilation Instructions.
- **Flash and Run:** After generating the image files, learn how to flash the image and run it on the hardware device. For further details, please refer to SDK Firmware Upgrade.

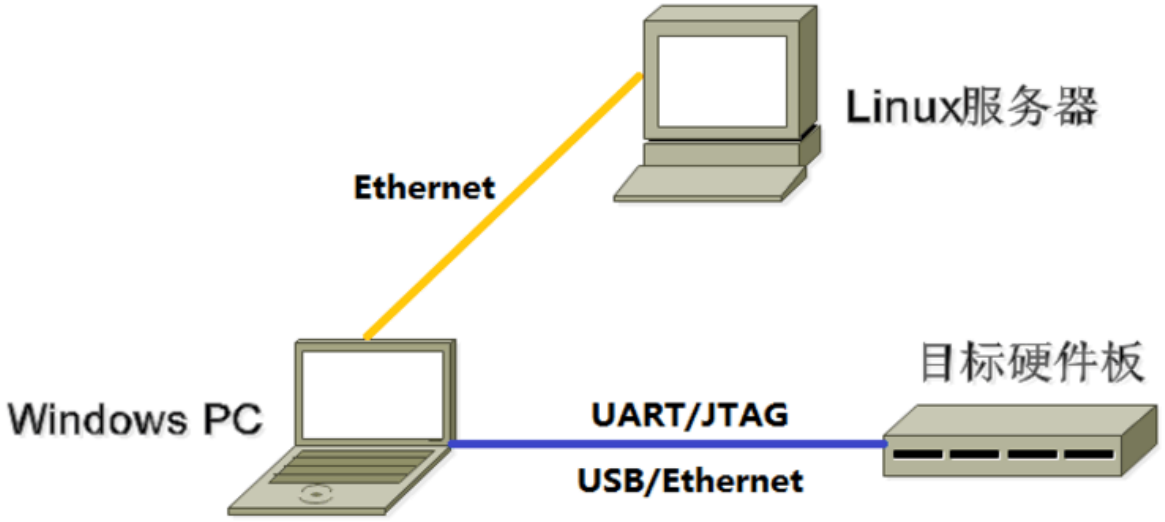
6. Chapter-6 SDK Development Environment Setup

6.1 Overview

This section mainly introduces how to set up a local compilation environment to build Rockchip Linux SDK source code. Currently, the SDK only supports compilation and secondary development in a Linux environment.

A typical embedded development environment typically includes a Linux server, PC, and target hardware board, as shown in the diagram below.

- Set up a cross-compilation environment on the Linux server to provide services such as code provision, updates downloading, and compilation for software development.
- Share programs between the PC and Linux server, and install Putty or Minicom. Remote login to the Linux server over the network is used for cross-compilation and development code debugging.
- Connect the PC to the target hardware board via serial ports and USB. The compiled image file can be flashed onto the target hardware board, and the system or application program can be debugged.



> Note: Windows PC is used in the development environment. In fact, many tasks can also be completed on Linux PC, such as using Minicom instead of Putty. Users can choose by themselves.

6.2 Preparation Work before SDK Development

The code and related documents of Rockchip Linux SDK are divided into several git repositories for version management. Developers can use repo to download, submit, switch branches, and other operations on these git repositories.

6.2.1 Install and Configure Git

Please configure your own git information before using repo, otherwise subsequent operations may encounter hook checking errors:

- Update sources and install git

```
sudo apt update && sudo apt-get install git
```

- Configure user information

```
git config --global user.name "your name"
git config --global user.email "your mail"
```

6.2.2 Install and Configure repo

repo is a script written by Google using Python script to call git. It is mainly used to download and manage the project's software repositories.

The installation path in the following commands takes "~/bin" as an example. Users should create the required directories themselves.

```
mkdir ~/bin
cd ~/bin
git clone ssh://git@www.rockchip.com.cn/repo/rk/tools/repo
chmod a+x ~/bin/repo/repo
export PATH=~/bin/repo:$PATH
```

In addition to the above methods, you can also use the following command to obtain repo

- In foreign regions, you can download the repo tool from the Google mirror site:

```
mkdir ~/bin
curl https://storage.googleapis.com/git-repo-downloads/repo -o ~/bin/repo
chmod a+x ~/bin/repo
export PATH=~/bin:$PATH
```

- In domestic areas, you can download the repo tool from the tsinghua.edu.cn site:

```
mkdir ~/bin
curl https://mirrors.tuna.tsinghua.edu.cn/git/git-repo -o ~/bin/repo
chmod a+x ~/bin/repo
export PATH=~/bin:$PATH
```

6.2.3 SDK Obtaining

The SDK is released through Rockchip code server. When customers apply for the SDK through Rockchip's technical window, they need an account to access the source code repository provided by Rockchip. In order to be able to obtain code synchronization, please provide SSH public key for server authentication and authorization when apply for SDK from Rockchip technical window. About Rockchip server SSH public key authorization, please refer to the Chapter SSH Public Key Operation Instructions in this document.

6.2.3.1 SDK Download Command

Rockchip Linux SDK can be adapted to different chip platforms, such as RK3588, RK3576,RK3562, RK3566, RK3568, RK3308, RK3288, RK3326/PX30, RK3399, etc. The source code of different chip platforms will be different to a certain extent. Developers need to declare the chip platform they want when downloading the source code, so as to avoid downloading code they don't need.

SDK uses different XML to declare the corresponding chip platform you want to download.

Please refer to the Chapter Download through code server in this document.

When the code start to download automatically, just wait patiently. The source code files will be located in the working directory under the corresponding project name. The initial sync operation will take an hour or more to complete.

6.2.3.2 Compressed Package of SDK Code

For quick access to SDK source code, Rockchip Technical Window usually provides corresponding version of SDK initial compression package. In this way, developers can get SDK source code through decompressing the initial compression package, which is the same as the one downloaded by repo.

Please refer to the Chapter Obtain by decompressing the local compressed package in this document.

6.2.3.3 Software Update History

The software release version upgrade can be viewed through the project xml. For example, the way for the RK3588 chip is as follows:

```
.repo/manifests$ realpath rk3588_linux6.1_release.xml
In the example: the printed version number is v1.3.0 and the update time is 20250620
<SDK>/.repo/manifests/release/rk3588_linux6.1_release_v1.3.0_20250620.xml
```

Software release version upgrade and update content can be viewed through the project text. Please refer to the project directory. For example, check the version of RK3588 as follows:

```
<SDK>/.repo/manifests/RK3588_Linux6.1_SDK_Note.md

<SDK>/docs/en/RK3588/RK3588_Linux6.1_SDK_Note.md
```

6.2.3.4 SDK Update

Developers can synchronize updates through commands according to the update instructions published regularly in the FAE window.

```
.repo/repo/repo sync -c
```

6.2.3.5 SDK Issue Feedback

In order to better assist customer development, the Rockchip bug system (Redmine) records the user problem handling process and status, making it easier for both parties to track at the same time, making issue processing more timely and efficient. SDK issues, specific technical issues, technical consultation, etc. can be submitted to this bug system, and Rockchip technical services will distribute, handle and track the issues in a timely manner. For more detailed instructions, please refer to the documentation

`<SDK>/docs/en/Others/Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf`。

6.3 Setting Up a Linux Development Environment

6.3.1 Preparing the Development Environment

We recommend using a system with Ubuntu 22.04 or a higher version for compilation. Other Linux versions may require corresponding adjustments to the software packages. In addition to system requirements, there are other hardware and software requirements.

Hardware Requirements: 64-bit system with hard disk space greater than 40GB. If you are performing multiple builds, you will need more hard disk space.

Software Requirements: System with Ubuntu 22.04 or a higher version.

6.3.2 Installing Libraries and Toolkits

When developing for devices using the command line, you can install the necessary libraries and tools required for compiling the SDK by following these steps.

Use the following `apt-get` commands to install the libraries and tools needed for subsequent operations:

```
sudo apt-get update && sudo apt-get install git ssh make gcc libssl-dev \
liblz4-tool expect expect-dev g++ patchelf chrpath gawk texinfo chrpath \
diffstat binfmt-support qemu-user-static live-build bison flex fakeroot \
cmake gcc-multilib g++-multilib unzip device-tree-compiler ncurses-dev \
libgucharmap-2-90-dev bzip2 expat gpgv2 cpp-aarch64-linux-gnu libgmp-dev \
libmpc-dev bc python-is-python3 python3-setuptools swig python3-dev \
gettext libc6-dev libncurses-dev rsync
```

> **Note:**
> The installation command is suitable for Ubuntu 22.04. For other versions, please use the corresponding installation command based on the package name. If you encounter errors during compilation, you can install the corresponding software packages based on the error messages. Specifically:
>
> - Python requires the installation of version 3.6 or higher, with version 3.6 used as an example here.
> - Make requires the installation of version 4.0 or higher, with version 4.2 used as an example here.
> - LZ4 requires the installation of version 1.7.3 or higher.
> - Compiling Yocto requires a VPN network.

6.3.2.1 Checking and Upgrading the Host's Python Version

The method to check and upgrade the host's Python version is as follows:

- Check the host's Python version

```
$ python3 --version
Python 3.10.6
```

If the requirement of Python version >= 3.6 is not met, you can upgrade by the following method:

- Upgrade to the new version of `Python 3.6.15`

```
PYTHON3_VER=3.6.15
echo "wget https://www.python.org/ftp/python/${PYTHON3_VER}/Python-${PYTHON3_VER}.tgz"
echo "tar xf Python-${PYTHON3_VER}.tgz"
echo "cd Python-${PYTHON3_VER}"
echo "sudo apt-get install libsqlite3-dev"
echo "./configure --enable-optimizations"
echo "sudo make install -j8"
```

6.3.2.2 Checking and Upgrading the Host's `make` Version

The method to check and upgrade the host's `make` version is as follows:

- Checking the host's `make` version

```
$ make -v
GNU Make 4.2
Built for x86_64-pc-linux-gnu
```

- Upgrading to a newer version of `make 4.2`

```
$ sudo apt update && sudo apt install -y autoconf autopoint

git clone https://gitee.com/mirrors/make.git
cd make
git checkout 4.2
git am $BUILDROOT_DIR/package/make/*.patch
autoreconf -f -i
./configure
make make -j8
sudo install -m 0755 make /usr/bin/make
```

6.3.2.3 Checking and Upgrading the Host's LZ4 Version

The method to check and upgrade the host's LZ4 version is as follows:

- Check the host's LZ4 version

```
$ lz4 -v
*** LZ4 command line interface 64-bits v1.9.3, by Yann Collet ***
```

- Upgrade to a new version of LZ4

```
git clone https://gitee.com/mirrors/LZ4_old1.git
cd LZ4_old1

make
sudo make install
sudo install -m 0755 lz4 /usr/bin/lz4
```

6.4 Window PC Development Environment Setup

6.4.1 Development Tool Installation

- Users should install editing software such as Vim and Notepad++ by themselves.
- Download Virtual-Box (software that provides a virtual development environment)
- Download XShell6 (used to establish interaction with Linux system)

6.4.2 Rockchip USB Driver Installation

During the development and debugging process, the device needs to be switched to Loader mode or Maskrom mode, and Rockusb driver should be installed, then the device can be recognized normally. Rockchip USB driver installation assistant is stored in tools/windows/DriverAssitant_v5.x.zip. support xp, win7_32, win7_64, win10_32, win10_64 and other operating systems.
The installation steps are as follows:





6.4.3 Windows Burning Tool Usage

The burning tool for Windows systems is released in `<SDK>/tools/windows/RKDevTool/RKDevTool_Release`, which can be used for development, debugging and firmware burning in Windows environment. Please refer to Development and programming tools for details.

6.4.4 Target Hardware Board Preparation

Please refer to the SDK software package applicable hardware list SDK software package applicable hardware list to select the corresponding hardware board for development and debugging. The corresponding hardware instruction document will introduce the hardware interface, usage instructions and burning operation methods.

6.5 Docker Environment Setup

To help developers quickly complete the complex development environment preparation mentioned above, we also provide a cross-compiler Docker image. This enables customers to rapidly verify and then shorten the build time of the compilation environment.

Before using the Docker environment, you can refer to the following document for instructions: `<SDK>/docs/en/Linux/Docker/Rockchip_Developer_Guide_Linux_Docker_Deploy_EN.pdf`.

The following systems have been verified:

Distribution Version	Docker Version	Image Loading	Firmware Compilation
ubuntu 22.04	24.0.5	pass	pass
ubuntu 21.10	20.10.12	pass	pass
ubuntu 21.04	20.10.7	pass	pass
ubuntu 18.04	20.10.7	pass	pass
fedora35	20.10.12	pass	NR (not run)

The Docker image can be obtained from the website Docker Image.

6.6 Introduction to Cross-compilation Tool Chain

Since Rockchip Linux SDK is currently only built in the Linux PC environment, we only provide the cross-compilation tool chain under Linux. The tool chain in the prebuilt directory is used by U-Boot and Kernel. Specifically, Rootfs needs to use its corresponding tool chain, or use a third-party tool chain for building.

6.6.1 U-Boot and Kernel Compilation Tool Chain

```
Linux4.4/4.19 SDK:
prebuilts/gcc/linux-x86/aarch64/gcc-linaro-6.3.1-2017.05-x86_64_aarch64-linux-
gnu/bin/aarch64-linux-gnu-

Linux5.10/6.1 SDK:
prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-
gnu/bin/aarch64-none-linux-gnu-

Linux6.12 SDK:
prebuilts/gcc/linux-x86/aarch64/arm-gnu-toolchain-14.3.rel1-x86_64-aarch64-none-linux-
gnu/bin/aarch64-none-linux-gnu-
```

prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu/bin/aarch64-none-linux-gnu-

```
Corresponding version:
```

Linux4.4/4.19 SDK:
gcc version 6.3.1 20170404 (Linaro GCC 6.3-2017.05)

Linux5.10/6.1 SDK:
gcc version 10.3.1 20210621 (GNU Toolchain for the A-profile Architecture 10.3-2021.07 (arm-10.29))

Linux6.12 SDK:
gcc version 14.3.1 20250527 (GNU Toolchain for the A-profile Architecture 14.3.Rel1 (arm-14.3))

```
> **Note:** Linux6.12 SDK uses Buildroot 2026.02, the built-in GCC version may vary, please
refer to the actual compilation output.



#### Buildroot Toolchain

##### Configuring the Compilation Environment

To compile individual modules or third-party applications, a cross-compilation environment
must be set up. For instance, for the RK3576, the cross-compilation tools are located in the
`buildroot/output/rockchip_rk3576/host/usr` directory. It is necessary to set the bin/
directory of the tools and the `aarch64-buildroot-linux-gnu/bin/` directory as environment
variables. Execute the script for automatic configuration of the environment variables in
the top-level directory:

```shell

source buildroot/envsetup.sh rockchip_rk3576
```

Enter the command to check:

```
cd buildroot/output/rockchip_rk3576/host/usr/bin
./aarch64-linux-gcc --version
```

At this point, the following information will be printed:

```
aarch64-linux-gcc.br_real (Buildroot -gc307c95550) 12.3.0
```

6.6.1.1 Packaging Toolchain

Buildroot supports the packaging of the built-in toolchain into a compressed archive for standalone compilation by third-party applications. For detailed information on how to package the toolchain, please refer to the Buildroot official documentation:
```

```
buildroot/docs/manual/using-buildroot-toolchain.txt
```

In the SDK, you can directly run the following command to generate the toolchain package:

```
./build.sh bmake:sdk
```

The generated toolchain package is located in the `buildroot/output/*/images/` directory, named `aarch64-buildroot-linux-gnu_sdk-buildroot.tar.gz`, for users who have the need. After extraction, the path to `gcc` will be:

```
./aarch64-buildroot-linux-gnu_sdk-buildroot/bin/aarch64-buildroot-linux-gnu-gcc

---


## Chapter-7 SDK Version and Update Instructions

The Linux SDK currently has different versions such as Linux 4.4, Linux 4.19, Linux 5.10,
Linux6.1, and Linux6.12. Use repo to manage the SDK and use tags to manage versions. The
various components and modules of the SDK are stored in different Git repositories, and the
repo tool is used to coordinate version and code synchronization between them. Each
component's Git repository has its own branch and commit history.

### SDK Naming Rules

#### SDK tag Naming Rules

System - System version - form - rkr Release version

For example: Linux6.1 SDK tag is `linux-6.1-stan-rkr1`, Linux6.12 SDK tag is `linux-6.12-
stan-rkr1`

Note:

- System: Linux, Android, RTOS, etc.
- System version: Linux5.10, Linux6.1 etc.
- Form: gen, common, standard, ipc, chip model, etc.
- Release version: external: alpha, beta, release; internal: rka1, rkb1, rkr1

#### SDK Release Document Naming Rules

Rockchip[1]_RK3399[2]_Linux5.10[3]_SDK[4]_Release[5]_V1.0.0[6]_20220920[7]_EN [8]

> Note:
> [1]: Rockchip, required. Documents released by RK must begin with Rockchip.
```

> [2]: Chip model. The initial letter of the chip is uniformly capitalized, the suffix depends on the chip and remains consistent with the official website definition. If there are two chips, connect them with an underscore, such as (RK3566_RK3568).
> [3]: Linux, Android, FreeRTOS, etc., the first letter is capitalized.
> [4]: Optional, indicating the target market. If it is a general SDK, this item is omitted.
> [5] and [6]: Release type (Alpha/Beta/Release) and version number. The version number style is defined as follows: "Vn.n.n" . The letter "V" remains capitalized and n represents an integer. It can be seen that the version number consists of three numbers with 2 dots in the middle.
>
> - Alpha version starts from V0.0.1, that is, the first two digits keep the number 0, and the last digit is incremented according to the actual testing/release situation;
> - The Beta version starts from V0.1.1, that is, the first digit is kept as 0, the increment of the second digit indicates the addition of functional modules or major updates of functions, and the increment of the third digit indicates bug correction;
> - Release version starts from V1.0.0, the first digit indicates a major update, and the second and third digits refer to the Beta version.
> [7]: Release date. Use the format 20220920 uniformly.
> [8]: Document language version information. CN: Chinese, EN: English

SDK Compressed Package Naming Rules

Chip platform - software version - [target market] + SDK + version number + release date

Example:
RK3399_LINUX5.10_SDK_RELEASE_V1.0.0_20220920
RK3572_LINUX6.12_SDK_RELEASE_V1.0.0_20260620

> Note:
>
> - All English letters are capitalized
> - Software platform: LINUX5.10, LINUX6.12, etc.
> - The target market is enclosed in square brackets to indicate optional options. If it is a general SDK, this item is omitted.
> - The version number of the SDK compressed package continues to retain 3 digits with two dots in the middle.

SDK Update Mechanism

SDK Update

If there is an update to the SDK, it will be updated around the 20th of each month, and a Change Notice will be sent out at the end of the month. The update mechanism is a one-month small update (such as V1.0.x), and the version has been fully tested by QA (such as V1.x.0).

SDK Patch Release

Important patches will be periodically released on the Redmine system. The link is as follows:

You can subscribe to receive email notifications for patch releases in real-time. The subscription steps are as follows:

```
#### Step 1: Log in to the Redmine System

Log in to the Redmine system using your Redmine account registered with Rockchip.

#### Step 2: Go to My Account

![My Account](resources/my_account.png)

#### Step 3: Select Email Notification Type

In the **Email Notifications** section, select **Receive all notifications for selected
projects** from the dropdown menu as shown below:

![Email Notification](resources/select_email.png)

#### Step 4: Select Project

Check the **Patch Release** project as shown in the image below, and click **Save**.

![Select Project](resources/select_patch.png)

Once these steps are completed, you will be subscribed to **Patch Release** notifications.
After subscribing, you will receive email notifications for any patch releases from Rockchip
to the email address registered on Redmine.

### How to Build a Server to Sync the SDK

How to build a Gitolite server and use it for management and maintenance of SDK code. Please
refer to the following documents for details.
```

/docs/en/Others/Rockchip_Developer_Guide_Repo_Mirror_Server_Deploy.pdf

```
### Linux6.12 SDK Version Introduction

Linux6.12 SDK is the latest SDK developed by Rockchip based on Linux Kernel 6.12 LTS
version. It currently supports RK3572, RK3588, and RK3576 chips. Compared to Linux6.1 SDK,
the main updates are as follows:

- **Kernel Upgrade**: Based on Linux Kernel 6.12 LTS, including the latest kernel features
and security patches
- **Toolchain Upgrade**: Cross-compilation toolchain upgraded from GCC 10.3 to GCC 14.3
- **Buildroot Upgrade**: Upgraded from Buildroot 2024.02 to Buildroot 2026.02
- **Debian Upgrade**: Upgraded from Debian 12 (Bookworm) to Debian 13 (Trixie)
- **Yocto Upgrade**: Upgraded from Yocto 5.0 (Scarthgap) to Yocto 6.0 (Wrynose)
- **U-Boot Upgrade**: Based on U-Boot 2025 version
- **New RK3572 Chip Support**: RK3572 is a new-generation chip designed for cleaning robots,
commercial robots, and other scenarios
- **AMP Multi-core Heterogeneous**: Supports AMP 5.2 version (under development)
- **RKNN3 Support**: Supports the new-generation NPU framework RKNN3
```

```
---


## Chapter-8 SDK Compilation Instructions

The SDK can be configured and compiled with `make` or `./build.sh` along with target
parameters. For specific references, see the compilation instructions in
`device/rockchip/common/README.md`.

To ensure SDK updates smoothly each time, it is recommended to clean previous compilation
products before updating. This avoids potential compatibility issues or compilation errors,
as old compilation products may not be suitable for the new version of the SDK. Run the
following command to clean these compilation products:  `./build.sh cleanall`.

### SDK Compilation Command

`make help` , for example:
```

$ make help
menuconfig - interactive curses-based configurator
oldconfig - resolve any unresolved symbols in .config
syncconfig - Same as oldconfig, but quietly, additionally update deps
olddefconfig - Same as syncconfig but sets new symbols to their default value
savedefconfig - Save current config to RK_DEFCONFIG (minimal config)
...

```
`make` is used to run `./build.sh`

That is, you can also run `./build.sh <target>` to build related functions. You can view the
detailed compilation commands through `./build.sh help`.

```shell
$ ./build.sh -h
############### Rockchip Linux SDK ###############

Manifest: rk3562_linux_release_v1.1.0_20231220.xml

Log colors: message notice warning error fatal

Usage: build.sh [OPTIONS]
Available options:
chip[:<chip>[:<config>]]            choose chip
defconfig[:<config>]                choose defconfig
config                              modify SDK defconfig
...
updateimg                           build update image
otapackage                          build A/B OTA update image
all                                 build images
release                             release images and build info
save                                alias of release
```
```

```
all-release                          build and release images
allsave                              alias of all-release
shell                                setup a shell for developing
cleanall                             cleanup
clean[:module[:module]]...           cleanup modules
post-rootfs <rootfs dir>             trigger post-rootfs hook scripts
help                                 usage

Default option is 'all'.
```

6.7 SDK Board Level Configuration

`make rockchip_defconfig`, the detailed board-level configuration instructions are as follows:

Enter the project `<SDK>/device/rockchip/<chipset_name>` directory:

Board Level Configuration	Description
rockchip_rk3572_evb1_v10_defconfig	Suitable for RK3572 EVB1 development board(Linux6.12)
rockchip_rk3572_evb2_v10_defconfig	Suitable for RK3572 EVB2 development board(Linux6.12)
rockchip_rk3572_evb3_v10_defconfig	Suitable for RK3572 EVB3 development board(Linux6.12)
rockchip_rk3572_evb1_robot_defconfig	Suitable for RK3572 EVB1 Robot development board(Linux6.12)
rockchip_rk3572_ipc_evb1_v10_defconfig	Suitable for RK3572 IPC EVB1 development board(Linux6.12)
rockchip_rk3588_evb1_lp4_v10_defconfig	Suitable for RK3588 EVB1 with LPDDR4 development board
rockchip_rk3588_evb7_lp4_v11_defconfig	Suitable for RK3588 EVB7 with LPDDR4 development board
rockchip_rk3588s_evb1_lp4x_v10_defconfig	Suitable for RK3588S EVB1 with LPDDR4 development board
rockchip_rk3562_evb1_lp4x_v10_defconfig	Suitable for RK3562 EVB1 with LPDDR4 development board
rockchip_rk3562_evb2_ddr4_v10_defconfig	Suitable for RK3562 EVB2 with DDR4 development board
rockchip_defconfig	The default configuration will be linked to the default board level configuration through `make lunch` or `./build.sh lunch` for configuration
Take RK3572 chip (Linux6.12 SDK) for example:	

```
$ ./build.sh lunch

You're building on Linux
Lunch menu...pick a combo:

1. rockchip_defconfig
2. rockchip_rk3562_evb1_lp4x_v10_defconfig
3. rockchip_rk3562_evb2_ddr4_v10_defconfig
Which would you like? [1]:
```

The configuration of other functions can be configured through `make menuconfig` to configure related properties.

6.8 SDK Customization Configuration

The SDK can be configured through `make menuconfig`, with the main components currently available for configuration being:

```
(rk3562) SoC
    Rootfs  --->
    Loader (u-boot)  --->
    RTOS  --->
    Kernel  --->
    Boot  --->
    Recovery (based on buildroot)  --->
    PCBA test (based on buildroot)  --->
    Security  --->
    Extra partitions  --->
    Firmware  --->
    Update (Rockchip update image)  --->
    Others configurations  --->
```

- Rootfs: This represents the "Root File System". Here, you can choose different root file systems like Buildroot, Yocto, Debian, etc.
- Loader (u-boot): This is the boot loader configuration, usually u-boot, which initializes hardware and loads the main operating system.
- RTOS: Real-Time Operating System (RTOS) configuration options, suitable for applications requiring real-time performance.
- Kernel: Configure kernel options here, customizing a Linux kernel suitable for your hardware and application needs.
- Boot: Configure Boot partition support formats here.
- Recovery (based on buildroot): This is the configuration for the recovery environment based on buildroot, used for system recovery and upgrade.
- PCBA test (based on buildroot): This is a configuration for a PCBA (Printed Circuit Board Assembly) testing environment based on buildroot.
- Security: Indicates that certain security features will depend on the buildroot configuration chosen for

Rootfs, mainly the Secureboot feature activation.

- Extra partitions: Used to configure additional partitions.
- Firmware: Configure firmware-related options here.
- Update (Rockchip update image): Used to configure options for Rockchip's complete firmware.
- Others configurations: Other additional configuration options.

The `make menuconfig` interface provides a text-based user interface to select and configure various options. Once configuration is complete, use the command `make savedefconfig` to save these settings, so that custom builds will be done based on these settings.

With the above configs, you can choose different Rootfs/Loader/Kernel configurations for various custom builds, allowing flexible selection and configuration of system components to meet specific needs.

6.9 SDK Environment Variable Configuration

Configurations for different chips and target functions can be set via `source envsetup.sh`.

```
$ source envsetup.sh
Top of tree: /home/wxt/linux-develop/rk3562

You're building on Linux
Pick a board:

1. rockchip_px30_32
2. rockchip_px30_64
3. rockchip_px30_recovery
4. rockchip_rk3036
5. rockchip_rk3036_recovery
6. rockchip_rk3126c
7. rockchip_rk312x
8. rockchip_rk312x_recovery
9. rockchip_rk3288
10. rockchip_rk3288_recovery
11. rockchip_rk3308_32_release
12. rockchip_rk3308_b_32_release
13. rockchip_rk3308_b_release
14. rockchip_rk3308_bs_32_release
15. rockchip_rk3308_bs_release
16. rockchip_rk3308_h_32_release
17. rockchip_rk3308_recovery
18. rockchip_rk3308_release
19. rockchip_rk3326_64
20. rockchip_rk3326_recovery
21. rockchip_rk3358_32
22. rockchip_rk3358_64
23. rockchip_rk3358_recovery
24. rockchip_rk3399
25. rockchip_rk3399_base
26. rockchip_rk3399_recovery
27. rockchip_rk3399pro
```

```
28. rockchip_rk3399pro-multi-cam
29. rockchip_rk3399pro-npu
30. rockchip_rk3399pro-npu-multi-cam
31. rockchip_rk3399pro_combine
32. rockchip_rk3399pro_recovery
33. rockchip_rk3528
34. rockchip_rk3528_recovery
35. rockchip_rk3562
36. rockchip_rk3562_32
37. rockchip_rk3562_recovery
38. rockchip_rk3566
39. rockchip_rk3566_32
40. rockchip_rk3566_recovery
41. rockchip_rk3566_rk3568_base
42. rockchip_rk3566_rk3568_ramboot
43. rockchip_rk3568
44. rockchip_rk3568_32
45. rockchip_rk3568_recovery
46. rockchip_rk3588
47. rockchip_rk3588_base
48. rockchip_rk3588_ramboot
49. rockchip_rk3588_recovery
Which would you like? [1]:
```

For example, select 35 `rockchip_rk3562`

Then go to the Buildroot directory of RK3562 and start compiling the relevant modules.

6.10 Fully Automatic Compilation

Enter the project root directory and execute the following command to automatically complete all compilation:

```
./build.sh all # Only build module code (U-Boot, kernel, Rootfs, Recovery)
               # Need to execute ./mkfirmware.sh again to package the firmware

./build.sh     # build module code (U-Boot, kernel, Rootfs, Recovery)
               # Package into update.img complete upgrade package
               # Copy and generate all compilation information to the out directory
```

It is Buildroot by default, you can specify different rootfs by setting the environment variable `RK_ROOTFS_SYSTEM`, which can set three systems currently: Buildroot, Debian, and Yocto.

For example, if you need Debian, you can generate it with the following command:

```
export RK_ROOTFS_SYSTEM=debian
./build.sh
or
RK_ROOTFS_SYSTEM=debian ./build.sh
```

6.11 Build Modules

6.11.1 Build U-Boot

Enter the SDK project. Run the following command to build:

```
<SDK>#./build.sh U-Boot
```

For detailed board-level build, please refer to the build instructions in the SDK release document.

6.11.2 Build Kernel

Enter the root directory of the project directory and execute the following command to automatically build and package kernel.

```
<SDK>#./build.sh kernel
```

For detailed board-level compilation, please refer to the release notes or the compilation instructions in Quick Start.

6.11.3 Build Recovery

Enter the root directory of the project and execute the following command to automatically complete the compilation and packaging of Recovery.

```
<SDK>#./build.sh recovery
```

After compilation, a recovery.img is generated in the Buildroot directory `output/rockchip_rk3562_recovery/images`.

> Note: recovery.img includes Kernel, so every time Kernel changes, Recovery needs to be repackaged. The method for repackaging Recovery is as follows:

```
<SDK>#source buildroot/envsetup.sh
<SDK>#cd buildroot
<SDK>#make recovery-reconfigure
<SDK>#cd -
<SDK>#./build.sh recovery
```

> Note: Recovery is not a mandatory feature, some board configurations might not set it

6.11.4 Build Buildroot

Enter the root directory of the project and execute the following command to automatically complete the compilation and packaging of Rootfs:

```
./build.sh rootfs
```

After compilation, different formats of images are generated in the Buildroot directory `output/rockchip_rk3562/images`, rootfs.ext4 format is used by default.

For more information, please refer to the Buildroot development document:

`<SDK>/docs/en/Linux/System/Rockchip_Developer_Guide_Buildroot_EN.pdf`

6.11.5 Build Debian

```
./build.sh debian
```

After compilation, a linaro-rootfs.img is generated in the debian directory.

> **Note: Related dependencies must be installed beforehand**
> sudo apt-get install binfmt-support qemu-user-static live-build
> sudo dpkg -i ubuntu-build-service/packages/*
> sudo apt-get install -f

For more information, please refer to the Debian development document:

`<SDK>/docs/en/Linux/System/Rockchip_Developer_Guide_Debian_EN.pdf`

6.11.6 Build Yocto

Enter the root directory of the project and execute the following command to automatically complete the compilation and packaging of Rootfs:

```
./build.sh yocto
```

After compilation, a rootfs.img is generated in the yocto directory build/latest.

The default login username is root. For more information on Yocto, please refer to Rockchip Wiki.

FAQ:

- If you encounter the following issue during compilation:

```
Please use a locale setting which supports UTF-8 (such as LANG=en_US.UTF-8).
Python can't change the filesystem locale after loading so we need a UTF-8
when Python starts or things won't work.
```

Solution:

```
locale-gen en_US.UTF-8
export LANG=en_US.UTF-8 LANGUAGE=en_US.en LC_ALL=en_US.UTF-8
```

Or refer to setup-locale-python3.

6.11.7 Cross-Compilation

6.11.7.1 SDK Directory Built-in Cross-Compilation

The SDK prebuilts directory built-in cross-compilation are as follows:

Contents	Description
prebuilts/gcc/linux-x86/aarch64/arm-gnu-toolchain-14.3.rel1-x86_64-aarch64-none-linux-gnu	gcc arm 14.3 64-bit toolchain (Linux6.12 SDK)
prebuilts/gcc/linux-x86/arm/arm-gnu-toolchain-14.3.rel1-x86_64-arm-none-linux-gnueabihf	gcc arm 14.3 32-bit toolchain (Linux6.12 SDK)
prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu	gcc arm 10.3.1 64-bit toolchain (Linux5.10/6.1 SDK)
prebuilts/gcc/linux-x86/arm/gcc-arm-10.3-2021.07-x86_64-arm-none-linux-gnueabihf	gcc arm 10.3.1 32-bit toolchain (Linux5.10/6.1 SDK)

The toolchain can be downloaded from the following address:

Click here

6.11.7.2 Built-in Cross Compilation in Buildroot

Different chip and target feature configurations can be set via `source buildroot/envsetup.sh`

```
$ source buildroot/envsetup.sh
Top of tree: rk3562

You're building on Linux
Lunch menu...pick a combo:

44. rockchip_rk3562
45. rockchip_rk3562_32
46. rockchip_rk3562_dictpen
47. rockchip_rk3562_ramboot
48. rockchip_rk3562_recovery
49. rockchip_rk3562_robot
...

Which would you like? [1]:
```

Default choice is 44, `rockchip_rk3562`. Then, enter the RK3562 Buildroot directory to start compiling related modules.

`rockchip_rk3562_32` is for compiling a 32-bit Buildroot system, and `rockchip_rk3562_recovery` is used to compile the Recovery module.

For example, to compile the rockchip-test module, commonly used compilation commands are as follows:

Enter the buildroot directory

```
SDK#cd buildroot
```

- Delete and recompile rockchip-test

```
buildroot#make rockchip-test-recreate
```

- Recompile rockchip-test

```
buildroot#make rockchip-test-rebuild
```

- Delete rockchip-test

```
buildroot#make rockchip-test-dirclean
or
buildroot#rm -rf output/rockchip_rk3562/build/rockchip-test-master/
```

To compile an individual module or third-party application, a cross-compilation environment needs to be configured. For instance, for RK3562, its cross-compilation tools are located in `buildroot/output/rockchip_rk3562/host/usr`. You need to set the bin/ directory of the tool and `aarch64-buildroot-linux-gnu/bin/` as environment variables and run the script for auto-configuring environment variables in the top-level directory:

```
source buildroot/envsetup.sh rockchip_rk3562
```

Enter the command to check:

```
cd buildroot/output/rockchip_rk3562/host/usr/bin
./aarch64-linux-gcc --version
```

The following information will be printed:

```
aarch64-linux-gcc.br_real (Buildroot) 12.3.0
```

> Save to the rootfs configuration file
> buildroot$ make update-defconfig

7. Chapter-9 Introduction to SDK Partition Configuration

Rockchip has a total of 3 types of partition tables that can be into storage partitions: MTD Partition, GPT, and RK partition.

For detailed information, please refer to the document "docs/en/Common/TOOL/Rockchip_Introduction_Partition_EN.pdf".

Partition	Description	Applicable Platform	Limitation
MTD Partition	Defined in the parameter file, passed through cmdline, uboot-next branch no longer supports	All AP*1	Requires a separate partition for storage
GPT	EFI General Partition Table, supported by uboot-next branch	All AP*2	Uses more resources
RK partition	Designed based on GPT, mainly used for small-capacity storage, saves resources	RV1107/8, MCU	RK custom, not universal

*1 Platforms using the uboot-next branch no longer support MTD partition. If needed, self-adaptation is required.

*2 Platforms using the uboot-next branch default to using GPT as the partition table. If other partition tables are needed, self-adaptation is required.

The parameter file is used to configure some system parameters, such as firmware version, storage partition information, etc. The parameter file is a very important system configuration file. It is best to understand the functions of each configuration clearly before making changes to avoid abnormal parameter file configurations that can cause the system to not work properly.

The size of the parameter file is limited, with a maximum of 64KB.

7.1 SDK Partition Introduction

Default partition description (below is the RK3572 EVB partition reference, Linux6.12 SDK)

Number	Start (sector)	End (sector)	Size	Name
1	16384	24575	4M	uboot
2	24576	32767	4M	misc
3	32768	163839	64M	boot
4	163840	294911	32M	recovery
5	294912	360447	32M	backup
6	360448	12943359	6144M	rootfs
7	12943360	12943359	128M	oem
8	13205504	61120478	22.8G	userdata

- uboot Partition: For the uboot.img compiled from uboot.
- misc Partition: For misc.img, used by recovery.
- boot Partition: For the boot.img compiled from the kernel.
- recovery Partition: For the recovery.img compiled from recovery.
- backup Partition: Reserved, not in use for now.
- rootfs Partition: For the rootfs.img compiled from buildroot, debian, or yocto.
- oem Partition: For manufacturers to use, storing manufacturer's APP or data. Mounted in the /oem directory.
- userdata Partition: For APPs to temporarily generate files or for end-users. Mounted in the /userdata directory.

7.2 Preview of Parameter File

The parameter file is mainly used to define the partition table and supports two partition formats: one is the GPT format, and the other is the traditional CMDLINE partition. The content of the parameter file will vary for different projects and platforms.
By default, Linux platforms use the GPT format.

This document uses the parameter file of the RK3576 platform as an example for illustration.

RK3576 GPT partition format parameter definition:

```
FIRMWARE_VER: 1.0
MACHINE_MODEL: RK3576
MACHINE_ID: 007
MANUFACTURER: RK3576
MAGIC: 0x5041524B
ATAG: 0x00200800
MACHINE: 0xffffffff
CHECK_MASK: 0x80
PWR_HLD: 0,0,A,0,1
```

```
TYPE: GPT
GROW_ALIGN: 0
CMDLINE:
mtdparts=:0x00002000@0x00004000(uboot),0x00002000@0x00006000(misc),0x00020000@0x00008000(boo
t),0x00040000@0x00028000(recovery),0x00010000@0x00068000(backup),0x01c00000@0x00078000(rootf
s),0x00040000@0x01c78000(oem),-@0x01cb8000(userdata:grow)
uuid:rootfs=614e0000-0000-4b53-8000-1d28000054a9
uuid:boot=7A3F0000-0000-446A-8000-702F00006273
```

The main differences between the GPT partition and the traditional partition in the parameter file are:

- Define **TYPE: GPT**
- The last partition adds a grow identifier, such as: **userdata:grow**
- Specify the UUID of the system or rootfs, such as: **uuid:rootfs=614e0000-0000-4b53-8000-1d28000054a9**
- The parameter file is not burned into the NVM (EMMC, NAND, etc.), and only the partition definition and UUID of mtdparts are used. Other information is defined only for compatibility with upgrade tools.

7.3 Explanation of Parameter File Content

7.3.1 FIRMWARE_VER:1.0

Entry	FIRMWARE_VER
Type	Decimal number, format: X.X
Value	0 - 255
Description	Used when packaging updata.img, the upgrade tool identifies the firmware version number based on this.

7.3.2 MACHINE_MODEL:RK3576

Entry	MACHINE_MODEL
Type	String
Length	255 (maximum)
Description	Machine model, used when packaging updata.img, different projects can be modified, used for upgrade tool display. Can be used in recovery to upgrade firmware to determine if the firmware and machine match.

7.3.3 MACHINE_ID:007

Entry	MACHINE_ID
Type	String
Length	255 (maximum)
Description	Product development ID, can be a combination of characters and numbers, used when packaging updata.img, different projects use different IDs, can be used to identify machine models. In recovery, it can be used to determine if the firmware matches.

7.3.4 MANUFACTURER:RK3576

Entry	MANUFACTURER
Type	String
Length	255 (maximum)
Description	Manufacturer information, used when packaging updata.img, can be modified, used for upgrade tool display.

7.3.5 MAGIC:0x5041524B

Entry	MAGIC
Type	Hexadecimal number
Value	0x5041524B (fixed)
Description	Magic number MAGIC, cannot be modified, some new APs use DTS, this item is useless, for compatibility, do not delete or modify.

7.3.6 ATAG:0x60000800

Entry	ATAG
Type	Hexadecimal number
Value	32bits DDR address
Description	ATAG DDR storage address, some new APs use DTS, this item is useless, for compatibility, do not delete or modify.

7.3.7 MACHINE:0xffffffff

Entry	MACHINE
Type	String
Length	255 (maximum)
Description	Used for kernel recognition, this definition matches the kernel. This item is useless in GPT format, for compatibility, do not delete or modify.

The following table lists the values for several platforms:

Chip	MACHINE
RK29xx	2929
RK292X	2928
RK3066	3066
RK3126C	3126c
RK3326	3326
RK3399	3399
RK3308	3308

7.3.8 CHECK_MASK:0x80

Entry	CHECK_MASK
Type	Hexadecimal number
Value	0x80(fixed)
Description	Reserved, do not modify.

7.3.9 TYPE:GPT

GPT partition table identifier, the burning tool will create a GPT partition table based on the partition defined in CMDLINE, the parameter file will not be burned into NVM (NAND, EMMC, etc.) storage devices.

7.3.10 GROW_ALIGN

GPT partition table identifier, can only choose 0 or 1, Linux defaults to 0; other platforms such as Android, the kernel DM verity is enabled (CONFIG_DM_VERITY=y), GROW_ALIGN needs to be set to 1. This way, the tool aligns the end position of the last partition to 32K.

7.3.11 CMDLINE

MTD partition definition explanation:

```
CMDLINE:
mtdparts=:0x00002000@0x00004000(uboot),0x00002000@0x00006000(misc),0x00020000@0x00008000(boo
t),0x00040000@0x00028000(recovery),0x00010000@0x00068000(backup),0x01c00000@0x00078000(rootf
s),0x00040000@0x01c78000(oem),-@0x01cb8000(userdata:grow)
```

Partition definition explanation:

1. Single partition explanation:

For example: 0x00020000@0x00008000(boot), the value before the @ symbol is the partition size, the value after the @ symbol is the partition start position, and the characters in parentheses are the partition name. All values are in sectors, with 1 sector being 512Bytes. In the example, the boot partition starts at the 0x8000 sector position and is 0x2000 sectors (16MB) in size.

2. For performance, each partition start address needs to be aligned to 32KB (64 sectors), and the size also needs to be a multiple of 32KB.
3. If using a sparse format image, data will be erased during the upgrade, for better compatibility, the corresponding partition should be aligned to 4MB, and the size should also be configured in multiples of 4MB.
4. When using GPT partitions, the addresses defined in the parameter are all real logical addresses (LBA). For example, if uboot is defined at 0x4000, then when burned into EMMC and NAND, the logical address is also 0x4000.

Name	Parameter Defined Address	EMMC Logical Address	NAND Logical Address	Size
GPT	--	0	0	32KB
LOADER	--	0x40	0x40	4MB-32KB
Reserved	--	0x2000	0x2000	4MB
UBOOT	0x4000	0x4000	0x4000	4MB
TRUST	0x6000	0x6000	0x6000	4MB

The last partition needs to specify the grow parameter, and the tool will allocate the remaining space to the last partition.

7.4 Parameter Partition Definition Specification

- All partitions that are written to by bootrom, pre-loader, uboot, trust are placed before the recovery partition;
- Any new partitions added by customers that are written to by pre-loader, uboot, trust should also be placed before the recovery partition;
- Currently, the partitions that rockchip pre-level firmware operates on include misc, vbmeta, security, all of which should be placed before recovery.

7.5 SDK Partition Configuration Modification Guide

7.5.1 Command Overview

The following are the relevant commands for partition operations in the SDK:

```
$ ./build.sh --help | grep parts
print-parts              Print partition information
list-parts             Alias for printing partition information
mod-parts                 Interactive modification of partition table
edit-parts               Edit the original partition configuration
new-parts                Re-create partitions
extra-parts            Package additional partition images
```

7.5.2 Partition Information Printing

Use the following command to print the current partition configuration:

```
<SDK>$ make print-parts
...
=========================================
          Partition Table
=========================================
1: uboot located at 0x00004000 size 0x00002000 (4M)
2: misc  located at 0x00006000 size 0x00002000 (4M)
3: boot  located at 0x00008000 size 0x00020000 (64M)
4: recovery located at 0x00028000 size 0x00040000 (128M)
5: backup located at 0x00068000 size 0x00010000 (32M)
6: rootfs located at 0x00078000 size 0x01c00000 (14G)
7: oem  located at 0x01c78000 size 0x00040000 (128M)
8: userdata growable partition

Traditional command line parameters:
```

```
0x00002000@0x00004000(uboot),0x00002000@0x00006000(misc),0x00020000@0x00008000(boot),...
```

7.5.3 Partition Information Modification

Enter the edit mode through the following command to modify the partition configuration:

```
$ make edit-parts

## Chapter-9 Partition name Size
<hidden> 0x00004000 # 8M
uboot 0x00002000 # 4M
misc 0x00002000 # 4M
boot 0x00020000 # 64M
recovery 0x00040000 # 128M
backup 0x00010000 # 32M
rootfs 0x01c00000 # 14G
oem 0x00040000 # 128M
userdata - # Growable

## Chapter-9 For example, increase the boot partition to 128M, change to:
boot 0x00080000 # 128M
## Chapter-9 If you need to add a new env partition, add:
env 0x00040000 # 64M
```

After modification, the configuration is automatically saved to the `device/rockchip/parameter.txt` file. Use `git diff` to view the changes:

```
device/rockchip$ git diff
...
```

7.5.4 Adding a New Partition

If you want to add an `env` partition in the SDK compilation, the steps are as follows:

1. Add an `env` directory in the `device/rockchip/common/extra-parts` directory.
2. Run `make menuconfig` to configure additional partitions.
3. After configuration, use `make savedefconfig` to save the configuration.
4. Add `env` partition configuration in `parameter.txt` through `make edit-parts`.
5. Compile and generate additional partition images using `make extra-parts`, and the corresponding files will be generated in the `output/extra-parts` directory.

```
$ tree -L 1 output/extra-parts
output/extra-parts/
├── env
├── env.fs
├── env.img
...
```

8. Chapter-10 SDK Firmware Upgrade

This chapter mainly introduces the process of building a complete image file (image), burning it and running it on a hardware device.
The introduction of several image burning tools provided by Rockchip platform is as follows. You can choose the appropriate burning method for burning. Before burning, you need to install the latest USB driver. For details, please see Rockchip USB Driver Installation.

Tools	Run System	Description
RKDevTool	Windows	Rockchip development tools, discrete firmware upgrade and entire update firmware upgrade tools
FactoryTool	Windows	Mass production upgrade tool, supports USB one-to-multiple burning
Linux_Upgrade_Tool	Linux	A tool developed under Linux to support firmware upgrades

8.1 Burning Mode Introduction

The several modes of Rockchip platform hardware operation are as shown in the following table. Firmware can only be burned or updated on the board when the device is in Maskrom or Loader mode.

Mode	Tool Burn	Description
Maskrom	Support	When firmware is not yet burned into the Flash, the Chip will enter Maskrom mode, allowing the initial firmware burning. During the development and debugging process, if Loader fails to start normally, firmware can also be burned in Maskrom mode.
Loader	Support	In Loader mode, firmware can be burned and upgraded. You can use tools to burn a certain partition image file separately to facilitate debugging.
Recovery	Not supported	System boot recovery starts, its main function is to upgrade and restore factory settings.
Normal Boot	Not supported	System boot rootfs starts, loads rootfs, most development is debugged in this mode.

There are several ways to enter the burning mode:

- The board has not been burned with firmware. Power on and enter Maskrom mode.
- The board has been burned with firmware, press and hold the recovery button to power on or reset,

and the system will enter the Loader firmware burning mode.

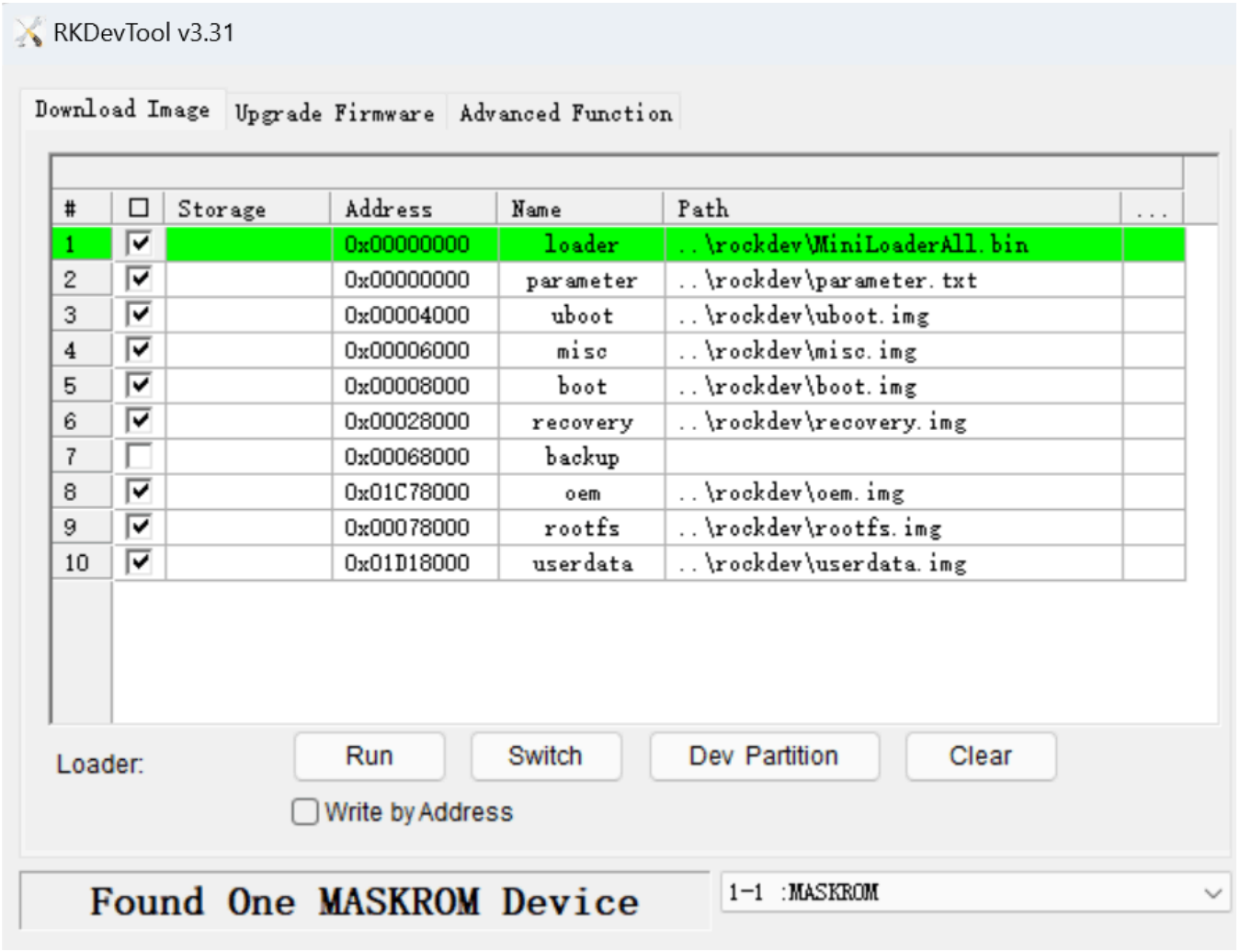
- The board has been burned with firmware, press and hold the Maskrom button to power on or reset, and the system will enter the MaskRom firmware burning mode.
- The board has been burned with firmware, and the board enters the system normally after powering on or resetting, Rockchip's development tool displays "An ADB device was found" or "An MSC device was found", and then click the "Switch" button on the tool to enter Loader mode.
- The board has been burned with firmware, you can enter `reboot loader` command in the serial port or ADB command line mode to enter the Loader mode.

8.1.1 Windows Flashing Instructions

Rockchip's SDK provides a Windows burning tool (the tool version requires V3.31 or above). The tool is located in the project root directory:

```
tools/
├── windows/RKDevTool
```

As shown in the figure below, after compiling and generating the corresponding firmware, the device needs to enter the MASKROM or BootROM burning mode.
After connecting the USB download cable, press and hold the "MASKROM" button and press the reset button "RST" and then release it to enter MASKROM mode, after loading the corresponding path of the compiled and generated firmware, click "Execute" to burn. You can also press and hold the "recovery" button and press the reset button "RST" and then release it to enter the loader mode for burning. The following is the partition offset and burning file in the MASKROM mode . (Note: Windows PC requires administrator rights to run the tool before it can be executed)



Note: Before burning, you need to install the latest USB driver. For driver details, please see:

```
<SDK>/tools/windows/DriverAssitant_v5.13.zip
```

8.1.2 Linux Flashing Instructions

The Linux burning tool is located in the tools/linux directory (Linux_Upgrade_Tool tool version requires V2.26 or above), please make sure your board is connected to MASKROM/loader rockusb. For example, if the generated firmware is in the rockdev directory, the upgrade command is as follows:

```
sudo ./upgrade_tool ul rockdev/MiniLoaderAll.bin -noreset
sudo ./upgrade_tool di -p rockdev/parameter.txt
sudo ./upgrade_tool di -u rockdev/uboot.img
sudo ./upgrade_tool di -trust rockdev/trust.img ##For new chips, trust has been merged into
the uboot partition
sudo ./upgrade_tool di -misc rockdev/misc.img
sudo ./upgrade_tool di -b rockdev/boot.img
sudo ./upgrade_tool di -recovery rockdev/recovery.img
sudo ./upgrade_tool di -oem rockdev/oem.img
sudo ./upgrade_tool di -rootfs rockdev/rootfs.img
sudo ./upgrade_tool di -userdata rockdev/userdata.img
sudo ./upgrade_tool rd
```

Or upgrade the packaged complete firmware:

```
sudo ./upgrade_tool uf rockdev/update.img
```

Or in the root directory, run the following command to upgrade when the device is in MASKROM mode:

```
./rkflash.sh
```

8.1.3 System Partition Introduction

Default partition description (below is the RK3572 EVB partition reference, Linux6.12 SDK)

Number	Start (sector)	End (sector)	Size	Name
1	16384	24575	4M	uboot
2	24576	32767	4M	misc
3	32768	163839	64M	boot
4	163840	294911	32M	recovery
5	294912	360447	32M	backup
6	360448	12943359	6144M	rootfs
7	12943360	12943359	128M	oem
8	13205504	61120478	22.8G	userdata

- uboot partition: uboot.img compiled by uboot.
- misc partition: for misc.img, used by recovery.
- boot partition: boot.img compiled by kernel.
- recovery partition: recovery.img compiled by recovery.
- backup partition: reserved, temporarily unused.
- rootfs partition: rootfs.img compiled by Buildroot, Debian or Yocto.
- oem partition: used by manufacturers to store their APP or data. Mounted in the /oem directory.
- userdata partition: used for APP to temporarily generate files or for end users, mounted in the /userdata directory.

9. Chapter-11 SDK Development

Core components development in SDK, such as U-Boot, Kernel, Recovery, Buildroot, Debian, Yocto, Multivideo, Graphics, Rootfs post-processing, Overlays, and other key component development.

9.1 Bootloader Development

Bootloader is a type of software that runs before the operating system starts. It is responsible for initializing hardware devices, configuring system parameters, and loading the operating system kernel. Bootloader development on the Rockchip platform typically involves two main components: U-Boot and UEFI (Unified Extensible Firmware Interface).

9.1.1 U-Boot Development

This section provides a brief introduction to the basic concepts of U-Boot and considerations for compilation, assisting customers in understanding the U-Boot framework for RK platforms. For specific details on U-Boot development, refer to `<SDK>/docs/en/Common/U-Boot/Rockchip-Developer-Guide-UBoot-*.pdf`.

9.1.1.1 Introduction to U-Boot

The v2017(next-dev) is a development branch derived from the official next-dev release of U-Boot by RK, which now supports all mainstream RK chips currently on the market. The main features supported include:

- Support for booting RK Android firmware;
- Support for booting Android AOSP firmware;
- Support for booting Linux Distro firmware;
- Support for Rockchip miniloader and two types of Pre-loaders, SPL/TPL;
- Support for display devices such as LVDS, EDP, MIPI, HDMI, CVBS, and RGB;
- Support for booting from storage devices including eMMC, Nand Flash, SPI Nand flash, SPI NOR flash, SD card, and USB flash drive;
- Support for FAT, EXT2, and EXT4 file systems;
- Support for GPT and RK parameter partition tables;
- Support for boot logo, charging animation, low battery management, and power management;
- Support for I2C, PMIC, CHARGE, FUEL GAUGE, USB, GPIO, PWM, GMAC, eMMC, NAND, and Interrupt;
- Support for Vendor storage to save user data and configurations;
- Support for RockUSB and Google Fastboot for USB gadget flashing of eMMC;
- Support for USB devices such as Mass storage, ethernet, and HID;
- Support for dynamically selecting kernel DTB based on hardware status.

9.1.1.2 Version

There are two versions of RK's U-Boot: the old version v2014 and the new version v2017, with internal names rkdevelop and next-dev, respectively. Users have two methods to confirm whether the current U-Boot is version v2017.

Method 1: Confirm whether the version number of the Makefile in the root directory is 2017.

```
#
### Chapter-1 SPDX-License-Identifier:      GPL-2.0+
#

VERSION = 2017
PATCHLEVEL = 09
SUBLEVEL =
EXTRAVERSION =
NAME =
......
```

Method 2: Confirm whether the first line of the formal print at boot is U-Boot next-dev.

```
U-Boot next-dev-01818-g11818ff-dirty (Nov 14 2019 - 11:11:47 +0800)
......
```

> Project Open Source: v2017 is open source and regularly updated to Github: https://github.com/rockchip-linux/u-boot
>
> Kernel Version: v2017 requires RK kernel version >= 4.4

9.1.1.3 Preliminary Preparation

- Download rkbin

  This is a repository of tools, used to store the non-open source bin, scripts, and packaging tools of RK. When compiling U-Boot, it indexes the relevant files from this repository to package and generate the loader, trust, and uboot firmware. The rkbin and U-Boot projects must maintain the same directory level relationship.

  > Repository Download: Please refer to the appendix section.

- Download GCC

  The GCC compiler uses gcc-arm-14.3 (Linux6.12 SDK) or gcc-arm-10.3 (Linux5.10/6.1 SDK), placed in the prebuilts directory. prebuilts and U-Boot are at the same directory level. as follows:

```
// 32-bit (Linux6.12 SDK):
prebuilts/gcc/linux-x86/arm/arm-gnu-toolchain-14.3.rel1-x86_64-arm-none-linux-
gnueabihf
// 64-bit (Linux6.12 SDK):
prebuilts/gcc/linux-x86/aarch64/arm-gnu-toolchain-14.3.rel1-x86_64-aarch64-none-linux-
gnu
// 32-bit (Linux5.10/6.1 SDK):
prebuilts/gcc/linux-x86/arm/gcc-arm-10.3-2021.07-x86_64-arm-none-linux-gnueabihf
// 64-bit (Linux5.10/6.1 SDK):
prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu
```

- Select defconfig

  The support status of defconfig for each platform (subject to the SDK release):

  > "[Chip]_defconfig" or "[Chip].config" are usually full-featured versions, and the rest are specific feature versions.

1. The support status of defconfig for each platform (subject to the SDK release):

> "[Chip]_defconfig" or "[Chip].config" are usually full-featured versions, and the rest are specific feature versions.

Chip	defconfig	Support for Kernel DTB	Description
RK3036	rk3036_defconfig	Yes	General version
RK3128x	rk3128x_defconfig	Yes	General version
RK3126	rk3126_defconfig	Yes	General version
RK3288	rk3288_defconfig	Yes	General version
RK3328	rk3328_defconfig	Yes	General version
RK3399	rk3399_defconfig	Yes	General version
RK3399Pro	rk3399pro_defconfig	Yes	General version
RK3399Pro-npu	rknpu-lion_defconfig	Yes	General version
RK3308	rk3308_defconfig rk3308-aarch32_defconfig	Yes	General version Supports aarch32 mode
PX30	px30_defconfig	Yes	General version
RK3326	rk3326_defconfig rk3326-aarch32_defconfig	Yes	General version Supports aarch32 mode
RK3568	rk3568_defconfig rk3568-dfu.config rk3568-nand.config rk3568-spl-spi-nand_defconfig rk3568-aarch32.config rk3568-usbplug.config	Yes	General version Supports DFU Supports MLC/TLC/eMMC SPI-nand specific SPL Supports aarch32 mode Supports usbplug mode
RK3566	rk3566.config rk3566-eink.config	Yes	General version E-ink version
RK3588	rk3588_defconfig rk3588-ramboot.config rk3588-sata.config rk3588-aarch32.config rk3588-ipc.config	Yes	General version No storage device (memory boot) Dual storage support for sata boot Supports aarch32 mode Used on IPC SDK
RK3562	rk3562_defconfig	Yes	General version

Chip	defconfig	Support for Kernel DTB	Description
RK3576	rk3576_defconfig rk3576-usbplug.config rk3576-car.config rk3576-ab-car.config rk3576-eink.config	Yes	General version Open source usbplug Car version Supports AB system car version E-ink version

> Note: If the table and the defconfig released by the SDK are different, please refer to the SDK.

9.1.1.4 Booting Process

The U-Boot booting process for the RK platform is as follows, with only some important steps listed:

```
start.s
    // Assembly environment
    => IRQ/FIQ/lowlevel/vbar/errata/cp15/gic   // ARM architecture-related lowlevel
initialization
    => _main
        => stack                               // Prepare the stack needed for the C
environment
        // [First Phase] C environment initialization, initiating a series of function calls
        => board_init_f: init_sequence_f[]
            initf_malloc
            arch_cpu_init                      // [SoC's lowlevel initialization]
            serial_init                        // Serial port initialization
            dram_init                          // [Obtaining DDR capacity information]
            reserve_mmu                        // Reserve memory from the end of the DDR
towards lower addresses
            reserve_video
            reserve_uboot
            reserve_malloc
            reserve_global_data
            reserve_fdt
            reserve_stacks
            dram_init_banksize
            sysmem_init
            setup_reloc                        // Determine the relocation address for U-
Boot itself
        // Assembly environment
        => relocate_code                       // Assembly implementation of U-Boot code
relocation
        // [Second Phase] C environment initialization, initiating a series of function
calls
        => board_init_r: init_sequence_r[]
            initr_caches                       // Enable MMU and I/D cache
            initr_malloc
            bidram_initr
```

```
            sysmem_initr
            initr_of_live                       // Initialize of_live
            initr_dm                            // Initialize the device model (dm) framework
            board_init                          // [Platform initialization, the core part]
                board_debug_uart_init           // Serial port I/O multiplexing, clock
configuration
                init_kernel_dtb                 // [Switch to the kernel device tree blob
(dtb)]!
               clks_probe                      // Initialize system clock frequencies
               regulators_enable_boot_on       // Initialize system power supply
                io_domain_init                  // I/O domain initialization
                set_armclk_rate                 // __weak, ARM frequency scaling (only
implemented if the platform requires it)
                dvfs_init                       // DVFS for wide-temperature chips
                rk_board_init                   // __weak, Implemented by each specific
platform
            console_init_r
            board_late_init                     // [Platform late initialization]
                rockchip_set_ethaddr            // Set MAC address
                rockchip_set_serialno           // Set serial number
                setup_boot_mode                 // Parse "reboot xxx" command,
                // Identify key presses and loader burn mode, recovery
                charge_display                  // U-Boot charging
                rockchip_show_logo              // Display boot logo
                soc_clk_dump                    // Print the clock tree
                rk_board_late_init              // __weak, Implemented by each specific
platform
            run_main_loop                       // [Enter command line mode, or execute boot
command]
```

9.1.1.5 Shortcut Keys

The RK platform offers a set of serial port combination keys to trigger certain events for debugging and flashing (if they do not trigger, please try multiple times; ineffective when secure-boot is enabled). **Press and hold at startup**:

- Ctrl+C: Enter the U-Boot command line mode;
- Ctrl+D: Enter the loader flashing mode;
- Ctrl+B: Enter the maskrom flashing mode;
- Ctrl+F: Enter the fastboot mode;
- Ctrl+M: Print the info of bidram/system;
- Ctrl+I: Enable the kernel initcall_debug;
- Ctrl+P: Print the cmdline info;
- Ctrl+S: Enter the U-Boot command line after "Starting kernel...";

9.1.2 UEFI Development

Currently, only the RK3588 chip supports UEFI development.

EDK2-RK3588 is an open-source project based on the UEFI Development Kit (EDK II), specifically designed for the Rockchip RK3588 SoC. This project provides a complete development environment to assist developers in firmware development, debugging, and optimization on RK3588.

To familiarize yourself with RK3588 UEFI development and debugging, you can obtain the "Rockchip_Developer_Guide_UEFI_EN.pdf" development guide, which is released with the SDK and located under `docs/en/RK3588/UEFI`.

9.2 Kernel Development

This section aims to provide you with a brief introduction to common kernel configuration modifications, focusing on Device Tree (DTS) configurations, to help you make simple modifications more quickly and conveniently. The following introduction is based on the Linux 5.10 kernel version.

9.2.1 Introduction to DTS

9.2.1.1 DTS Overview

The Device Tree Source (DTS) is a data structure that describes hardware devices, used to define and configure hardware devices within the Linux kernel. The DTS employs a text-like format, representing hardware devices and their relationships through hierarchical structures and property descriptions.

The DTS is compiled into a Device Tree Blob (DTB), which the kernel loads and parses at startup, extracting information about the hardware devices for initialization and configuration.

The introduction of the device tree addresses the inflexibility and maintainability issues of the traditional "hard-coding" approach when dealing with different hardware configurations. With the device tree, hardware descriptions and configuration information are abstracted, allowing the kernel to adapt to different hardware platforms without modifying the kernel code. In this way, the same kernel can run on multiple different hardware devices, simply by loading the corresponding device tree.

This chapter aims to introduce how to add a new board DTS configuration and some common syntax. For more detailed information on DTS, see: devicetree-specifications and devicetree-bindings.

9.2.1.2 Adding a New Product DTS

- Creating a dts file

The Linux Kernel currently supports multiple platforms using dts files, and the dts files for the Rockchip platform are stored at:

```
ARM 32-bit Kernel: arch/arm/boot/dts/
ARM 64-bit Kernel: arch/arm64/boot/dts/rockchip
```

The general naming convention for dts files is "soc-board-name.dts", such as rk3399-evb-ind-lpddr4-linux.dts. soc refers to the chip model, and board_name is generally named according to the silk screen on the board. If your board is an all-in-one board, you only need one dts file to describe it.

If the hardware design consists of a core board and a bottom board, or if the product has multiple product forms, you can place the common hardware descriptions in the dtsi file, and the dts file describes different hardware modules, and include the common hardware descriptions by including "xxx.dtsi".

```
├──rk3399-evb-ind-lpddr4-linux.dts
│  ├── rk3399-evb-ind.dtsi
│  └── rk3399-linux.dtsi
```

- Modify the Makefile in the directory where the dts is located

```
--- a/arch/arm64/boot/dts/rockchip/Makefile
+++ b/arch/arm64/boot/dts/rockchip/Makefile
@@ -50,6 +50,7 @@ dtb-$(CONFIG_ARCH_ROCKCHIP) += rk3368-tablet.dtb
 dtb-$(CONFIG_ARCH_ROCKCHIP) += rk3399-evb.dtb
 dtb-$(CONFIG_ARCH_ROCKCHIP) += rk3399-evb-ind-lpddr4-android.dtb
 dtb-$(CONFIG_ARCH_ROCKCHIP) += rk3399-evb-ind-lpddr4-android-avb.dtb
+dtb-$(CONFIG_ARCH_ROCKCHIP) += rk3399-evb-ind-lpddr4-linux.dtb
```

When compiling the Kernel, you can directly `make dts-name.img` (such as rk3399-evb-ind-lpddr4-linux.img) to generate the corresponding boot.img (including dtb data).

- Several descriptions of dts syntax

The dts syntax can include other common dts data like c/c++ by using #include xxx.dtsi. The dts file will inherit all the properties and values of the device nodes included in the dtsi file. If a property is defined in multiple dts/dtsi files, its value will be the definition in the dts. All chip-related controller nodes will be defined in soc.dtsi, and if you need to enable the device function, you need to set its status to "okay" in the dts file. To disable the device, you need to set its status to "disabled" in the dts file.

```
/dts-v1/;

#include "rk3399-evb-ind.dtsi"
#include "rk3399-linux.dtsi"
...
&i2s2 {
        #sound-dai-cells = <0>;
        status = "okay";
};

&hdmi_sound {
        status = "okay";
};
```

9.2.2 Module Development Documentation

The directory `<SDK>/docs/en/Common/` contains development documents categorized by functional modules. This section primarily serves as an index for these development documents. When encountering issues during actual development, refer to the following table to read and learn the corresponding development guides. They can be obtained under `docs/en/Common` and will be continuously improved and updated. For more details, see Documentation Instructions.

9.2.3 Common Module Commands

9.2.3.1 CPU-Related Commands

9.2.3.1.1 CPU Frequency Locking Operation

- **Non-Heterogeneous Core Platforms**:

Switch the CPU frequency control policy to `userspace`:

```
    echo userspace > /sys/devices/system/cpu/cpufreq/policy0/scaling_governor
```

Set the CPU frequency to 216MHz:

```
    echo 216000 > /sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
```

- **Heterogeneous Core Platforms**:

Switch the little core CPU frequency control policy to `userspace`:

```
    echo userspace > /sys/devices/system/cpu/cpufreq/policy0/scaling_governor
```

Set the little core CPU frequency to 216MHz:

```
    echo 216000 > /sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
```

Switch the big core CPU frequency control policy to `userspace`:

```
    echo userspace > /sys/devices/system/cpu/cpufreq/policy4/scaling_governor
```

Set the big core CPU frequency to 408MHz:

```
    echo 408000 > /sys/devices/system/cpu/cpufreq/policy4/scaling_setspeed
```

9.2.3.1.2 View Current CPU Frequency

- **Non-Heterogeneous Core Platform**:

Using the CPUFreq user-space interface:

```
    cat /sys/devices/system/cpu/cpufreq/policy0/scaling_cur_freq
```

Using the Clock Debug interface:

```
    cat /sys/kernel/debug/clk/armclk/clk_rate
```

- **Heterogeneous Core Platform**:

Using the CPUFreq user-space interface to view the frequency of little cores and big cores:

```
    cat /sys/devices/system/cpu/cpufreq/policy0/scaling_cur_freq  # Frequency of little
cores
    cat /sys/devices/system/cpu/cpufreq/policy4/scaling_cur_freq  # Frequency of big cores
```

Using the Clock Debug interface to view the frequency of little cores and big cores:

```
    cat /sys/kernel/debug/clk/armclkl/clk_rate  # Frequency of little cores
    cat /sys/kernel/debug/clk/armclkb/clk_rate  # Frequency of big cores
```

9.2.3.1.3 Checking Current CPU Voltage

- **Non-Heterogeneous Core Platforms**:

```
  cat /sys/kernel/debug/regulator/vdd_core/voltage  # Note: vdd_core may need to be modified
according to actual configuration
```

- **Heterogeneous Core Platforms**:

```
  cat /sys/kernel/debug/regulator/vdd_core_l/voltage  # Little core voltage
  cat /sys/kernel/debug/regulator/vdd_core_b/voltage  # Big core voltage
```

9.2.3.1.4 CPU Independent Frequency and Voltage Adjustment

- When adjusting frequency and voltage, please follow the sequence below:

  Increase Frequency: First adjust the voltage, then increase the frequency.
  Decrease Frequency: First decrease the frequency, then adjust the voltage.

- **Setting CPU Frequency via Clock Debug Interface**:

For non-heterogeneous core platforms (e.g., RK3288), set to 216MHz:

```
    echo 216000000 > /sys/kernel/debug/clk/armclk/clk_rate
    cat /sys/kernel/debug/clk/armclk/clk_rate
```

For heterogeneous core platforms (e.g., RK3399), set the little cores to 216MHz and the big cores to 408MHz:

```
echo 216000000 > /sys/kernel/debug/clk/armclkl/clk_rate
cat /sys/kernel/debug/clk/armclkl/clk_rate
echo 408000000 > /sys/kernel/debug/clk/armclkb/clk_rate
cat /sys/kernel/debug/clk/armclkb/clk_rate
```

- **Setting CPU Voltage via Regulator Debug Interface**:

For non-heterogeneous core platforms (e.g., RK3288), set to 950mV:

```
echo 950000 > /sys/kernel/debug/regulator/vdd_core/voltage
cat /sys/kernel/debug/regulator/vdd_core/voltage
```

For heterogeneous core platforms (e.g., RK3399), set the little cores to 950mV and the big cores to 1000mV:

```
echo 950000 > /sys/kernel/debug/regulator/vdd_core_l/voltage
cat /sys/kernel/debug/regulator/vdd_core_l/voltage
echo 1000000 > /sys/kernel/debug/regulator/vdd_core_b/voltage
cat /sys/kernel/debug/regulator/vdd_core_b/voltage
```

9.2.3.1.5 How to View Frequency and Voltage Table

```
cat /sys/kernel/debug/opp/opp_summary
```

9.2.3.1.6 How to Check CPU Temperature

```
cat /sys/class/thermal/thermal_zone0/temp
```

For more details, please refer to the document:
`docs/en/Common/DVFS\Rockchip_Developer_Guide_CPUFreq_EN.pdf`

9.2.3.2 GPU-Related Commands

9.2.3.2.1 GPU Clock Setting Operation

- Switch the GPU to user space control mode (the path `ff400000.gpu` may vary depending on the platform):

```
echo userspace > /sys/class/devfreq/ff400000.gpu/governor
```

- Set the GPU frequency to 400MHz:

```
echo 400000000 > /sys/class/devfreq/ff400000.gpu/userspace/set_freq
```

- Check the current GPU frequency:

```
cat /sys/class/devfreq/ff400000.gpu/cur_freq
```

9.2.3.2.2 View Current GPU Frequency

- Method One: Through the devfreq user-space interface (the path `ff400000.gpu` may vary depending on the platform):

```
cat /sys/class/devfreq/ff400000.gpu/cur_freq
```

- Method Two: Through the clock debug interface (`aclk_gpu` may vary according to actual configuration):

```
cat /sys/kernel/debug/clk/aclk_gpu/clk_rate
```

9.2.3.2.3 Inspect Current GPU Voltage

- Inspect GPU voltage (`vdd_logic` may vary based on actual regulator configuration):

```
cat /sys/kernel/debug/regulator/vdd_logic/voltage
```

9.2.3.2.4 GPU Frequency and Voltage Adjustment

- **Important**: When increasing frequency, adjust the voltage first; when decreasing frequency, adjust the frequency first.
- Disable GPU automatic frequency scaling (the path `ff400000.gpu` may vary depending on the platform):

```
echo userspace > /sys/class/devfreq/ff400000.gpu/governor
```

- Adjust GPU frequency (`aclk_gpu` may vary depending on the actual configuration):

```
echo 400000000 > /sys/kernel/debug/clk/aclk_gpu/clk_rate
cat /sys/kernel/debug/clk/aclk_gpu/clk_rate
```

- Adjust GPU voltage (`vdd_logic` may vary depending on the actual regulator configuration):

```
echo 1000000 > /sys/kernel/debug/regulator/vdd_logic/voltage
cat /sys/kernel/debug/regulator/vdd_logic/voltage
```

9.2.3.2.5 Inspect GPU Utilization

- Inspect GPU Utilization (the path `fb000000.gpu` may vary depending on the platform):

```
cat /sys/class/devfreq/*.gpu/load
```

9.2.3.2.6 Inspect GPU Temperature

- Inspect GPU temperature (the `thermal_zone1` may need to be modified according to the actual hardware configuration):

```
cat /sys/class/thermal/thermal_zone1/temp
```

9.2.3.3 DDR Related Commands

9.2.3.3.1 Inspect DDR Frequency

- Use the following commands to view DDR frequency information:

```
cat /sys/kernel/debug/clk/clk_summary | grep ddr
cat /sys/class/devfreq/dmc/cur_freq
```

9.2.3.3.2 DDR Frequency Setting Operation

- Switch the DDR frequency control strategy to `userspace`:

```
echo userspace > /sys/class/devfreq/dmc/governor
```

- List available DDR frequencies:

```
cat /sys/class/devfreq/dmc/available_frequencies
```

- Set the DDR frequency to 1560 MHz:

```
echo 1560000000 > /sys/class/devfreq/dmc/userspace/set_freq
```

9.2.3.3.3 View DDR Bandwidth Utilization

- To check the current utilization of DDR bandwidth:

```
cat /sys/class/devfreq/dmc/load
```

9.2.3.3.4 DDR Bandwidth Statistics

- Utilize the DDR bandwidth statistics tool `rk-msch-probe` provided by FAE:

```
./rk-msch-probe-for-user-64bit -c rk3576
```

9.2.3.3.5 Memory Debugging

- Viewing Memory Information:

```
cat /proc/meminfo
```

- Viewing Virtual Memory Usage:

```
cat /proc/vmallocinfo
```

- Enabling CMA Debugging Features, which requires setting in the configuration:

```
CONFIG_CMA_DEBUGFS=y  # Enable debug file nodes
CONFIG_CMA_DEBUG=y    # Print CMA log information
```

- Viewing CMA Related Information:

```
ls /sys/kernel/debug/cma/cma-reserved
```

- Viewing DMA Buffer Information:

```
cat /sys/kernel/debug/dma_buf/bufinfo
cat /proc/rk_dmabuf/size
```

9.2.3.3.6 Memory Stress Test

Download DDR related materials from Rockchip Redmine:
DDR Related Materials Execute memory stress test commands:

```
stressapptest -s 43200 -i 4 -C 4 -W --stop_on_errors -M 128
memtester 128m > /data/memtester_log.txt &
```

For detailed reference:

```
docs/en/Common/DDR\***
docs/en/Common/MEMORY\***
```

9.2.3.4 NPU-related Commands

9.2.3.4.1 Viewing NPU Frequency

Use the following commands to view the NPU frequency (note: the device path `fdab0000.npu` may vary depending on the platform):

```
cat /sys/kernel/debug/clk/clk_summary | grep npu
cat /sys/class/devfreq/*.npu/cur_freq
```

9.2.3.4.2 NPU Frequency Setting Operation

Perform frequency configuration for NPU devices (the device path `*.npu` may need to be modified according to the platform):

```
echo userspace > /sys/class/devfreq/*.npu/governor
echo 1000000000 > /sys/class/devfreq/*.npu/userspace/set_freq
cat /sys/class/devfreq/*.npu/cur_freq
```

9.2.3.4.3 NPU Support Query Settings

The following commands are applicable for the RKNPU2 platform with a driver version above 0.7.2:

- Query NPU Utilization:

```
cat /sys/kernel/debug/rknpu/load
cat /proc/debug/rknpu/load
```

- Query NPU Driver Version:

```
cat /sys/kernel/debug/rknpu/driver_version
cat /proc/debug/rknpu/driver_version
```

- Manage NPU Power State:

```
echo on > /sys/kernel/debug/rknpu/power  # Turn on NPU power
echo off > /sys/kernel/debug/rknpu/power # Turn off NPU power
```

- Query NPU Operating Voltage:

```
cat /sys/kernel/debug/rknpu/volt
```

- Dynamically Manage NPU Power and Delay Shutdown Time (unit: ms):

```
cat /sys/kernel/debug/rknpu/delayms  # Query power delay shutdown time
echo 2000 > /sys/kernel/debug/rknpu/delayms  # Set power delay shutdown time to 2000ms
```

9.2.3.4.4 Related Materials

- **RKNPU Development Materials**

Tools: rknn-toolkit
Runtime: rknpu, RK3399Pro_npu

- **RKNPU2 Development Materials**

Tools: rknn-toolkit2
Runtime: rknpu2
rknn_model_zoo: rknn_model_zoo

9.2.3.5 RGA-related Commands

9.2.3.5.1 Query RGA Frequency

```
cat /sys/kernel/debug/clk/clk_summary | grep rga  # Query RGA frequency, including aclk
frequency
```

9.2.3.5.2 Modifying RGA Frequency

```
echo 400000000 > /sys/kernel/debug/clk/aclk_rga/clk_rate  # Modify the frequency value to
the desired frequency
```

9.2.3.5.3 RGA Driver Version Query

```
cat /sys/kernel/debug/rkrga/driver_version
cat /proc/rkrga/driver_version
```

9.2.3.5.4 Query the version number of the librga library

- **Linux System**

```
strings /usr/lib/librga.so | grep rga_api | grep version
```

9.2.3.5.5 Enable librga Logging

- **Linux System (librga version 1.9.0 and above)**

```
  export ROCKCHIP_RGA_LOG=1
  export ROCKCHIP_RGA_LOG_LEVEL=6
```

9.2.3.5.6 RGA Debug Node

```
cat /sys/kernel/debug/ampak/debug
```

9.2.3.5.7 RGA Load Query

```
cat /sys/kernel/debug/AMPAK/load
```

9.2.3.5.8 RGA Memory Manager Inquiry

```
cat /sys/kernel/debug/rkrga/mm_session
```

9.2.3.5.9 RGA Task Request Inquiry

```
cat sys/kernel/debug/AMPAK/request_manager
```

9.2.3.5.10 RGA Hardware Information Inquiry

```
cat /sys/kernel/debug/rkrga/hardware
```

9.2.3.5.11 Dump Runtime Data

```
/*Use the following command to dump runtime data for debugging purposes*/
echo /data/rga_image > sys/kernel/debug/rkrga/dump_path
echo 1 > sys/kernel/debug/rkrga/dump_image
```

For specific details, refer to the documentation:

```
  `docs/en/Common/RGA\***` or https://github.com/airockchip/librga/tree/main/docs
```

9.2.3.6 VPU-related Commands

9.2.3.6.1 Querying VPU Driver Version

Use the following command to query the version information of the VPU driver:

```
cat /proc/mpp_service/version
```

9.2.3.6.2 Checking VPU Frame Rate

To view the frame rate data of the VPU, use the following command:

```
cat /proc/mpp_service/sessions-summary
```

9.2.3.6.3 Viewing Frequency

To view frequency information related to rkvenc, execute the following command:

```
cat /sys/kernel/debug/clk/clk_summary | grep rkvenc
```

9.2.3.6.4 Modify Frequency

To modify the frequency of the VPU to 600MHz, you can use the following command:

```
echo 600000000 > /proc/mpp_service/rkvenc/clk_core
```

9.2.3.6.5 Enabling VPU Debug Print

To enable the debug print feature of VPU:

```
echo 0x100 > /sys/module/rk_vcodec/parameters/mpp_dev_debug
```

For more information on VPU documentation, please refer to:

```
docs/en/Common/MPP\Rockchip_Developer_Guide_MPP_EN.pdf
```

9.2.4 Pin Configuration Control

9.2.4.1 General-Purpose Input/Output (GPIO)

For instance, the RK3399/RK3399Pro chip offers 5 sets of GPIOs (GPIO0 to GPIO4), totaling 122 GPIO pins. All GPIOs can serve as interrupt sources, with GPIO0 and GPIO1 also capable of acting as system wake-up pins. Each GPIO can be configured via software for pull-up or pull-down, with the default state being input mode. Additionally, the driving capability of the GPIO can also be configured by software.

Regarding the correspondence between GPIO on the schematic and GPIO in the Device Tree Source (DTS), for example, GPIO4C0 should be represented as "gpio4 16" in the DTS. This is because GPIO4A has 8 pins, and GPIO4B also has 8 pins, with counting starting from 0, so the C0 pin corresponds to the 16th pin, the C1 pin corresponds to the 17th pin, and so on.

9.2.4.1.1 I/O Multiplexing (IOMUX)

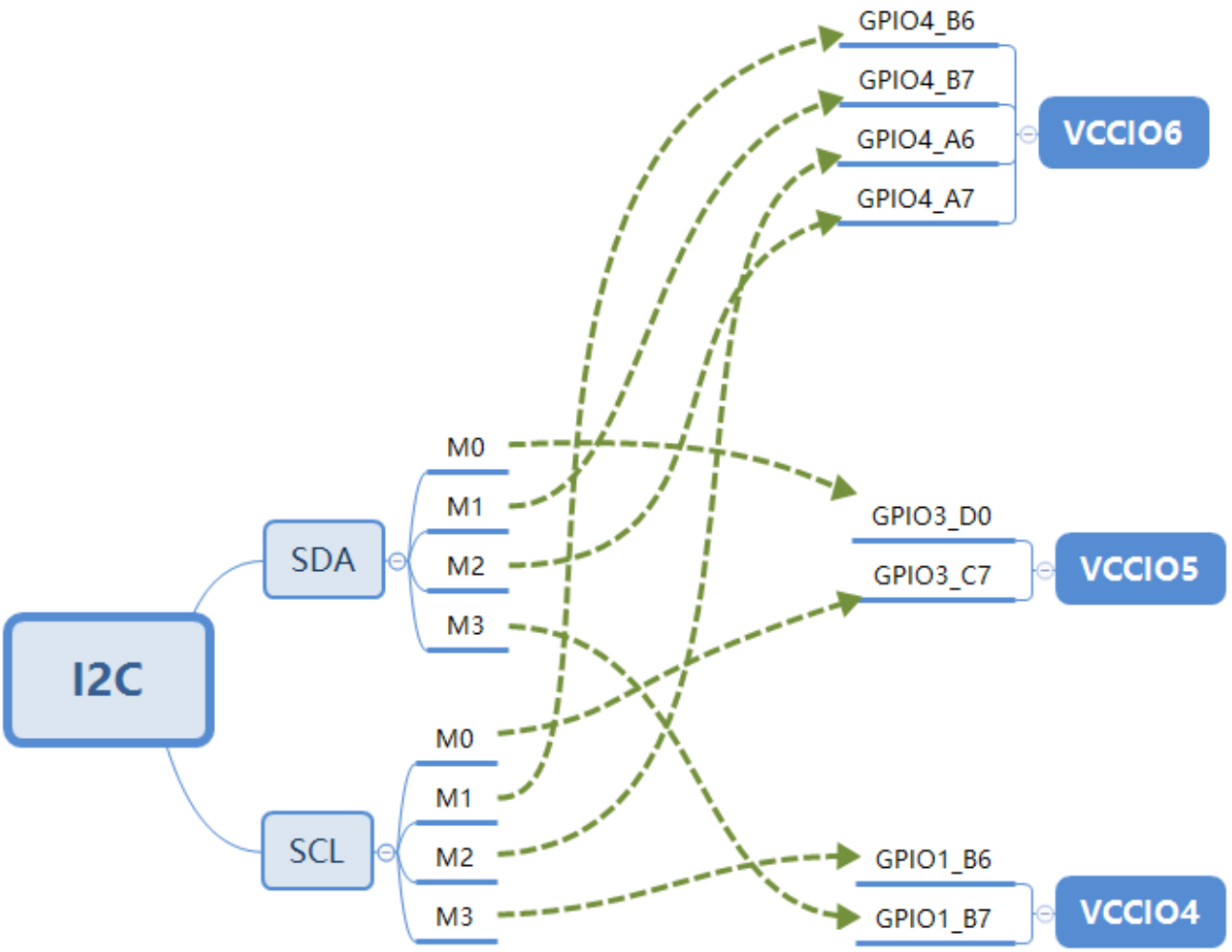
The SOC pins of Rockchip can be multiplexed into various functions. If a controller has multiple multiplexing pins, they are generally referred to as m0, m1, m2, etc. For instance, the I2C controller has two sets of multiplexing pins, which are named 2cm0 and i2cm1 respectively.

The configuration registers for pin multiplexing are located in GRF/PMUGRF (on RK3588, it is referred to as IOC).

As an example, the RK3588 BUS_IOC_GPIO1B_IOMUX_SEL_H Address: Operational Base + offset (0x002C)

> gpio1b7_sel
> 4'h0: GPIO
> 4'h2: MIPI_CAMERA2_CLK_M0
> 4'h3: SPDIF1_TX_M0
> 4'h4: PCIE30X2_PERSTN_M3
> 4'h5: HDMI_RX_CEC_M2
> 4'h6: SATA2_ACT_LED_M1
> 4'h9: I2C5_SDA_M3
> 4'ha: UART1_RX_M1
> 4'hb: PWM13_M2

The following is the IOMUX for RK3588 I2C5:



Multiplexing support provides greater flexibility in hardware design. When the peripheral working voltage is 1.8V or 3.3V, one can select pins with different voltage domains VCCIO.

Note: The register configuration for multi-path multiplexing is ineffective for TX-type pins and effective for RX-type pins.

9.2.4.2 PULL (Port Pull-Up and Pull-Down)

The bias of Rockchip IO PAD generally supports three modes:

- bias-disable
- bias-pull-up
- bias-pull-down

Pull-up and pull-down configurations are applied to the IO PAD and take effect for both GPIO and IOMUX.

9.2.4.3 DRIVE-STRENGTH (Port Drive Strength)

The drive strength of Rockchip IO PAD supports different intensity configurations based on various processes. For chips before RK3399, the drive strength configuration is set in units of mA, while for chips after RK1808, it is generally set in levels, with the numerical value of the gear representing the register configuration value.

For example, in the RK3588 TRM, the drive strength levels for GPIO0_C7 are as follows:

> gpio0c7_ds
> GPIO0C7 DS control Driver Strength Selection
> 3'b000: 100ohm
> 3'b100: 66ohm
> 3'b010: 50ohm
> 3'b110: 40ohm
> 3'b001: 33ohm
> 3'b101: 25ohm

The software driver still processes according to the level, that is, the above register description corresponds to:

> 3'b000: Level0
> 3'b100: Level4
> 3'b010: Level2
> 3'b110: Level6
> 3'b001: Level1
> 3'b101: Level5

In the DTS, `drive-strength=<5>;` indicates the configuration is set to Level5, which means writing `3'b101` to the register.

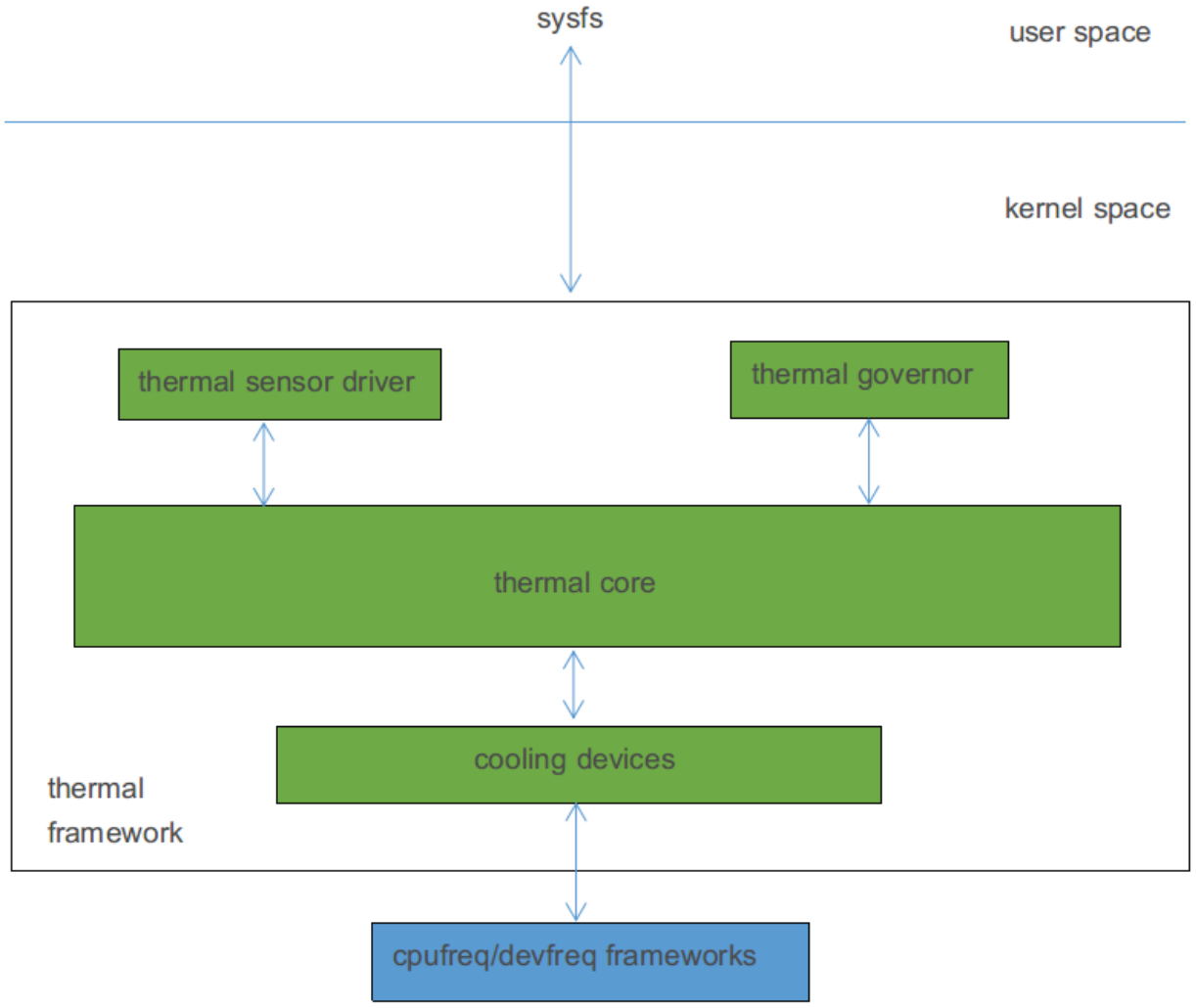
9.2.4.4 SMT (Synchronous Metastability Trigger)

Most Rockchip IO PAD chips support the SMT function, which is disabled by default; enabling SMT can eliminate edge jitter, increase the voltage range of VIH and VIL, and enhance the signal stability of IOs. Generally, the I2C SCL/SDA will have the SMT function enabled by default.

For more details on Pinctrl usage, please refer to the document `<SDK>/docs/en/Common/PINCTRL/Rockchip_Developer_Guide_Linux_Pinctrl_EN.pdf`.

9.2.5 Thermal Control Development

The Thermal framework is a model defined by kernel developers to support the control of system temperature based on a specified governor, in order to prevent the chip from overheating. The Thermal framework consists of governors, cores, cooling devices, and sensor drivers, with the software architecture as follows:



Thermal Governor: It is used to determine whether the cooling device needs to reduce frequency and to what extent. The following types of governors are included in Linux versions 4.4 and above:

- power_allocator: Introduces PID (Proportional-Integral-Derivative) control, dynamically allocates power to each cooling device based on the current temperature, and converts power into frequency, thereby achieving the effect of limiting frequency according to temperature.
- step_wise: The cooling device reduces frequency step by step according to the current temperature.
- fair_share: Cooling devices with more frequency bands are prioritized for frequency reduction.
- userspace: Does not limit frequency.

Thermal Core: It encapsulates and abstracts thermal governors and thermal drivers, and defines a clear interface.

Thermal Sensor Driver: A sensor driver used to obtain temperature, such as tsadc.

Thermal Cooling Device: A device that generates heat or can reduce temperature, such as CPU, GPU, DDR, etc.

The Rockchip chip has thermal control sensors on its ARM core and GPU core, which can monitor the temperature of the CPU and GPU in real time, and control the frequency of the CPU and GPU through algorithms to control the temperature of the CPU and GPU. The hardware design and mold of each product are different, and the corresponding heat dissipation conditions are also different. Appropriate adjustments to thermal control parameters can be made through the following configurations in dts to adapt to the product:

Set the temperature at which thermal control is activated:

```
&threshold {
    temperature = ; /* millicelsius */
};
```

Set the maximum temperature for thermal control:

```
&target {
    temperature = ; /* millicelsius */
};
```

Set the software shutdown temperature:

```
&soc_crit {
    temperature = ; /* millicelsius */
};
```

Configure the hardware shutdown temperature:

```
&tsadc {
    rockchip,hw-tshut-mode = ; /* tshut mode 0:CRU 1:GPIO */
    rockchip,hw-tshut-polarity = ; /* tshut polarity 0:LOW 1:HIGH */
    rockchip,hw-tshut-temp = ;
    status = "okay";
};
```

For detailed information on thermal control, refer to the relevant documentation in the `<SDK>/docs/en/Common/THERMAL` directory.

9.2.6 DDR Development Guide

During the DDR development process, you may encounter various issues. Below are some key development resources and steps to help you efficiently resolve problems:

- Frequently Asked Questions: Understand the typical issues that may arise in DDR development and their solutions.
- Troubleshooting Methods: Master the systematic problem diagnosis process to quickly locate the root cause of the problem.
- Particle Verification Process: Follow detailed verification steps to ensure the performance and compatibility of DDR particles.
- PCB Design Considerations: Pay attention to the key elements of PCB design to avoid potential layout issues.
- Eye Diagram Analysis Tools: Learn how to use eye diagram tools for signal integrity analysis to optimize DDR performance.
- Bandwidth Statistics Tools: Utilize bandwidth statistics tools to assess the data transmission capabilities

of DDR.

For more detailed information and guidance, please refer to the documents under the path `<SDK>/docs/en/Common/DDR`. These documents will provide you with comprehensive development instructions and practical guides.

9.2.7 SD Card Configuration

For specific instructions on SD card configuration, please refer to the relevant documents in the directory `<SDK>/docs/en/Common/MMC`.

For some chips, such as RK3326/RK3399PRO, the UART for debugging and the SD card are multiplexed by default. The default configuration enables debugging. If you want to use the SD card, you need the following configuration:

```
&fiq_debugger {
    status = "disabled";
    pinctrl-0 = <&uart2a_xfer>;
};
&sdmmc {
    ...
    sd-uhs-sdr104;
    status = "okay";
};
```

9.3 Recovery Development

9.3.1 Introduction

The development of the Recovery mechanism, akin to the Recovery feature in Android, primarily serves to erase user data and facilitate system upgrades. Currently, it mainly supports upgrades in Recovery mode, upgrades in A/B partition mode, and factory reset.

In Linux, Recovery mode involves an additional Recovery partition on the device, composed of the kernel, resources, and ramdisk, mainly used for upgrade operations. The u-boot will determine whether to boot into the Normal system or the Recovery system based on the fields stored in the misc partition. Due to the system's independence, Recovery mode ensures the integrity of the upgrade process, meaning that if the upgrade is interrupted, such as by an abnormal power loss, the upgrade can still continue to be executed.

9.3.2 Debugging

Common debugging methods involve enabling debug mode.

Create a hidden file named `.rkdebug` in the directory `buildroot/output/rockchip_xxx_recovery/target`:

```
touch .rkdebug
```

Logs for upgrades in Recovery mode are printed out through the serial port. Another method is to inspect the file `userdata/recovery/Log`.

9.3.3 Recovery Feature Summary

In summary, all Recovery features are summarized in the following table:

Feature	Description	Involved Components
Support for passing commands to recovery upgrade in normal mode	Enter recovery mode through shell commands in normal mode to complete partition upgrades	update/updateEngine
Direct system upgrade in normal mode, Linux A/B partition upgrade	Directly complete upgrades in normal mode, requiring the enablement of A/B partitions, Linux A/B, which means preparing two independent system firmwares stored separately on flash, the system can boot from one slot, if the current slot fails to boot, it can boot from the other slot, directly upgrade the system in Normal mode without entering the system upgrade mode, just restart the system to enter the upgraded system	updateEngine
bootcontrol	The bootcontrol program can be set in two boot modes, successful_boot and reset retry, used to set which partition is bootable	updateEngine
Upgrade recovery partition in normal mode	To prevent issues with other partitions not being able to upgrade normally due to power loss during the upgrade of recovery.img, this partition upgrade is placed under the normal system upgrade, that is, during the upgrade, it will first detect whether the update.img upgrade package includes a packaged recovery.img, if so, then upgrade the recovery partition, and then enter Recovery mode to upgrade other partition firmwares	update/updateEngine
Print log information	The upgrade log can be output through the serial port or stored under userdata, if it is a device with a screen, you can view the log print through the UI interface	Recovery
Command to obtain version information	Obtain the current upgrade program version information	update/updateEngine
Obtain data from the misc partition	Obtain data from the misc partition, successful upgrade can verify successful data acquisition	Recovery
Parse commands within the misc partition	Parse data within the misc partition, successful upgrade can verify successful data parsing	Recovery
Reboot Recovery command parsing	The upgrade program supports rebooting	Recovery
Provide command usage instructions	Obtain detailed instructions for the current command usage	update/updateEngine
Full package upgrade package production	Produce update.img upgrade packages	update/updateEngine
Differential package upgrade package production	Differential upgrade package production	update/updateEngine
A/B package	Produce A/B partition upgrade packages	update/updateEngine
Upgrade via USB	Store the upgrade package in a USB flash drive, and perform partition upgrades through the upgrade program	Recovery

Feature	Description	Involved Components
Upgrade via SD card	Create a bootable disk upgrade using SDDiskTool, can normally read firmware from SD, write to Flash, and start normally (the capacity of the SD card must not exceed 32G, otherwise formatting will fail), if it is in A/B mode, you need to export RK_UPDATE_SDCARD_ENABLE_FOR_AB=true, and then compile	Recovery
Unpack (support for update.img, A/B, differential)	During the upgrade process, the img upgrade file will be unpacked and analyzed	update/updateEngine
Verify upgrade data	After a successful upgrade, the partition will be verified	rkupdate/updateEngine
Block device	Upgrade of storage devices such as EMMC and other block devices	rkupdate/updateEngine
MTD device	MTD is a device for managing flash storage, such as NAND flash and NOR flash	rkupdate/updateEngine
Partition verification after upgrade completion	After the upgrade is completed, the partition needs to be verified	rkupdate/updateEngine
Factory reset	Read and write configuration files are stored in the userdata partition, the factory firmware will default some configuration parameters, users will generate or modify configuration files after using for a period of time, sometimes users need to clear this data, we need to restore to factory settings	Recovery
Implement a simple UI display	UI display	Recovery
Compatible with projects without UI	Compatible with projects without screens	Recovery
UI prints upgrade log	UI display	Recovery
Dual screen display	Can display on dual screens	Recovery
Screen rotation	Implement UI screen rotation	Recovery
Support for serial port printing to Console	Allow log output to the console	Recovery
Log storage in userdata/recovery/log (enable this feature through commands)	Store logs in the userdata partition, which can be viewed in normal mode after a successful upgrade	Recovery
Abnormal power off	Ensure that the upgrade can continue after restarting in case of unexpected power off during the upgrade process	Recovery
SD card abnormally removed	Directly remove the SD card during the upgrade process started via SD card	Recovery
USB flash drive abnormally removed	Abnormally remove the USB flash drive during the upgrade process via USB flash drive	Recovery

9.4 Buildroot Development

Buildroot is a tool that facilitates the construction of a complete Linux system for embedded systems through cross-compilation, offering a simple and automated process.

On the native Buildroot, Rockchip has integrated the BSP configuration for related chips, configurations for hardware module acceleration features, and in-depth customization development of third-party packages, making it convenient for customers to deeply customize and develop their products.

For specific Buildroot development documentation, refer to:

`<SDK>/docs/en/Linux/System/Rockchip_Developer_Guide_Buildroot_EN.pdf`

9.5 Debian Development

Rockchip offers support for Debian 10/11 based on the X11 display architecture and is based on the Linaro version. This support also includes graphics and video acceleration capabilities. The relevant packages include libmali, xserver, and gstreamer rockchip, among others. These packages can be compiled by setting up an environment in Docker, and the generated deb packages are stored in the `<SDK>/debian/packages/*` directory.

For detailed steps on how to use Docker to compile deb packages, please refer to the document:

`<SDK>/docs/en/Linux/Docker/Rockchip_Developer_Guide_Debian_Docker_EN.pdf`

Debian Development Documentation Reference

`<SDK>/docs/en/Linux/System/Rockchip_Developer_Guide_Debian_EN.pdf`

9.6 Yocto Development

For more information, please refer to: http://opensource.rock-chips.com/wiki_Yocto

9.7 Audio Development

9.7.1 Kernel Audio Driver Development

The kernel driver utilizes `soc/rockchip/rockchip_multicodecs.c` to replace `soc/generic/simple-card.c`, as the simple card in the kernel driver is merely for simplicity and cannot meet the complex requirements of some products.

The rockchip_multicodecs driver differs from the simple-card mainly in the following aspects:

- Primarily, it is compatible with the support for one sound card with one or multiple codecs.
- It supports ADC for headphones and other detections, whereas the simple-card commonly uses GPIO.
- The I2S TDM controller has separate TX/RX, and multicodecs support them independently, which would affect third-party implementations if added to the simple-card.

9.7.2 Audio Pulseaudio Path Adaptation

The default audio system utilizes Pulseaudio, and typically, it is only necessary to configure `/etc/pulse/default.pa`.

For example, in RK3588, adaptation for ES8388 and RK809 codecs is performed as follows:

```
+set-default-source alsa_input.platform-es8388-sound.HiFi__hw_rockchipes8388__source
+set-default-sink alsa_output.platform-es8388-sound.HiFi__hw_rockchipes8388__sink
+set-default-source alsa_input.platform-rk809-sound.HiFi__hw_rockchiprk809__source
+set-default-sink alsa_output.platform-rk809-sound.HiFi__hw_rockchiprk809__sink
```

To add support for additional codecs, obtain relevant information through the following commands:

```
pactl list sinks short
pactl list sources short
```

For detailed reference, consult the Debian official PulseAudio and `<SDK>/docs/en/Common/AUDIO/Rockchip_Developer_Guide_PulseAudio_EN.pdf`.

9.8 Multimedia Development

Development of multimedia on the Rockchip platform through GStreamer/Rockit.

The general pathway is as follows:

```
vpu_service  -->  mpp --> GStreamer/Rockit --> app

vpu_service: Driver
mpp: The video codec middleware for the Rockchip platform, refer to the mpp documentation
for details
GStreamer/Rockit: Component interfacing with the app
```

The complete solution currently provided by the Rockchip Linux general SDK is based on GStreamer. The advantage of using GStreamer is the convenience of building players, encoders, and other applications based on the pipeline approach. For customized development based on Rockit, refer to the relevant Rockit release documentation.

For specific materials, refer to:

```
<SDK>/docs/en/Linux/Multimedia
├── Rockchip_Developer_Guide_Linux_RKADK_EN.pdf
├── Rockchip_User_Guide_Linux_Gstreamer_EN.pdf
└── Rockchip_User_Guide_Linux_Rockit_EN.pdf

<SDK>/docs/en/Common/MPP/Rockchip_Developer_Guide_MPP_PDF
```

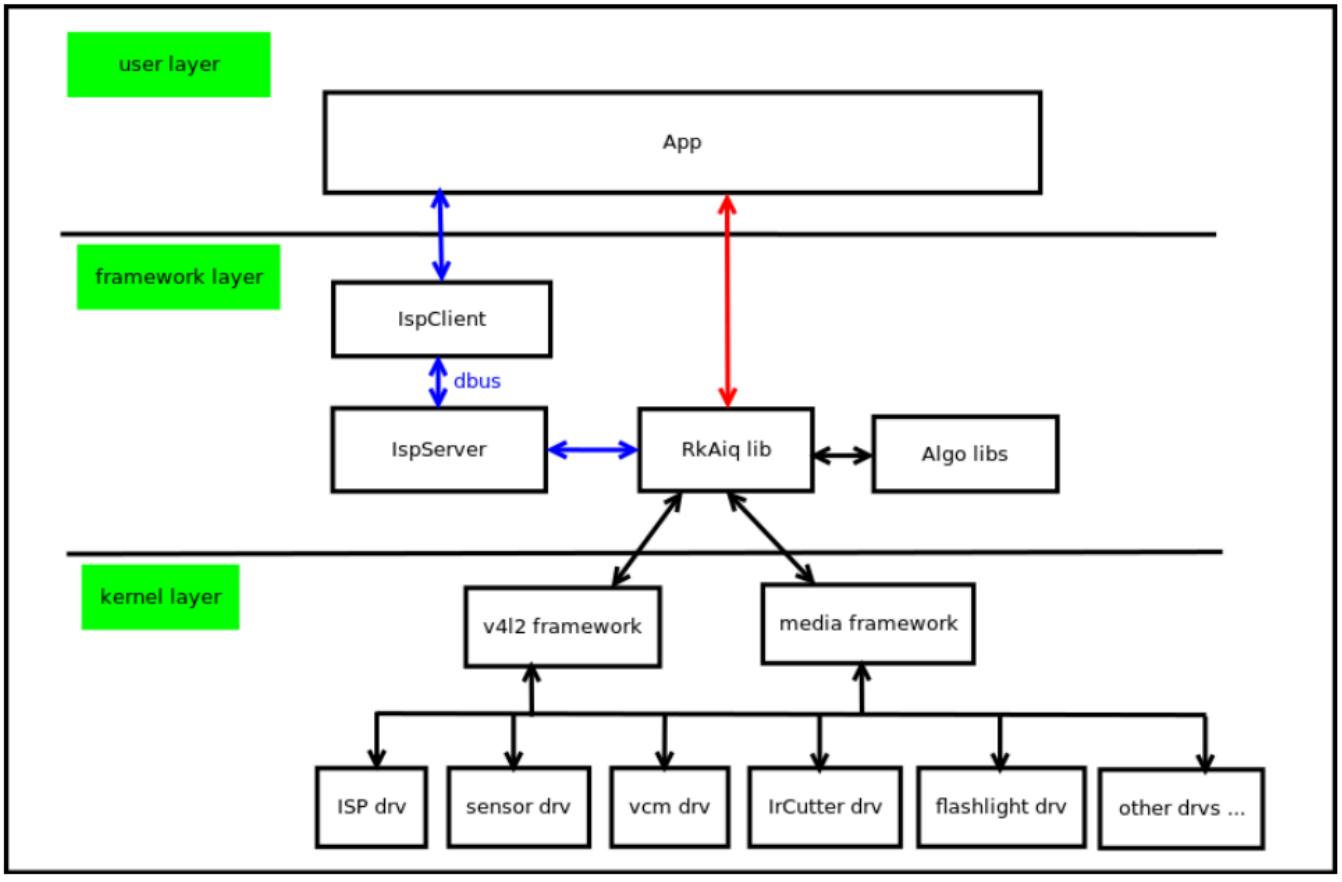
9.9 Camera Development

9.9.1 Image Processing Module (ISP)

The ISP encompasses a series of image processing algorithm modules, primarily including dark current correction, hot pixel correction, 3A, HDR, lens shading correction, lens distortion correction, 3DLUT, denoising (including RAW domain denoising, multi-frame noise reduction, color denoising, etc.), and sharpening. It includes both hardware algorithm implementation and software logic control parts, with RKAIQ being the implementation of the software logic control part.

The RkAIQ software module mainly implements the functionality of obtaining image statistics from the ISP driver, combining IQ Tuning parameters, and using a series of algorithms to calculate new ISP, Sensor, and other hardware parameters. This process is iterated continuously until the optimal image effect is achieved.

On the Rockchip platform, the RKISP or RKAIQ (Rockchip Auto Image Quality) module is mainly used for ISP function development. The ISP software framework is referenced as follows:



ISP has different versions, where:
ISP1.X is primarily applicable to RK3399/RK3288/PX30/RK3326/RK1808, etc.
ISP21 is primarily applicable to RK3566_RK3568, etc.
ISP30 is primarily applicable to RK3588, etc.
ISP32-lite is primarily applicable to RK3562, etc.
ISP39 is primarily applicable to RK3576, etc.

ISP development documentation includes ISP development documentation, VI driver development documentation, IQ Tool development documentation, debugging documentation, and color debugging documentation. The specific documents are as follows:

```
docs/en/Common/ISP/
├── ISP1.X
├── ISP21
├── ISP30
├── ISP32-lite
├── ISP39
└── The-Latest-Camera-Documents-Link.txt
```

9.9.2 USB CAMERA

- Supports standard UVC Camera functionality, up to 4k preview (Rockchip RK356X)
- Supports stable transmission with USB composite devices
- Compatible with smart TVs or PCs for various terminal device previews

Enable this configuration in Buildroot

```
+BR2_PACKAGE_ROCKIT_UVC=y
+BR2_PACKAGE_ROCKCHIP_UVC_APP=y
+BR2_PACKAGE_ROCKIT=y
```

You can refer to the following methods for testing:

```
Board-side configuration enabled:
root@rk3588-buildroot:/# /etc/init.d/S50usbdevice.sh stop
root@rk3588-buildroot:/# usb_config.sh
root@rk3588-buildroot:/# rockit_uvc
```

On the PC side, use applications like AMCap or PotPlayer for USB camera, and open it to see the preview. Switch formats or resolutions according to the application used on the upper machine.

Rockchip RK356X supports UVC Camera functionality and requires the following patch to be applied to the kernel:

```
diff --git a/arch/arm64/boot/dts/rockchip/rk3568-evb.dtsi
b/arch/arm64/boot/dts/rockchip/rk3568-evb.dtsi
index 18d6115341cd..268f5d2b1c05 100644
--- a/arch/arm64/boot/dts/rockchip/rk3568-evb.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rk3568-evb.dtsi
@@ -1778,6 +1778,8 @@
 &usbdrd_dwc3 {
        dr_mode = "otg";
+       snps,tx-fifo-resize;
+       snps,dis-u1u2-quirk;
        extcon = <&usb2phy0>;
        status = "okay";
 };
```

If the memory exceeds 4G, it needs to be limited to 4G because the USB controller can only access memory within 4G.

```
--- a/arch/arm64/boot/dts/rockchip/rk3568-linux.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rk3568-linux.dtsi
@@ -13,7 +13,7 @@ aliases {
        };

        chosen: chosen {
-               bootargs = "earlycon=uart8250,mmio32,0xfe660000 console=ttyFIQ0
root=PARTUUID=614e0000-0000 rw rootwait";
+               bootargs = "earlycon=uart8250,mmio32,0xfe660000 console=ttyFIQ0
root=PARTUUID=614e0000-0000 mem=4g rw rootwait";
        };
```

9.10 Graphics Development

Rockchip Linux graphics system development involves utilizing DRM (Direct Rendering Manager) and DMA-BUF (Direct Memory Access - Buffers) technologies for ARM Linux graphics development on the Rockchip Linux platform. The advantage of this platform lies in its universal architecture, which makes custom development based on this architecture relatively easy and allows for the full utilization of existing components. Currently, many basic open-source projects have begun to adopt the Rockchip platform as an adaptation platform for the ARM end.

However, there are also some challenges in this field. Many developers may not be familiar enough with the concepts and applications of DRM and DMA-BUF, which requires them to invest a certain amount of time and energy to learn and master. To gain an in-depth understanding of Rockchip graphics system development, you can refer to the Rockchip wiki and the following related technical documents.

```
<SDK>/docs/en/Linux/Graphics/
├── Rockchip_Developer_Guide_Buildroot_Weston_EN.pdf
├── Rockchip_Developer_Guide_Linux_Graphics_EN.pdf
└── Rockchip_Developer_Guide_Linux_LVGL_EN.pdf

<SDK>/docs/en/Common/DISPLAY/
├── DP
├── HDMI
├── MIPI
├── RK628
├── Rockchip_BT656_TX_AND_BT1120_TX_Developer_Guide_EN.pdf
├── Rockchip_DRM_Panel_Porting_Guide_V1.6_20190228.pdf
├── Rockchip_Develop_Guide_DRM_Direct_Show_EN.pdf
├── Rockchip_Developer_Guide_Baseparameter_Format_Define_And_Use_EN.pdf
├── Rockchip_Developer_Guide_DRM_Display_Driver_EN.pdf
├── Rockchip_Developer_Guide_DisplayPort_EN.pdf
├── Rockchip_Developer_Guide_HDMI_EN.pdf
├── Rockchip_Developer_Guide_RGB_MCU_EN.pdf
├── Rockchip_Developer_Guide_RK628_For_All_Porting_EN.pdf
├── Rockchip_RK3588_Developer_Guide_MIPI_DSI2_EN.pdf
```

```
└── Rockchip_RK3588_Developer_Guide_Vsync_Adjust_EN.pdf
```

9.10.1 Boot Screen Development

9.10.1.1 Boot Logo Development

- Enable U-Boot Logo

  The logo is controlled through the corresponding display interface's route_xxx node in the Linux kernel dts (the U-Boot display module and the Linux kernel share the same dtb), where xxx can be dsi, edp, hdmi, lvds, etc. For specifics, please search for the keyword "route" in the dts.

  Taking MIPI DSI0 as an example, find the route_dsi0 node in the corresponding board-level dts file and set the status to "okay":

  ```
  &route_dsi0 {
          status = "okay";
          connect = <&vp3_out_dsi0>;
  };
  ```

  The `connect` property is detailed in the subsequent section **DTS Configuration**.

- Configure U-Boot Logo for Full-Screen Display

  The logo is displayed centered by default. If a full-screen display is required, modify the route node for the corresponding interface.

  For example, to display the logo on DSI0 in full-screen mode, modify the logo,mode attribute of route_dsi0 to "fullscreen".

  ```
  --- a/arch/arm64/boot/dts/rockchip/rk3588s.dtsi
  +++ b/arch/arm64/boot/dts/rockchip/rk3588s.dtsi
  @@ -1065,7 +1065,7 @@
                                  status = "disabled";
                                  logo,uboot = "logo.bmp";
                                  logo,kernel = "logo_kernel.bmp";
  -                               logo,mode = "center";
  +                               logo,mode = "fullscreen";
                                  charge_logo,mode = "center";
                                  connect = <&vp3_out_dsi0>;
                          };
  ```

- Logo Image Requirements

  The U-Boot logo and the Linux Kernel logo have the same resolution, and the resolution must be an even number;

  It only supports bmp images with 8bit, 16bit, 24bit, and 32bit;

  Both the U-Boot logo and the Linux Kernel logo must be enabled simultaneously, meaning logo.bmp and logo_kernel.bmp must be provided together, not just one.

- Boot Log Confirmation

  If the logo function is properly enabled, you will see logs similar to the following during the U-Boot stage:

  ```
  Rockchip UBOOT DRM driver version: v1.0.1
  vp0 have layer nr:2[0 2 ], primary plane: 2
  ```

```
vp1 have layer nr:2[1 3 ], primary plane: 3
vp2 have layer nr:2[6 8 ], primary plane: 8
vp3 have layer nr:2[7 9 ], primary plane: 9
Using display timing dts
dsi@fde20000:  detailed mode clock 132000 kHz, flags[a]
    H: 1080 1095 1099 1129
    V: 1920 1935 1937 1952
bus_format: 100e
VOP update mode to: 1080x1920p0, type: MIPI0 for VP3
VOP VP3 enable Esmart3[654x270->654x270@213x825] fmt[2] addr[0xedf04000]
final DSI-Link bandwidth: 880000 Kbps x 4
......
hdmi_select_link_config use tmds mode
mode:1920x1080 bus_format:0x100a
hdmi@fde80000:  detailed mode clock 148500 kHz, flags[5]
    H: 1920 2008 2052 2200
    V: 1080 1084 1089 1125
bus_format: 100a
VOP update mode to: 1920x1080p0, type: HDMI0 for VP0
......
VOP VP0 enable Esmart0[654x270->654x270@633x405] fmt[2] addr[0xedf04000]
......
CLK: (uboot. arm: enter 1200000 KHz, init 1200000 KHz, kernel 0N/A)
```

It can be observed that the logo display has been initiated on both VP0 and VP3.

The display resolution for VP0 is 1920 x 1080, with the display interface being HDMI.

The display resolution for VP3 is 1080 x 1920, with the display interface being MIPI DSI.

The displayed logo size is 654 x 270.

9.10.1.2 Boot Animation Development

Utilize the SDK's make config to configure RK_ROOTFS_BOOTANIM.
Relevant settings include: RK_ROOTFS_BOOTANIM_TIMEOUT for animation timeout, default is 3 seconds.
`device/rockchip/common/overlays/rootfs/bootanim/etc/bootanim.d/gst-video.sh` serves as an example script for playing animations, which by default plays videos located in /etc/bootanim.d (if no video is available, it plays snowflakes) to the first screen in full screen.

For special requirements such as multi-screen, you can also modify the gst-video.sh script to achieve this. The process involves disabling Weston display at boot time, then calling this script to display the animation, waiting for 3 seconds before killing the script and restoring Weston display.

```
| Symbol: RK_ROOTFS_BOOTANIM [=y]
| Type  : bool
| Prompt: bootanim (Boot-time animation)
|   Location:
|     -> Rootfs
|       -> rootfs (RK_ROOTFS [=y])
|         -> Post rootfs installs
|           -> rootfs overlay (RK_ROOTFS_OVERLAY [=y])
```

- FAQ

If issues arise during the boot animation process, you can add the following line to the video playback script: echo 0xff > /sys/module/drm/parameters/debug
This enables the debug log of the display driver (prints all relevant upper-level call information). After reproducing the issue, package and send the /var/log directory.

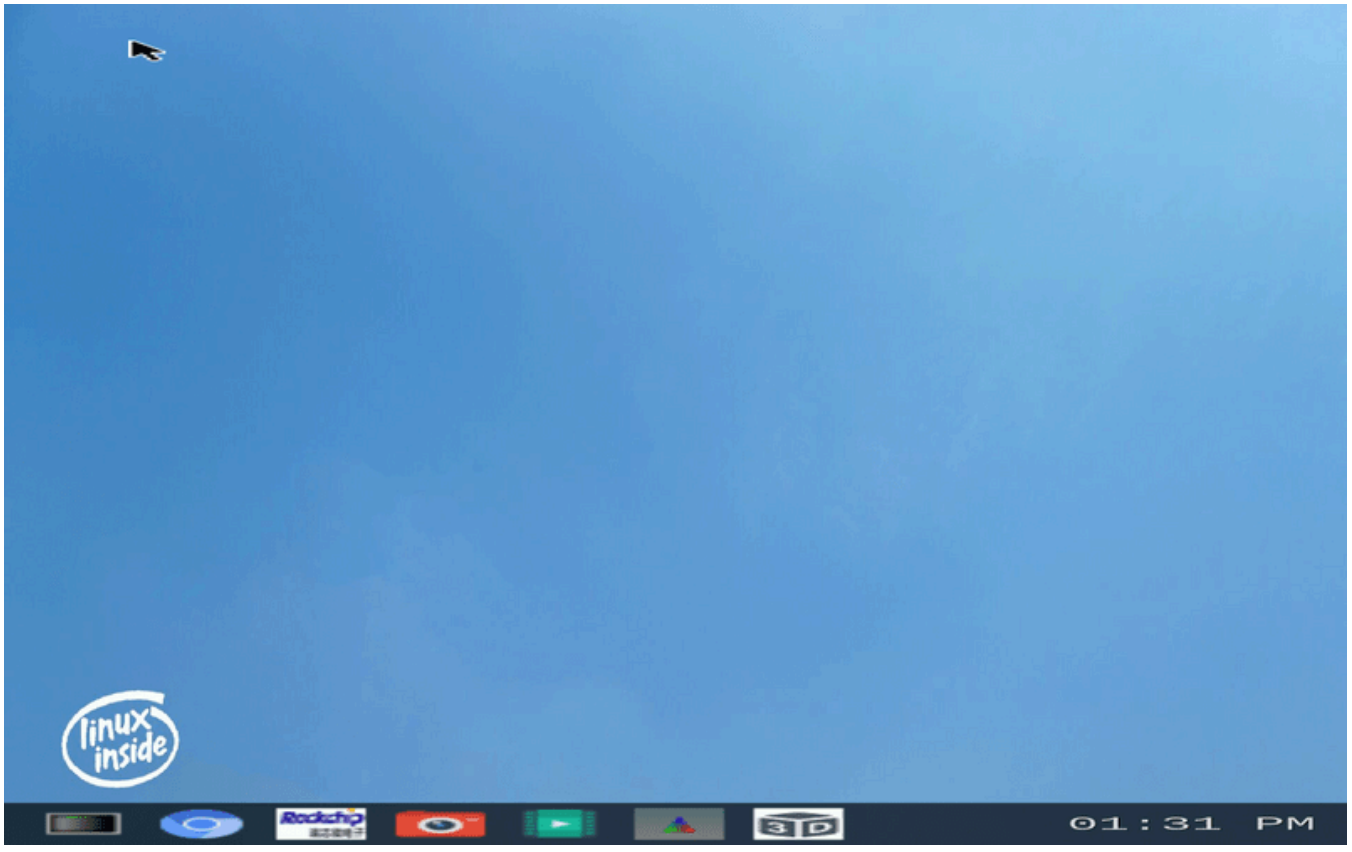
The general logic of the boot animation mechanism is as follows:

1. bootanim creates a marker file to block Weston from sending display.
2. bootanim plays the animation.
3. Weston starts, draws, but does not send display.
4. When bootanim exits, it pauses the animation, notifies Weston to redraw and send display.
5. bootanim ends the animation.

Some issues may be related to the boot logo. You can add `echo 3 > /sys/class/graphics/fb0/blank` at the beginning of the animation playback script to disable the logo display.

9.10.2 Desktop Application Development

The SDK uses desktop applications based on the X11 protocol such as XFCE, LXDE, GNOME, etc., by default in Debian, while Buildroot/Yocto uses Weston DRM based on the Wayland protocol as the display backend by default. As shown in the figure below:



These Weston applications provide basic functionality configurations such as status bars, backgrounds, Chromium browser, Terminal terminal, Launchers configuration, camera preview, multi-channel video, GPU, mouse, and other demos. For more demos, they can be added through /etc/xdg/weston/weston.ini.d/* configuration. For more Weston parameter configurations and usage, refer to Weston Development.

> Note: Third-party UI frameworks may involve infringement risks without commercial authorization. If using UI frameworks such as Buildroot QT/Enlightenment/Minigui on the Rockchip platform, third-party authorization and technical support are required, as Rockchip officially does not provide related technical support and maintenance.

The SDK provides three customized UI options: Weston+Wayland, LVGL UI, EFL, Flutter, etc., for desktop development.

9.10.2.1 Weston Desktop Development

Applications related to the Weston client used on Wayland:

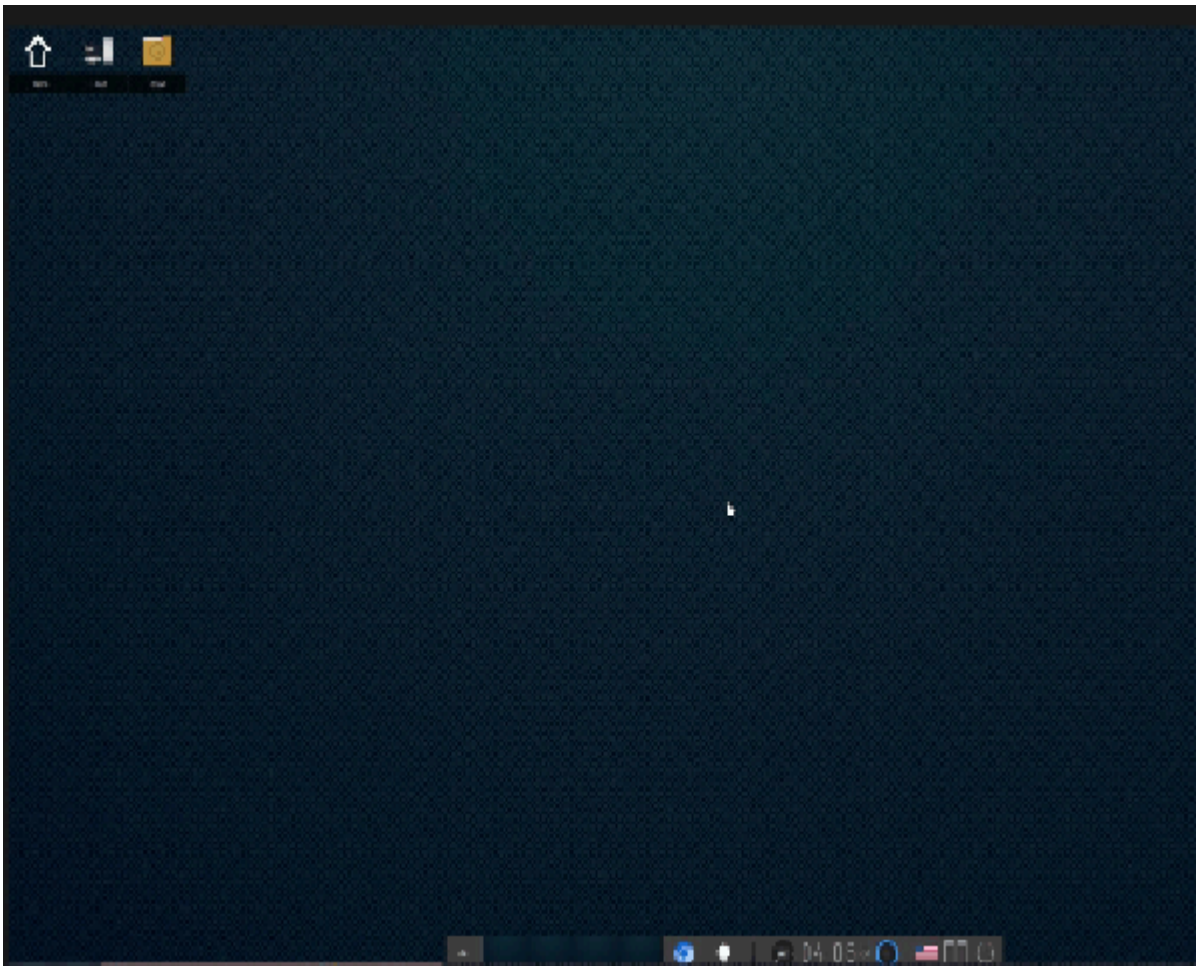
```
/usr/bin/#
├── weston-calibrator
├── weston-clickdot
├── weston-cliptest
├── weston-confine
├── weston-content_protection
├── weston-debug
├── weston-dnd
├── weston-editor
├── weston-eventdemo
├── weston-flower
├── weston-fullscreen
├── weston-image
├── weston-multi-resource
├── weston-presentation-shm
├── weston-resizor
├── weston-scaler
├── weston-screenshooter
├── weston-simple-damage
├── weston-simple-dmabuf-egl
├── weston-simple-dmabuf-v4l
├── weston-simple-egl
├── weston-simple-shm
├── weston-simple-touch
├── weston-smoke
├── weston-stacking
├── weston-subsurfaces
├── weston-tablet
├── weston-terminal
├── weston-touch-calibrator
├── weston-transformed
```

9.10.2.2 Enlightenment Desktop Development

To enable the necessary configurations, simply set them as follows:

```
BR2_PACKAGE_EFL=y
BR2_PACKAGE_LUAJIT=y
BR2_PACKAGE_EFL_GSTREAMER1=y
BR2_PACKAGE_ENLIGHTENMENT=y
BR2_PACKAGE_EFL_WAYLAND=y
```

The actual running desktop is shown in the figure below:



If your PC prompts this error:

```
2024-03-06T15:20:29 ERR<1307441>:emile ../src/lib/emile/emile_image.c:705
_emile_image_jpeg_error_exit_cb() Wrong JPEG library version: library is 80, caller expects
90

Solution:
sudo apt remove libjpeg8-dev
It appears there is a conflict with your PC's libjpeg.so*. You can try deleting this library
or the corresponding package libjpeg-turbo8.

$ whereis libjpeg.so
Find libjpeg.so and create a soft link to libjpeg.so.9
/usr/lib/x86_64-linux-gnu$ ln -s libjpeg.so.9 libjpeg.so
```

9.10.2.3 LVGL Desktop Development

Common configurations are stored in the `<SDK>/buildroot/configs/rockchip/lvgl/` directory.

```
<SDK>/buildroot/configs/rockchip/lvgl/
├── lvgl_drm.config     # Soft link to the V8 version configuration, compatible with old
file paths
├── lvgl_rkadk.config   # Soft link to the V8 version configuration, compatible with old
file paths
├── lvgl_sdl.config     # Soft link to the V8 version configuration, compatible with old
file paths
├── music.config        # LVGL official music player example
├── rk_demo.config      # Rockchip appliance display control example
├── v8                  # V8 version configuration files
│   ├── base.config     # Base configuration file for LVGL features, functions, etc.
│   ├── lvgl_drm.config # DRM configuration file, using DRM for display output
│   ├── lvgl_rkadk.config   # RKADK configuration file, using RKADK for display output
│   └── lvgl_sdl.config # SDL configuration file, using SDL for display output
├── v9                  # V9 version configuration files
│   ├── base.config
│   ├── lvgl_drm.config
│   ├── lvgl_rkadk.config
│   └── lvgl_sdl.config
└── widgets.config      # LVGL official widget example
```

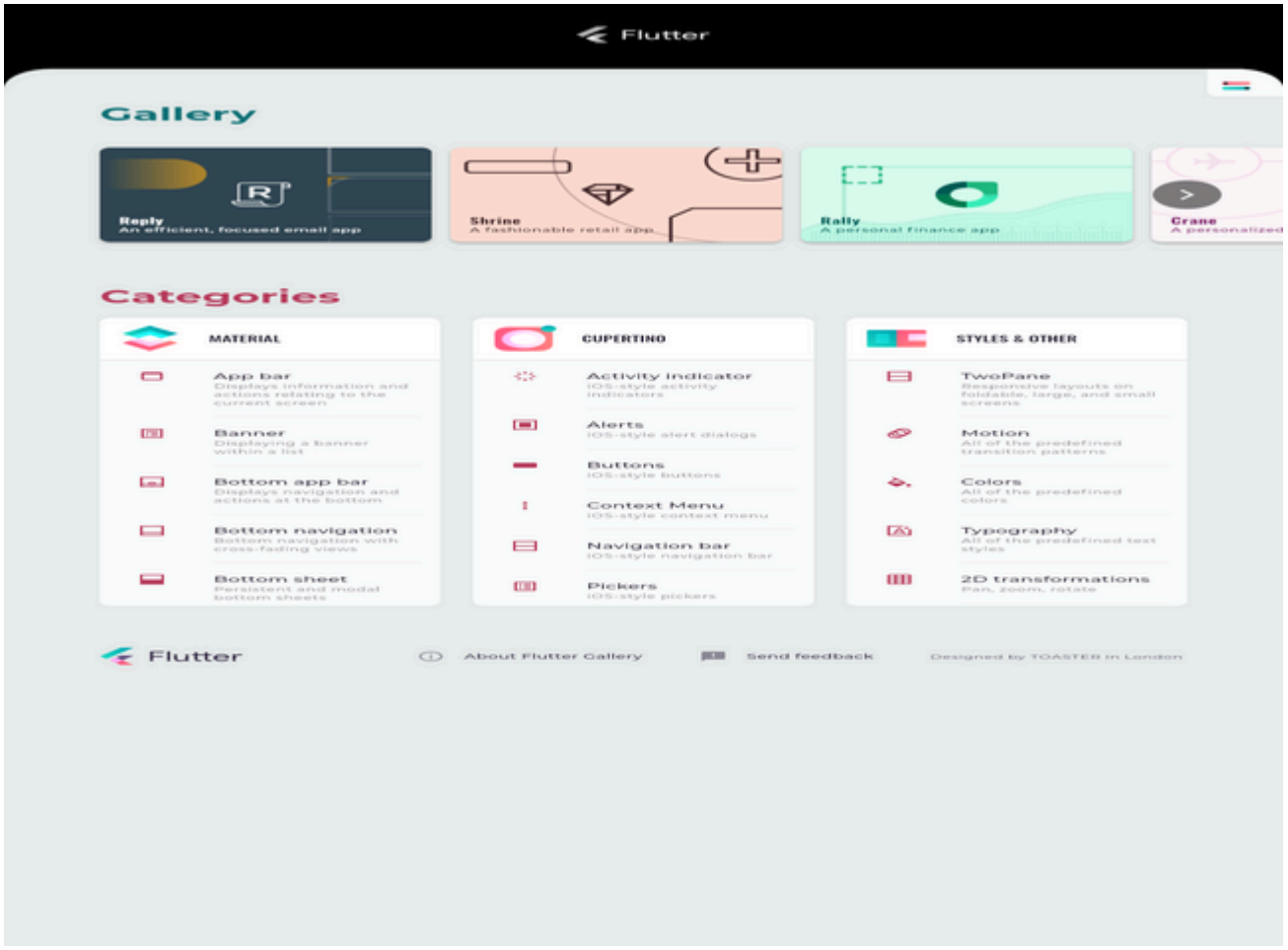


9.10.2.4 Flutter Linux Desktop Development

Ensure that the relevant configurations are enabled, such as:

```
BR2_PACKAGE_FLUTTER_EMBEDDED_LINUX=y
BR2_PACKAGE_FLUTTER_GALLERY=y
```

```
## Chapter-11 flutter-client -f -b /usr/share/flutter/gallery/release/
arm_release_ver: g13p0-01eac0, rk_so_ver: 10
[11:38:01.054] seeing the first app
```



9.11 Secure Development

Security solutions include hardware security (AMPAK, DICE), boot security (Secure Boot, UBM), firmware security (CFI, OP-TEE, CA/TA), secure storage (OPTEE REE, OP-TEE Secure, OP-TEE RPMB), secure verification (Dm-Verity), integrity and anti-tampering (Anti-Rollback, Anti-Copy, dual-key signature verification), hardware isolation (FIREWALL), and other functional developments. For details, refer to the documents in the `<SDK>/docs/en/Linux/Security` directory.

9.11.1 Current Security Features Supported by Each Chip

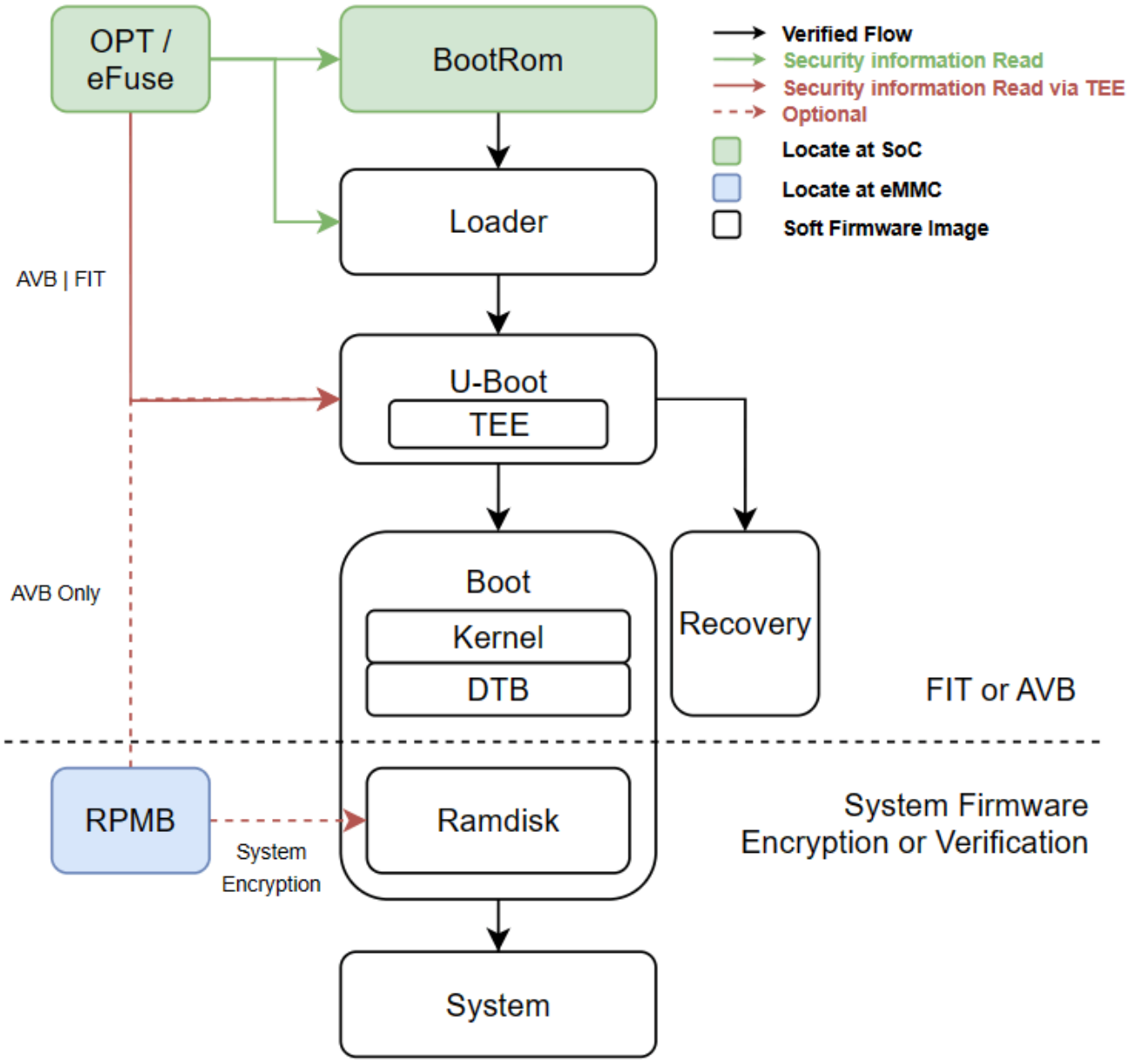
Feature	RK3588	RK3576	RK3568/RK3566	RK3562	Remarks
Secure Boot	√	√	√	√	Supports FIT and AVB schemes, full chain verification. Requires SecureBoot + AB system (mandatory)
CFI	-	-	-	-	Bootrom Control Flow Integrity, prevents hijacking control flow attacks
KEYLAD	-	-	-	-	Hardware Protect Key, CPU cannot read the key
Anti-Rollback	-	-	-	-	Anti-rollback protection (including Bootrom, TA, Uboot, etc., at all levels)
Anti-Copy	√	√	√	√	Anti-cloning protection (secure OTP writes private data, software verification, independent of SecureBoot)
Dual-key Signature Verification	√	√	√	√	BL_KEY0 verifies BL_KEY1, BL_KEY1 verifies the firmware
DICE	√	√	√	√	Device Identifier Composition Engine, only implemented in bootrom
UBM	-	-	-	-	Bootrom Unique Boot Mode OTP configures unique boot storage
FIREWALL	-	√	-	-	Configures secure and non-secure isolation, such as isolating OP-TEE secure memory, non-secure cannot access
OP-TEE	√	√	√	√	Version 3.13.0 (tls implemented in teeos, customers share related documents)
OP-TEE REE Secure Storage	√	√	√	√	-
OP-TEE RPMB Secure Storage	√	√	√	√	-
OP-TEE Secure Partition Secure Storage	√	√	√	√	Suitable for flash without RPMB, such as nand/nor flash OP-TEE secure memory, non-secure cannot access

Feature	RK3588	RK3576	RK3568/RK3566	RK3562	Remarks
TA/CA	√	√	√	√	Version 3.13.0 (tls implemented in teeos, customers share related documents)
TA Encryption	√	√	√	√	-
TA Anti-Rollback	√	√	√	√	-
KEYBOX	√	√	√	√	System encryption: ramdisk reads the secret key (maintain current)
DM-VERIFY	√	√	√	√	-
DM-CRYPT	√	√	√	√	-
Partition Encryption	√	√	√	√	Only supports system partition encryption, Loader/Uboot/Kernel: TBD
SELinux	-	-	-	-	Kernel support, SDK not deployed

PS: A checkmark for RK3588 indicates that the chip supports it; black highlighting indicates that the current SDK has been supported.

9.11.2 Secure Boot Initialization

Secure Boot is a mechanism designed to protect systems from malicious software and unauthorized modifications, ensuring that only software signed by the manufacturer or developer can run at startup. Below are the general steps of the Secure Boot process.



By default, Secure Boot is not enabled. If you need to verify, follow these steps:

- SDK to enable security configuration

```
$ make menuconfig
Enable  Security  --->  [*] security feature
   --- Security feature (secureboot, encryption, verity, etc.)
        security check method (base|system-encryption|system-verity) (system-encryption)  --
->
```

There are three security mechanisms by default: base, system-encryption, and system-verity. Base is the basic version without encryption, system-encryption provides encryption functionality, and system-verity provides verification functionality.

- SDK to save security configuration

```
$ make savedefconfig

For example, using RK3576 as a reference:

--- a/.chips/rk3576/rockchip_rk3576_evb1_v10_defconfig
+++ b/.chips/rk3576/rockchip_rk3576_evb1_v10_defconfig
@@ -1,4 +1,5 @@
 RK_ROOTFS_PREBUILT_TOOLS=y
-RK_UBOOT_SPL=y
 RK_KERNEL_DTS_NAME="rk3576-evb1-v10-linux"
+RK_SECURITY=y
+RK_SECURITY_CHECK_SYSTEM_ENCRYPTION=y
```

9.11.3 Compiling Secureboot

- Directly run `./build.sh` and follow the compilation error messages to proceed step by step.

```
./build.sh
```

```
=========================================
system-encryption
ERROR: No root passwd(u-boot/keys/root_passwd) found in u-boot
       echo your root key for sudo to u-boot/keys/root_passwd
       some operations need super user permission when create encrypt image

Write the PC root password into `u-boot/keys/root_passwd`

For example, if the PC password is test0000:
echo -n "test0000" > u-boot/keys/root_passwd
```

```
./build.sh
```

```
=========================================
system-encryption
ERROR: No enc key(u-boot/keys/system_enc_key) found in u-boot
       Create it by ./build.sh security-createkeys or move your key to it
```

```
./build.sh security-createkeys
```

```
./build.sh
```

```

Security: No found config CONFIG_BLK_DEV_DM in
kernel/arch/arm64/configs/rockchip_linux_defconfig
make sure your config include this list
---------------------------------------
```

```
CONFIG_BLK_DEV_DM
CONFIG_DM_CRYPT
CONFIG_DM_VERITY
CONFIG_TEE
CONFIG_OPTEE

Add the relevant configurations to the kernel

kernel$ git diff
diff --git a/arch/arm64/configs/rockchip_linux_defconfig
b/arch/arm64/configs/rockchip_linux_defconfig
index d8757f713ec4..7beca18172e0 100644
--- a/arch/arm64/configs/rockchip_linux_defconfig
+++ b/arch/arm64/configs/rockchip_linux_defconfig
@@ -590,3 +590,9 @@ CONFIG_RCU_CPU_STALL_TIMEOUT=60
 CONFIG_FUNCTION_TRACER=y
 CONFIG_BLK_DEV_IO_TRACE=y
 CONFIG_LKDTM=y
+CONFIG_BLK_DEV_DM=y
+CONFIG_DM_CRYPT=y
+CONFIG_DM_VERITY=y
+CONFIG_TEE=y
+CONFIG_OPTEE=y
```

```
./build.sh
```

```
Security: No found config CONFIG_FIT_SIGNATURE in /home/wxt/linux-develop/rockchip-linux/u-
boot/.config
make sure your config include this list
---------------------------------------
CONFIG_FIT_SIGNATURE
CONFIG_SPL_FIT_SIGNATURE

Add the relevant configurations to u-boot:

u-boot$ git diff
diff --git a/configs/rk3576_defconfig b/configs/rk3576_defconfig
index e9b5dbf1210..35925f8d5df 100644
--- a/configs/rk3576_defconfig
+++ b/configs/rk3576_defconfig
@@ -23,6 +23,8 @@ CONFIG_DEBUG_UART=y
 CONFIG_FIT=y
 CONFIG_FIT_IMAGE_POST_PROCESS=y
 CONFIG_FIT_HW_CRYPTO=y
+CONFIG_FIT_SIGNATURE=y
+CONFIG_SPL_FIT_SIGNATURE=y
 CONFIG_SPL_LOAD_FIT=y
 CONFIG_SPL_FIT_IMAGE_POST_PROCESS=y
 CONFIG_SPL_FIT_HW_CRYPTO=y
```

```
./build.sh
```

```
CONFIG_DM_VERITY is enabled in kernel!
Please set "GROW_ALIGN: 1" in rockchip-linux/device/rockchip/.chip/parameter.txt:

device/rockchip$ git diff
diff --git a/.chips/rk3576/parameter.txt b/.chips/rk3576/parameter.txt
index a2c21a9d..dec0979d 100644
--- a/.chips/rk3576/parameter.txt
+++ b/.chips/rk3576/parameter.txt
@@ -8,7 +8,7 @@ MACHINE: 0xffffffff
 CHECK_MASK: 0x80
 PWR_HLD: 0,0,A,0,1
 TYPE: GPT
-GROW_ALIGN: 0
+GROW_ALIGN: 1
```

```
./build.sh
```

```
No found optee node in dts
Please add:
        optee: optee {
                compatible = "linaro,optee-tz";
                method = "smc";
                status = "okay";
        };
To kernel dts

--- a/arch/arm64/boot/dts/rockchip/rk3576-linux.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rk3576-linux.dtsi
@@ -22,6 +22,13 @@ fiq_debugger: fiq-debugger {
                status = "okay";
        };

+       firmware {
+                optee: optee {
+                        compatible = "linaro,optee-tz";
+                        method = "smc";
+                };
+        };
+
```

```
./build.sh
```

```
Security: No found BR2_ROOTFS_OVERLAY+="board/rockchip/common/security-system-overlay/" in
system config
```

```
Add the relevant configuration to buildroot:

buildroot$ git diff
diff --git a/configs/rockchip_rk3576_defconfig b/configs/rockchip_rk3576_defconfig
index 5b47b07c43..9dd11af1cc 100644
--- a/configs/rockchip_rk3576_defconfig
+++ b/configs/rockchip_rk3576_defconfig
@@ -23,3 +23,4 @@
 #include "npu2.config"
 #include "powermanager.config"
 #include "weston.config"
+BR2_ROOTFS_OVERLAY="board/rockchip/common/security-system-overlay"

Security: No found config BR2_PACKAGE_RECOVERY in rockchip-
linux/buildroot/output/rockchip_rk3576_ramboot/.config

Add the relevant configuration to buildroot:

diff --git a/configs/rockchip_rk3576_ramboot_defconfig
b/configs/rockchip_rk3576_ramboot_defconfig
index 46a27b3a9e..9b344004fe 100644
--- a/configs/rockchip_rk3576_ramboot_defconfig
+++ b/configs/rockchip_rk3576_ramboot_defconfig
@@ -1,6 +1,9 @@
 #include "chips/rk3576_aarch64.config"
 #include "base/kernel.config"
+#include "tee_aarch64_v2.config"
 BR2_TARGET_ROOTFS_CPIO=y
 BR2_TARGET_ROOTFS_CPIO_GZIP=y
 BR2_PACKAGE_LUKSMETA=y
+BR2_PACKAGE_RECOVERY=y
+BR2_PACKAGE_RECOVERY_UPDATEENGINEBIN=y
```

```
./build.sh
```

After running the `./build.sh` command, the generated update.img firmware in the rockdev directory can be used for system updates. Simply start the system normally, and as long as the vroot partition is properly mounted, the secureboot feature is enabled. You can check the mounting status of the vroot partition by executing the `df -h` command:

```
root@rk3576-buildroot:/# df -h
Filesystem         Size  Used Avail Use% Mounted on
/dev/mapper/vroot  924M  799M   62M  93% /
```

- FAQ

1. Environment Installation

```
If the compilation reports an error No module named Crypto.Signature, this is due to the
development computer not having the python cryptographic library installed. Execute the
following commands:
pip uninstall Crypto
pip uninstall pycrypto
pip install pycrypto

If the following error occurs: ModuleNotFoundError: No module named 'Cryptodome'
Please install the python package on the development host: pip3 install [--user]
pycryptodomex
```

2. When verifying, find a machine that has not been burned with rpmb. If there is an encryption key in the machine, use the encryption key in the machine first. If the machine's encryption key is different from the one we compiled, the system will fail to start.
3. The encryption key is placed in the rpmb area and can be rewritten repeatedly. If you encounter a situation where the key is different and the system cannot be loaded, you can open FORCE_KEY_WRITE=true in `buildroot/board/rockchip/common/security-ramdisk-overlay/init.in` to force an update of the key.

For more details, refer to:
Linux Secureboot Software Development Guide:
`<SDK>/docs/en/Linux/Security/Rockchip_Developer_Guide_Linux_Secure_Boot_EN.pdf`
U-Boot Secureboot Software Development Guide:
`<SDK>/docs/en/Common/SECURITY/Rockchip_Developer_Guide_Secure_Boot_for_UBoot_Next_Dev_EN.pdf`
TEE Development Guide:
`<SDK>/docs/en/Common/SECURITY/Rockchip_Developer_Guide_TEE_SDK_EN.pdf`

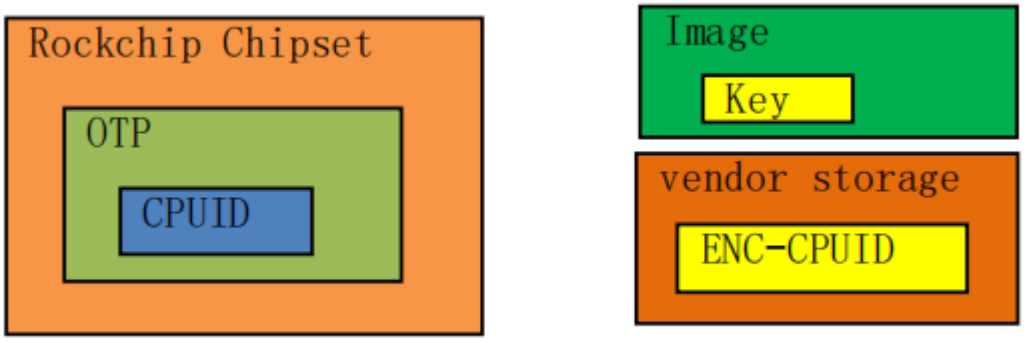
9.11.4 Rockchip Anti-Cloning Feature

Rockchip chips offer anti-cloning technology to protect customers' firmware, private data, and core code. This technology is primarily used to prevent unauthorized users from illegally copying and using customers' firmware and private data, thus avoiding commercial losses due to cloning.

Currently, there are three schemes provided for anti-cloning technology:

- Basic Scheme

During production, Rockchip chips burn a CPUID into the OTP, and each chip has a unique CPUID. Customers can read the CPUID, encrypt it with a symmetric key Key to obtain ENC-CPUID, and store the ENC-CPUID in the vendor storage partition. After the system boots up, it reads the ENC-CPUID from the vendor storage partition, decrypts it to obtain DEC-CPUID, and compares DEC-CPUID with CPUID. If the match fails, it is considered illegal, and the device is restarted to achieve the anti-cloning purpose.

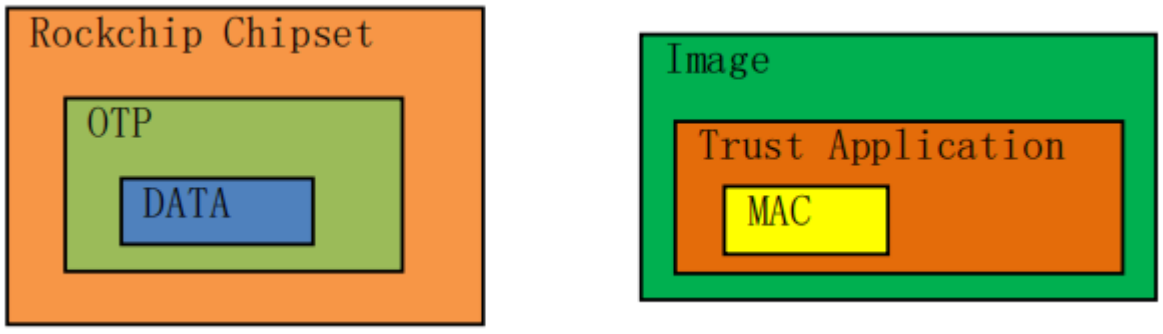


The security of this scheme depends on the protection of the key Key. Customers should avoid hardcoding the plaintext key directly in the code. It is recommended to obfuscate the key before hardcoding it into the code to prevent illegal users from obtaining the plaintext key through disassembly.

Additionally, since ENC-CPUID is stored in the vendor storage partition, erasing the flash will clear the vendor storage partition and result in the loss of ENC-CPUID. Therefore, after erasing the flash, it is necessary to rewrite the ENC-CPUID.

- Intermediate Scheme

Rockchip chips have reserved space in the OTP for customers to store anti-cloning private data. Customers generate custom private data DATA and MAC, and a custom-specific function func is used, where DATA and MAC satisfy a specific functional transformation relationship, such as MAC = func(DATA). Customers can write anti-cloning private data DATA into the OTP area and hardcode MAC into the trusted application TA (Trusted Application) code. After the system boots up, the trusted application TA reads the anti-cloning private data DATA from the OTP, and the trusted application TA checks if there is a given functional transformation relationship between DATA and MAC. If the match fails, it is considered illegal, and the device is restarted to achieve the anti-cloning purpose.



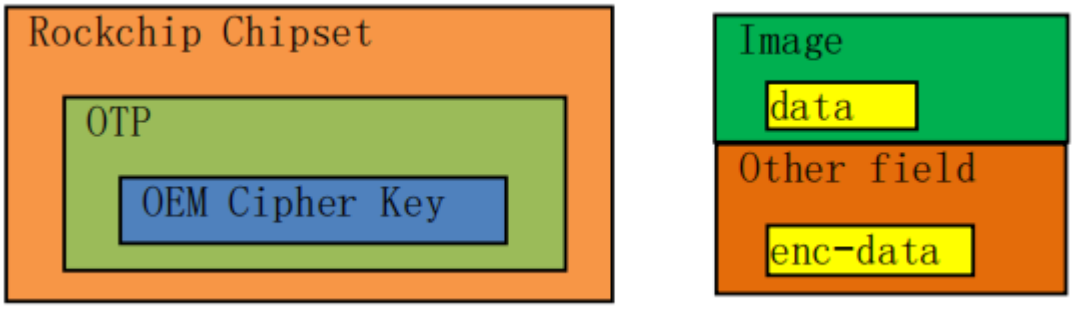
The security of this scheme depends on the protection of the anti-cloning private data DATA. The kernel cannot read this data, only the trusted application TA (Trusted Application) can read it. Customers should refer to the "TA Signature" section in the document "Rockchip_Developer_Guide_TEE_SDK_EN.md" and sign the trusted application TA (Trusted Application) with their own key to prevent illegal TA from running and causing leakage of anti-cloning private data DATA.

- Advanced Scheme

Rockchip chips have reserved space in the OTP for customers to store custom private keys. For usage, please refer to the "OEM Cipher Key" section in `docs/en/Common/SECURITY/Rockchip_Developer_Guide_OTP_EN.pdf`.

The anti-cloning technology relies on the customer's private key OEM Cipher Key stored in the chip's OTP. This key is written during the factory production phase. To ensure the key is not leaked, the system only provides a writing interface without a reading interface. Some platforms also support the Hardware Read feature to lock the key, making it unreadable by the CPU, thus offering higher security.

Customers can customize plaintext data data, use the private key OEM Cipher Key, and encrypt data with the AES algorithm to obtain encrypted data enc-data, which is then stored in a specified location on the flash or hardcoded in the code. The application code sets plaintext data data. After the system boots up, it first decrypts the encrypted data enc-data with the private key OEM Cipher Key to obtain decrypted data dec-data and compares it with the plaintext data data in the application code. If the plaintext data data or encrypted data enc-data is tampered with, or if the flash firmware is illegally copied to other unauthorized chip platforms, the decrypted data dec-data will not match the plaintext data data. A match failure is considered illegal, and the device is restarted to achieve the anti-cloning purpose.



The advantage of this scheme is that the customer's private key OEM Cipher Key, once locked, cannot be read by the CPU, offering higher confidentiality, integrity, and non-tampering security features.

Customers can freely choose to use any one of the schemes.

For more details, please refer to the anti-cloning document `docs/en/Common/SECURITY/Rockchip_Developer_Guide_Anti_Copy_Board_EN.pdf`.

The anti-cloning tool BoardProofTool is located at `tools/windows/BoardProofTool_v1.0_20231201.zip`.

9.12 WIFI/BT Development

Refer to the documentation in the `<SDK>/docs/en/Linux/Wifibt` directory.

9.13 Post-Processing Development for ROOTFS

The Post rootfs option is primarily used for post-processing of the root file system, including performance tuning, feature support, and debugging assistance, to meet various development and deployment needs. Functional configuration can be done through the SDK menuconfig. Below is an example of using menuconfig in the SDK directory:

```
<SDK>$ make menuconfig
  [*] Rootfs (Buildroot|Debian|Yocto)  --->
    Post rootfs installs  --->
```

```
        [*] create /etc/ld.so.cache
                hostname (auto)  --->
                locale (auto)  --->
        [*] strip kernel modules
    [*] async-commit DRM driver hack
    [*] debug information dir (/info/)
    [*] Rockchip udev rules
    [*] Wi-Fi/BT (Kernel modules, firmwares and scripts)  --->
            disk helpers (Boot-time mounting and resizing) (auto)  --->
    [ ] format extra partitions when needed
    [ ] bypass boot time fsck
    [*] log guardian (Truncate logs when disk is full)  --->
```

Rockchip Post rootfs development options provide a variety of features for customizing and optimizing the root file system, mainly including:

1. System Configuration

- Create dynamic link library cache to accelerate program loading
- Set hostname and localization environment
- Remove debugging symbols from kernel modules to reduce module size

2. Driver Support

- Apply a temporary fix for DRM drivers
- Add udev rules for Rockchip devices to enable device recognition and configuration
- Install Wi-Fi and Bluetooth related kernel modules, firmware, and scripts

3. File System Management

- Create a debug information directory to store related files
- Disk helper for mounting and resizing file systems at boot time
- Option to format additional partitions when needed
- Option to bypass file system integrity check at boot time

4. Log Management

- Log monitoring feature to truncate log files when disk space is insufficient

Through these options, you can customize and optimize the root file system on Rockchip devices according to actual needs, improving system performance, feature support, and debugging capabilities. These options provide developers with flexible customization space, helping to meet special needs in different scenarios.

9.14 Overlays Development

Rockchip Overlays offers a variety of feature options, including USB device support, fonts, input event handling, boot animation, interrupt balancing, file system optimization, virtual terminals, prebuilt tools, and custom overlays. You can configure these features through the SDK menuconfig.
Here is an example of using menuconfig in the SDK directory:

$ make menuconfig

```
[*] Rootfs (Buildroot|Debian|Yocto)  --->
  [*]   Overlays  --->
        [*]   USB gadget  --->
                extra fonts (auto)  --->
                input-event-daemon (power-key handling) (auto)  --->
                [ ]   bootanim (Boot-time animation)
                [*]   irqbalance (Balance hardware IRQs)
                [*]   fstrim (Discard unused blocks on all filesystems)  --->
                [ ]   frecon virtual terminal (VT)
                [*]   prebuilt tools
                ()    extra overlay dirs
```

For instance, if you need to enable the MTP feature in USB device support, you can configure it as follows:

```
[*] Rootfs (Buildroot|Debian|Yocto)  --->
  [*]   Overlays  --->
        [*]   USB gadget  --->
              [*]   Media Transfer Protocol (MTP)  --->
                    (devicon.ico) device icon (NEW)
                    (umtprd.conf) umtprd config file
```

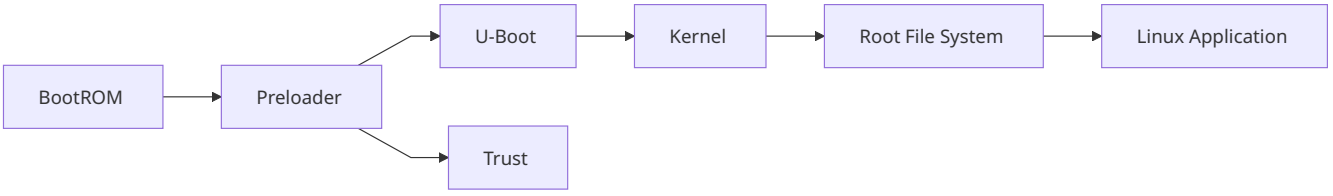
After the configuration is complete, save the configuration and compile:
make savedefconfig && ./build.sh

In this way, you can enable or disable different Overlays features according to your needs and make corresponding configurations to meet specific development requirements.

9.15 SDK Launch Method

9.15.1 Device Boot Process

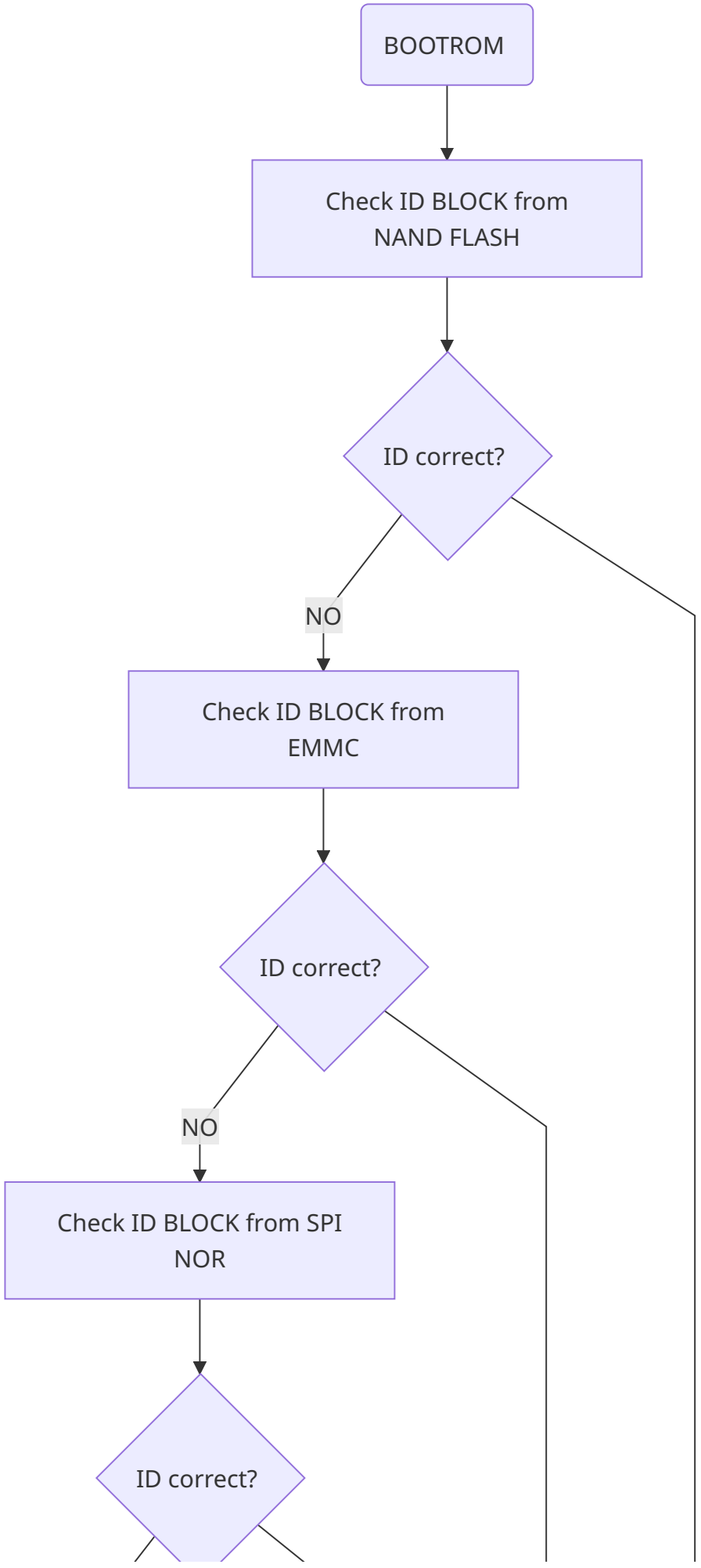
The boot process refers to the software procedure from system power-on to system startup completion. Below is the Linux system boot process:

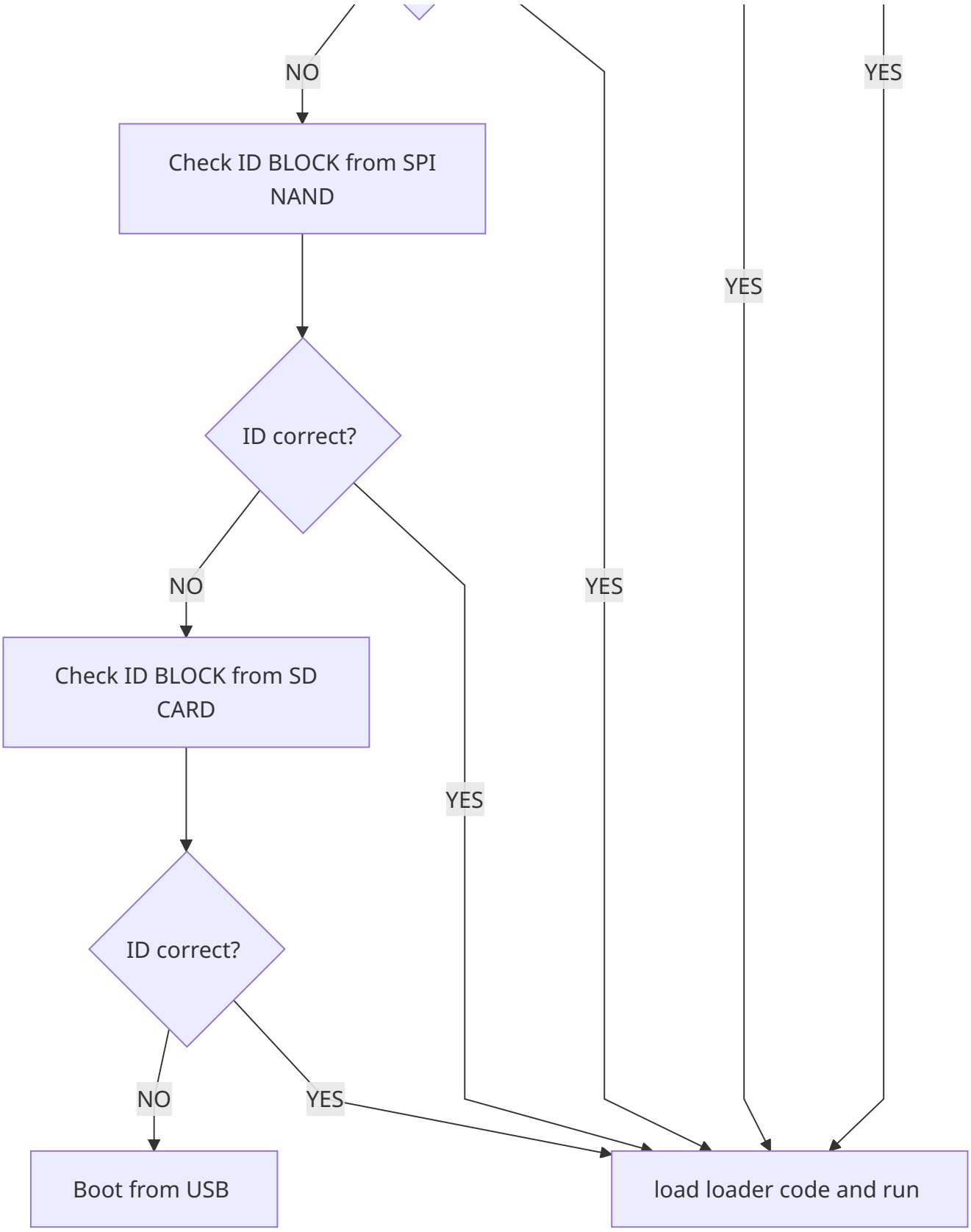


9.15.2 BOOTROM Process

After power-on, the latest executable code on the Rockchip chip is the mask code integrated within the chip, which cannot be changed. This is the BOOTROM code, which is integrated into both the AP and MCU. The BOOTROM code runs first upon system power-up, and then it detects peripheral storage devices and loads the Loader code from them.

Different chips have different sequences for detecting peripheral storage devices. The following diagram is an example of the BOOTROM boot process:


BOOTROM
Check ID BLOCK from NAND FLASH
ID correct?
NO
Check ID BLOCK from EMMC
ID correct?
NO
Check ID BLOCK from SPI NOR
ID correct?



Explanation:

- Some chips support ADC Key with different input voltage levels to specify the storage device detection order in BOOTROM. If the storage detection fails, the device directly enters maskrom mode.
- Detecting storage devices is usually done by detecting the device ID to confirm the presence of an external device.
- If all devices fail to detect valid firmware, the device enters maskrom mode and waits for specific

interfaces such as USB/UART to download firmware. Not all chips support USB/UART interface upgrades.

BootROM Boot Order for Various Chips

AP	No.1	No.2	No.3	No.4	No.5	No.6
RK3128	NAND	EMMC	SPI NOR(SFC)	SPI NAND(SFC)	SD0	USB
RK3126(B)	NAND	EMMC	SPI NOR(SFC)	SPI NAND(SFC)	SD0	USB
RK3036	NAND	EMMC	SPI NOR(SFC)	SPI NAND(SFC)	SD0	USB
RK3288	NAND	EMMC	SPI NOR(SPI2)	SPI NAND(SPI2)	SD0	USB
RK3399	SPI NOR(SPI2)	SPI NAND(SPI2)	EMMC	SD0	USB	--
RK3328	EMMC	SPI NOR(SPI2)	SPI NAND(SPI2)	SD0	USB	--
RK3128X/H	NAND	EMMC	SPI NOR(SPI2)	SPI NAND(SPI2)	SD0	USB
RK3308	NAND	EMMC	SPI NOR(SFC)	SPI NAND(SFC)	SD0	USB
RK3326	NAND	EMMC	SPI NOR(SFC)	SPI NAND(SFC)	SD0	USB
RK1808	SPI SLAVE	SPI NOR(SFC)	SPI NAND(SFC)	EMMC	USB	--
RK3399PRO	SPI NOR(SPI2)	SPI NAND(SPI2)	EMMC	SD0	USB	--
RK3568	SPI NOR(SFC)	SPI NAND(SFC)	NAND	EMMC	SD0	USB
RK3566	SPI NOR(SFC)	SPI NAND(SFC)	NAND	EMMC	SD0	USB
RK3588	SPI NOR(SFC)	SPI NAND(SFC)	EMMC--	SD0	USB	
RK3528	SPI NOR(SFC)	SPI NAND(SFC)	EMMC--	SD0	USB	
RK3562	SPI NOR(SFC)	SPI NAND(SFC)	EMMC--	SD0	USB	

9.15.3 System Boot Methods

Currently, the Linux SDK system provides three boot methods: Sysv, Busybox, and Systemd init.

Yocto systems default to using the Sysv init method for managing boot startup, utilizing the `update-rc.d.bbclass` to manage service boot startup configurations.

Buildroot systems default to using the Busybox init method for managing boot startup, which reads the `/etc/inittab` file after booting.

Debian systems default to using the Systemd method for managing boot startup. Systemd init reads related services in the `/etc/systemd/system/` directory after booting.

The SDK has different processing for different startup services. For example:

```
<SDK>/device/rockchip/common/scripts/post-disk.sh

if [ "$POST_INIT_BUSYBOX" ]; then
```

```
        mkdir -p "$TARGET_DIR/etc/init.d"
        install -m 0755 external/rkscript/$SCRIPT "$TARGET_DIR/etc/init.d/"
fi

if [ "$POST_INIT_SYSTEMD" ]; then
        mkdir -p "$TARGET_DIR/lib/systemd/system"
        install -m 0755 external/rkscript/$DISK_HELPER_TYPE-all.service
"$TARGET_DIR/lib/systemd/system/"
        mkdir -p "$TARGET_DIR/etc/systemd/system/sysinit.target.wants"
        ln -sf /lib/systemd/system/$DISK_HELPER_TYPE-all.service
"$TARGET_DIR/etc/systemd/system/sysinit.target.wants/"
fi

if [ "$POST_INIT_SYSV" ]; then
        mkdir -p "$TARGET_DIR/etc/init.d"
        install -m 0755 external/rkscript/$SCRIPT
"$TARGET_DIR/etc/init.d/${DISK_HELPER_TYPE}all.sh"
        mkdir -p "$TARGET_DIR/etc/rcS.d"
        ln -sf ../init.d/${DISK_HELPER_TYPE}all.sh
"$TARGET_DIR/etc/rcS.d/$SCRIPT"
fi
```

9.16 Shared Remote Desktop Development

Remote desktop is a technology that allows users to access and control the desktop of another computer (server) through a computer (client). This technology enables users to run applications on the remote computer, manipulate files, and use devices as if they were operating the computer directly.

9.16.1 Controlling Remote Desktop with wlfreerdp

wlfreerdp is a Wayland Remote Desktop Protocol (RDP) client. It enables users to remotely connect to Windows servers or personal computers from Linux systems, allowing them to access the Windows desktop environment remotely.

PC End: Ubuntu22.04 xrdp (0.9.17)
Device: RK3588 + buildroot + wlfreerdp (2.11.5)

PC End: Share --> Remote Desktop, enable remote desktop, the device name is domain, the username is Username, and the password is Password.

root@rk3588-buildroot:/# wlfreerdp /f /v:172.16.15.86 /u:wxt /p:test0000 /d:pc
[08:00:27:928] [1872:1872] [WARN][com.freerdp.crypto] - Certificate verification failure 'self-signed certificate (18)' at stack position 0
[08:00:27:928] [1872:1872] [WARN][com.freerdp.crypto] - CN = GNOME, C = US
[08:00:28:237] [1872:1872] [INFO][com.freerdp.gdi] - Local framebuffer format PIXEL_FORMAT_BGRA32
[08:00:28:237] [1872:1872] [INFO][com.freerdp.gdi] - Remote framebuffer format PIXEL_FORMAT_BGRA32
[08:00:28:258] [1872:1872] [INFO][com.freerdp.channels.rdpsnd.client] - [static] Loaded fake backend for rdpsnd
[08:00:28:259] [1872:1872] [INFO][com.freerdp.channels.drdynvc.client] - Loading Dynamic Virtual Channel

rdpgfx
..

In buildroot, enabling `BR2_PACKAGE_FREERDP=y` is sufficient.

9.16.2 Weston VNC Remote Desktop Access

- Enable Weston VNC

To enable Weston VNC in Buildroot:

Ensure that the configuration option BR2_PACKAGE_WESTON_VNC is enabled in Buildroot.
Start the Weston VNC service:

On your system development board, run the following command to start the Weston VNC service:

```
weston -B drm,vnc &
```

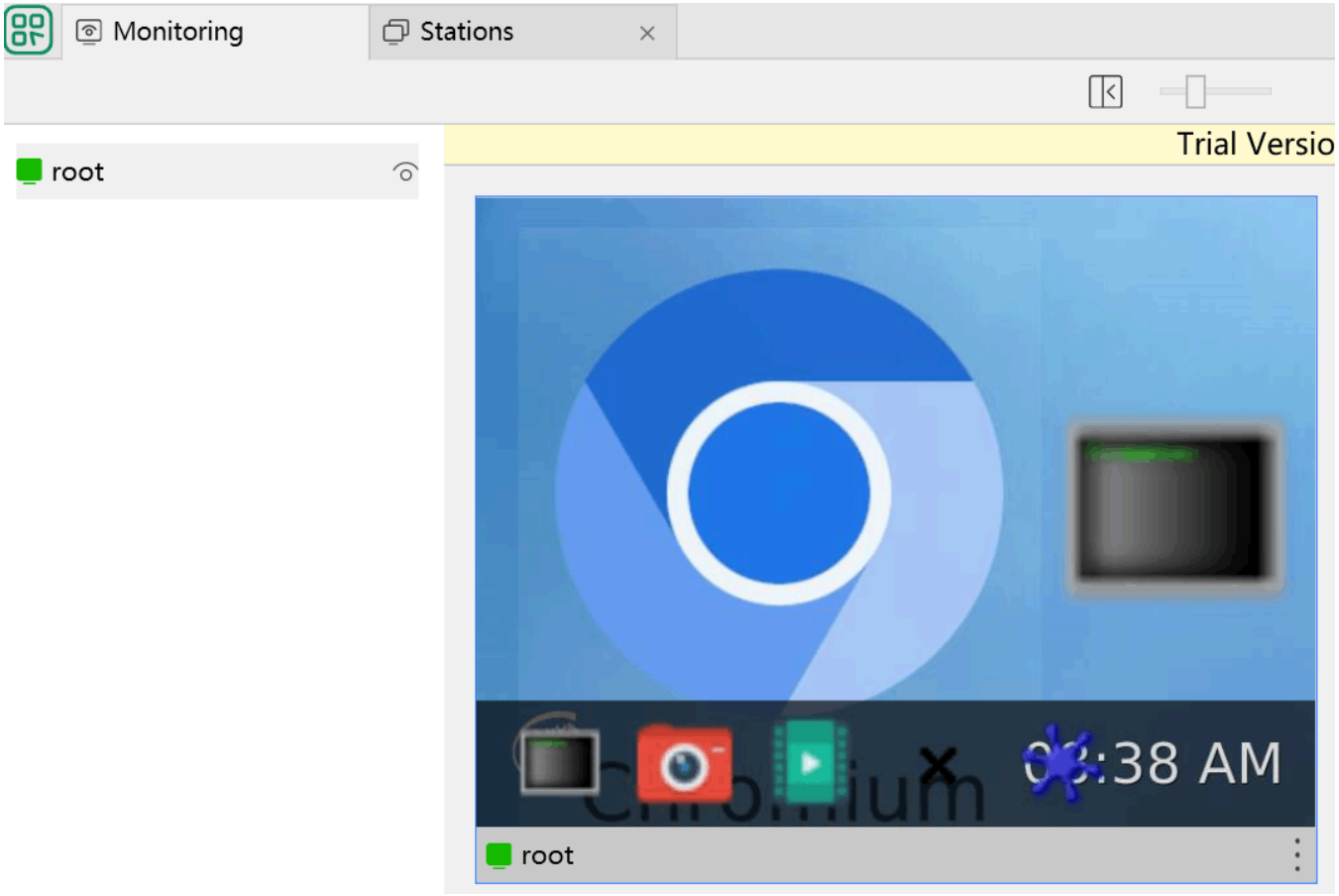
This will launch the Weston display server and attach the VNC service.

- Remote Access:

To access the Weston VNC service from a remote location, you can install the MightyViewer software on your Windows system, which is a VNC client.

- Connection and Control:

Use MightyViewer to connect to the IP address of your development board and begin remote control and viewing of the desktop environment.
Screenshot Example
Below is a screenshot example of connecting to the Weston VNC service using MightyViewer:



9.17 SDK Testing

9.17.1 Integrating Rockchip Stress Test Scripts

The `rockchip_test` integrates functionality, stress, and performance-related testing.

```
       ROCKCHIPS TEST TOOLS
172.16.15.86
ddr test:              1 (ddr stress test)
cpu test:              2 (cpu stress test)
gpu test:              3 (gpu stress test)
npu test:              4 (npu stress test)
suspend_resume test:   5 (suspend resume)
reboot test:           6 (auto reboot test)
power lost test:       7 (power lost test)
flash stress test:     8 (flash stress test)
recovery test:         9 (recovery wipe all test)
audio test:           10 (audio test)
camera test:          11 (camera test)
video test:           12 (video test)
bluetooth test:       13 (bluetooth on off test)
wifi test:            14 (wifi on off test)
chromium test:        15 (chromium with video test)
**************************************************
```

```
please input your test module:
```

9.17.2 Benchmark Testing

The following are reference data for common benchmark tests. You can find more information in the following test documents:

```
<SDK>/docs/en/Linux/Profile/Rockchip_Introduction_Linux_Benchmark_KPI_EN.pdf
```

9.17.3 Rockchip Modules and Stress Testing

Below are some common module functionalities and stress testing methods. You can find more detailed information in the following documentation:

```
<SDK>/docs/en/Linux/Profile/Rockchip_User_Guide_Linux_Software_Test_EN.pdf
```

10. Chapter-12 SDK System Debugging Tools Introduction

There are some commonly used debugging tools for static compilation are pre-installed in the SDK , as follows:

```
device/rockchip/common/tools/
├── adbd
├── busybox
├── gdb
├── io
├── kmsgrab
├── modetest
├── perf-4.19
├── perf-4.4
├── perf-5.10
├── pmap
├── procrank
├── ps
├── slabtop
├── strace
├── top
├── update
├── vendor_storage
├── vmstat
└── watch
```

Just `make menuconfig` in the SDK directory and select the relevant configuration.

```
  | Symbol: RK_ROOTFS_PREBUILT_TOOLS [=y]
                                     |
  | Type  : bool
                                    |
  | Prompt: prebuilt tools
                                    |
  |   Location:
                                     |
  |     -> Rootfs
                                     |
  |       -> Post rootfs installs
                                     |
  |   Defined at Config.in.post-rootfs:263
```

Save the configuration, for example, the detailed modifications of RK3588 EVB1 are as follows:

```
device/rockchip/rk3588/rockchip_rk3588_evb1_lp4_v10_defconfig
@@ -1,4 +1,7 @@
 RK_YOCTO_CFG="rockchip-rk3588-evb"
 RK_WIFIBT_TTY="ttyS8"
+RK_ROOTFS_PREBUILT_TOOLS=y
```

10.1 ADB Tool

You can open the following macros in Buildroot to build and obtain:

```
BR2_PACKAGE_ANDROID_TOOLS=y
BR2_PACKAGE_ANDROID_TOOLS_STATIC=y
```

For details usage of ADB tools, please refer to ADB official guidance document.

10.1.1 Overview

- Run the device's shell (command line)
- Manage port mapping for emulators or devices
- Upload/download files between computer and device
- Install local software to Debian, Buildroot and other Linux devices
- ADB is a "client-server" program, where the client mainly refers to a PC and a server is an entity device or a virtual device of Debian. Depending on how the PC connects to Debian devices, ADB can be divided into two types:

Network ADB: The host is connected to the STB device via a wired/wireless network (same LAN)
USB ADB: The host is connected to the STB device through a USB cable

10.1.2 USB ADB Usage Instructions

USB adb usage has the following restrictions:

- Only supports USB OTG port
- Does not support using of multiple clients at the same time
- Only supports the host to connect to one device, does not support connecting to multiple devices. To test whether the connection is successful, run the "adb devices" command. If the serial number of the device is displayed, the connection is successful.

10.2 Busybox Tool

You can open the following macros in Buildroot to build and obtain:

```
BR2_PACKAGE_BUSYBOX=y
BR2_PACKAGE_BUSYBOX_STATIC=y
```

Busybox is similar to a Linux toolbox. It integrates and compresses many tools and commands of Linux.

```
## Chapter-12 ./busybox
BusyBox v1.36.0 (2023-05-20 11:25:39 CST) multi-call binary.
BusyBox is copyrighted by many authors between 1998-2015.
Licensed under GPLv2. See source distribution for detailed
copyright notices.

Usage: busybox [function [arguments]...]
   or: busybox --list[-full]
   or: busybox --show SCRIPT
   or: busybox --install [-s] [DIR]
   or: function [arguments]...

        BusyBox is a multi-call binary that combines many common Unix
        utilities into a single executable.  Most people will create a
        link to busybox for each function they wish to use and BusyBox
        will act like whatever it was invoked as.

Currently defined functions:
        [, [[, addgroup, adduser, ar, arch, arp, arping, ascii, ash, awk,
        base32, base64, basename, bc, blkid, bunzip2, bzcat, cat, chattr,
        chgrp, chmod, chown, chroot, chrt, chvt, cksum, clear, cmp, cp, cpio,
        crc32, crond, crontab, cut, date, dc, dd, deallocvt, delgroup, deluser,
        devmem, df, diff, dirname, dmesg, dnsd, dnsdomainname, dos2unix, du,
        dumpkmap, echo, egrep, eject, env, ether-wake, expr, factor, fallocate,
        false, fbset, fdflush, fdformat, fdisk, fgrep, find, flock, fold, free,
        freeramdisk, fsck, fsfreeze, fstrim, fuser, getopt, getty, grep,
        groups, gunzip, gzip, halt, hdparm, head, hexdump, hexedit, hostid,
        hostname, hwclock, i2cdetect, i2cdump, i2cget, i2cset, i2ctransfer, id,
        ifconfig, ifdown, ifup, inetd, init, insmod, install, ip, ipaddr,
        ipcrm, ipcs, iplink, ipneigh, iproute, iprule, iptunnel, kill, killall,
        killall5, klogd, last, less, link, linux32, linux64, linuxrc, ln,
        loadfont, loadkmap, logger, login, logname, losetup, ls, lsattr, lsmod,
        lsof, lspci, lsscsi, lsusb, lzcat, lzma, lzopcat, makedevs, md5sum,
        mdev, mesg, microcom, mim, mkdir, mkdosfs, mke2fs, mkfifo, mknod,
        mkpasswd, mkswap, mktemp, modprobe, more, mount, mountpoint, mt, mv,
        nameif, netstat, nice, nl, nohup, nologin, nproc, nslookup, nuke, od,
        openvt, partprobe, passwd, paste, patch, pidof, ping, pipe_progress,
        pivot_root, pmap, poweroff, printenv, printf, ps, pwd, rdate, readlink,
        readprofile, realpath, reboot, renice, reset, resize, resume, rm,
        rmdir, rmmod, route, run-init, run-parts, runlevel, sed, seedrng, seq,
        setarch, setconsole, setfattr, setkeycodes, setlogcons, setpriv,
        setserial, setsid, sh, sha1sum, sha256sum, sha3sum, sha512sum, shred,
```

```
        sleep, sort, start-stop-daemon, strings, stty, su, sulogin, svc, svok,
        swapoff, swapon, switch_root, sync, sysctl, syslogd, tail, tar,
        taskset, tc, tee, telnet, test, tftp, time, timeout, top, touch, tr,
        traceroute, tree, true, truncate, ts, tsort, tty, ubirename, udhcpc,
        uevent, umount, uname, uniq, unix2dos, unlink, unlzma, unlzop, unxz,
        unzip, uptime, usleep, uudecode, uuencode, vconfig, vi, vlock, w,
        watch, watchdog, wc, wget, which, who, whoami, xargs, xxd, xz, xzcat,
        yes, zcat
```

10.3 GDB Tool

You can open the following macros in Buildroot to build and obtain:

```
BR2_PACKAGE_GDB=y
BR2_PACKAGE_GDB_STATIC=y
BR2_PACKAGE_GDB_DEBUGGER=y
```

GDB, a commonly used debugging tool in Linux, is a powerful program debugger. Please refer to GDB for more usage instructions.

10.4 IO Tool

You can open the following macros in Buildroot to build and obtain:

```
BR2_PACKAGE_RKTOOLKIT=y
BR2_PACKAGE_RKTOOLKIT_STATIC=y
```

Memory can be accessed through /dev/mem, with permission to read and write chip registers.

10.5 kmsgrab Tool

In the SDK project, you can build and obtain it as follows.

```
For building kmsgrab:
1/ ./build.sh shell
2/ source ./buildroot/output/$RK_BUILDROOT_CFG/host/environment-setup
3/ $CC $RK_DATA_DIR/kmsgrab.c $(pkg-config --cflags --libs libdrm) -static -o kmsgrab
```

Captures the KMS scan-out framebuffer associated with the specified CRTC or screen as a DRM object that can be passed to other hardware functions.

10.6 modetest Tool

Modetest is a debugging tool that comes with the libdrm source code, which can perform some basic debugging on drm.

The help information of modetest is as follows:

```
(shell)# modetest -h
usage: modetest [-cDdefMPpsCvw]
 Query options:
    -c    list connectors
    -e    list encoders
    -f    list framebuffers
    -p    list CRTCs and planes (pipes)
 Test options:
    -P <crtc_id>:<w>x<h>[+<x>+<y>][*<scale>][@<format>]    set a plane
    -s <connector_id>[,<connector_id>][@<crtc_id>]:<mode>[-<vrefresh>][@<format>]    set a
mode
    -C    test hw cursor
    -v    test vsynced page flipping
    -w <obj_id>:<prop_name>:<value>    set property
 Generic options:
    -d    drop master after mode set
    -M module    use the given driver
    -D device    use the given device
    Default is to dump all info.
```

10.7 Perf Tool

You can open the following macros in Buildroot to build and obtain:

```
For building perf:
1/ Build dynamic version firstly
2/ Enable BR2_PACKAGE_LINUX_TOOLS_PERF_STATIC
3/ Run: make linux-tools-reconfigure
```

Perf is a Linux performance analysis tool.

10.8 Pmap Tool

To see how much memory a process uses, pmap provides a memory mapping of a process. The pmap command is used to display the memory status of one or more processes. It reports the address space and memory status information of the process.

```
#pmap PID
```

10.9 ProcrankTool

You can open the following macros in Buildroot to build and obtain:

```
BR2_PACKAGE_PROCRANK_LINUX=y
BR2_PACKAGE_PROCRANK_LINUX_STATIC=y
```

Use procrank to analyze memory utilization and analyze source code. procrank is a tool that counts memory usage, including detailed parameters of VSS, PSS, PSS and USS. It is commonly used memory analysis tool.

10.10 PS Tool

The Linux PS (process status) command is used to display the status of the current process, similar to the Windows Task Manager.

There are many parameters of PS. Here we only list a few commonly used parameters and briefly introduce them.
-A: lists all processes
-w: display more information by widening the display
-au: displays more detailed information
-aux: displays all processes containing other users

```
au(x) output format:
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND

USER: process owner
PID: pid
%CPU: occupied CPU usage
%MEM: occupied memory usage
VSZ: size of virtual memory occupied
RSS: memory size occupied
TTY: minor device number of tty
STAT: The status of the process:

D: Uninterruptible sleep state (usually, it is IO process)
R: Running
S: stationary state
T: pause execution
Z: Does not exist but cannot be eliminated temporarily
W: Not enough memory pages available to allocate
<: High priority itinerary
N: low priority itinerary
L: There are memory pages allocated and locked in the memory
START: The start time of the process
TIME: execution time
COMMAND: command executed
```

10.11 Slabtop Tool

Display kernel slab buffer information in real time.

Refresh the slab buffer information every ten seconds:

```
slabtop -d 10
```

10.12 Strace Tool

Strace is a Linux user space tracer that can be used for tracking and debugging.

10.13 Top Tools

The top command is a commonly used performance analysis tool under Linux. It can display the resource usage of each process in the system in real time and is often used for server-side performance analysis.

10.14 Update Tool

Enter the recovery program to update and format, then reboot into the normal system.

10.15 Vendor_storage Tool

Read and write on the Vendor partition.

```
root@rk3588-buildroot:/armhf# ./vendor_storage --help
vendor storage tool - Revision: 2.0

./vendor_storage [-r/w <vendor_id> -t <pr_type> -i <input>] [-R]
  -r           Read specify vendor_id
  -R           Read common vendor_id
  -w           Write specify vendor_id
  -t           print type
  -i           input string
  <vendor_id>  There are 16 types
               "VENDOR_SN_ID"
               "VENDOR_WIFI_MAC_ID"
               "VENDOR_LAN_MAC_ID"
               "VENDOR_BT_MAC_ID"
               "VENDOR_HDCP_14_HDMI_ID"
               "VENDOR_HDCP_14_DP_ID"
               "VENDOR_HDCP_2x_ID"
               "VENDOR_DRM_KEY_ID"
               "VENDOR_PLAYREADY_Cert_ID"
```

```
                "VENDOR_ATTENTION_KEY_ID"
                "VENDOR_PLAYREADY_ROOT_KEY_0_ID"
                "VENDOR_PLAYREADY_ROOT_KEY_1_ID"
                "VENDOR_SENSOR_CALIBRATION_ID"
                "VENODR_RESERVE_ID_14"
                "VENDOR_IMEI_ID"
                "VENDOR_CUSTOM_ID"
                And custom can define other id like
                VENDOR_CUSTOM_ID_1A (define ID = 26)
  <pr_type>     In write case, used with -i <input>
                There are 3 types
                "hex": <input> must be hex form like 0123
                "string": <input> must be ASCII string
                "file": <input> must be path to file
                Note: If use "file" and -i with read, it means save storage to file
Examples:
  ./vendor_storage -w VENDOR_CUSTOM_ID_1A -t file -i /userdata/test.bin
                       write userdata/test.bin to storage
                       Or -t string -i test_storage
                       write string "test_storage" to storage
                       ID = 26
  ./vendor_storage -r VENDOR_CUSTOM_ID_1A -t file -i /userdata/read.bin
                       read storage(ID=26) to userdata/read.bin
                       Or -t string
                       print storage(ID=26) with ASCII string
```

10.16 Vmstat Tool

The vmstat command reports statistics on kernel threads, virtual memory, disks, hypervisor pages, traps, and processor activity.

10.17 Watch Tool

Watch can help monitor the results of a command.

```
## Chapter-12 watch

Usage:
 watch [options] command

Options:
  -b, --beep             beep if command has a non-zero exit
  -c, --color            interpret ANSI color and style sequences
  -d, --differences[=<permanent>]
                         highlight changes between updates
  -e, --errexit          exit if command has a non-zero exit
  -g, --chgexit          exit when output from command changes
  -n, --interval <secs>  seconds to wait between updates
```

```
  -p, --precise          attempt run command in precise intervals
  -t, --no-title         turn off header
  -w, --no-wrap          turn off line wrapping
  -x, --exec             pass command to exec instead of "sh -c"

 -h, --help     display this help and exit
 -v, --version  output version information and exit
```

10.18 weston Debugging Method

root@rk3288-buildroot:/# WAYLAND_DEBUG=1 weston --debug

Weston related parameters are as follows:

```
root@rk3288-buildroot:/# weston --help
Usage: weston [OPTIONS]

This is weston version 12.0.1, the Wayland reference compositor.
Weston supports multiple backends, and depending on which backend is in use
different options will be accepted.

Core options:

  --version             Print weston version
  -B, --backend=BACKEND Backend module, one of
                                drm
  --renderer=NAME       Renderer to use, one of
                                auto    Automatic selection of one of the below renderers
                                gl      OpenGL ES
                                noop    No-op renderer for testing only
                                pixman  Pixman software renderer
  --shell=NAME  Shell to load, defaults to desktop
  -S, --socket=NAME     Name of socket to listen on
  -i, --idle-time=SECS  Idle time in seconds
  --modules             Load the comma-separated list of modules
  --log=FILE            Log to the given file
  -c, --config=FILE     Config file to load, defaults to weston.ini
  --no-config           Do not read weston.ini
  --wait-for-debugger   Raise SIGSTOP on start-up
  --debug               Enable debug extension
  -l, --logger-scopes=SCOPE
                        Specify log scopes to subscribe to.
                        Can specify multiple scopes, each followed by comma
  -f, --flight-rec-scopes=SCOPE
                        Specify log scopes to subscribe to.
                        Can specify multiple scopes, each followed by comma
  -w, --warm-up         Hold display for the first app
  -h, --help            This help message

Options for drm:
```

```
  --seat=SEAT           The seat that weston should run on, instead of the seat defined in
XDG_SEAT
  --drm-device=CARD     The DRM device to use, e.g. "card0".
  --use-pixman          Use the pixman (CPU) renderer (deprecated alias for --
renderer=pixman)
  --current-mode        Prefer current KMS mode over EDID preferred mode
  --continue-without-input      Allow the compositor to start without input devices
```

10.19 Obtain System Log Information Automatically

The log information will be automatically obtained in the /info directory. The main log information is as follows:

```
/info/
├── clk_summary -> /sys/kernel/debug/clk/clk_summary
├── cmdline -> /proc/cmdline
├── cpuinfo -> /proc/cpuinfo
├── device-tree -> /proc/device-tree
├── diskstats -> /proc/diskstats
├── dma_buf -> /sys/kernel/debug/dma_buf
├── dri -> /sys/kernel/debug/dri
├── fstab -> /etc/fstab
├── gpio -> /sys/kernel/debug/gpio
├── interrupts -> /proc/interrupts
├── iomem -> /proc/iomem
├── kallsyms -> /proc/kallsyms
├── log -> /var/log
├── meminfo -> /proc/meminfo
├── mountall.log -> /tmp/mountall.log
├── os-release -> /etc/os-release
├── partitions -> /proc/partitions
├── pinctrl -> /sys/kernel/debug/pinctrl/
├── rkcif-mipi-lvds -> /proc/rkcif-mipi-lvds
├── rk_dmabuf -> /proc/rk_dmabuf
├── rkisp0-vir0 -> /proc/rkisp0-vir0
├── slabinfo -> /proc/slabinfo
├── softirqs -> /proc/softirqs
├── version -> /proc/version
└── wakeup_sources -> /sys/kernel/debug/wakeup_sources
...
```

11. Chapter-13 SDK Software License Instructions

Before customers obtain the SDK, they need to sign the SDK agreement. This agreement states specific rights and obligations.

11.1 Copyright Detection Tool

11.1.1 Buildroot

```
make legal-info
```

The SDK project can automatically generate relevant License information by running the `make legal-info` command, Like RK3566 project:

```
$ source buildroot/envsetup.sh rockchip_rk3566
buildroot$ make legal-info
```

Executing the above command will generate the following information:

```
buildroot$ tree -L 1 output/rockchip_rk3566/legal-info/
output/rockchip_rk3566/legal-info/
├── buildroot.config
├── host-licenses
├── host-manifest.csv
├── host-sources
├── legal-info.sha256
├── licenses
├── manifest.csv
├── README
└── sources
```

The `manifest.csv` is the license information of the relevant repository used by RK3566 project.
The `host-manifest.csv` is the license information of the third-party tools of the PC used by RK3566 project.

11.1.2 Debian

For license detection, please refer to Official Tool

```
licensecheck --check '.*' --recursive --copyright --deb-fmt \
        --lines 0 * | /usr/lib/cdbs/licensecheck2dep5
```

The relevant copyright instructions for each source code package are located in `/usr/share/doc/*/copyright`

11.1.3 Yocto

The relevant copyright instructions for each source code package are located in `build/tmp/deploy/licenses/*/recipeinfo`

11.2 License List

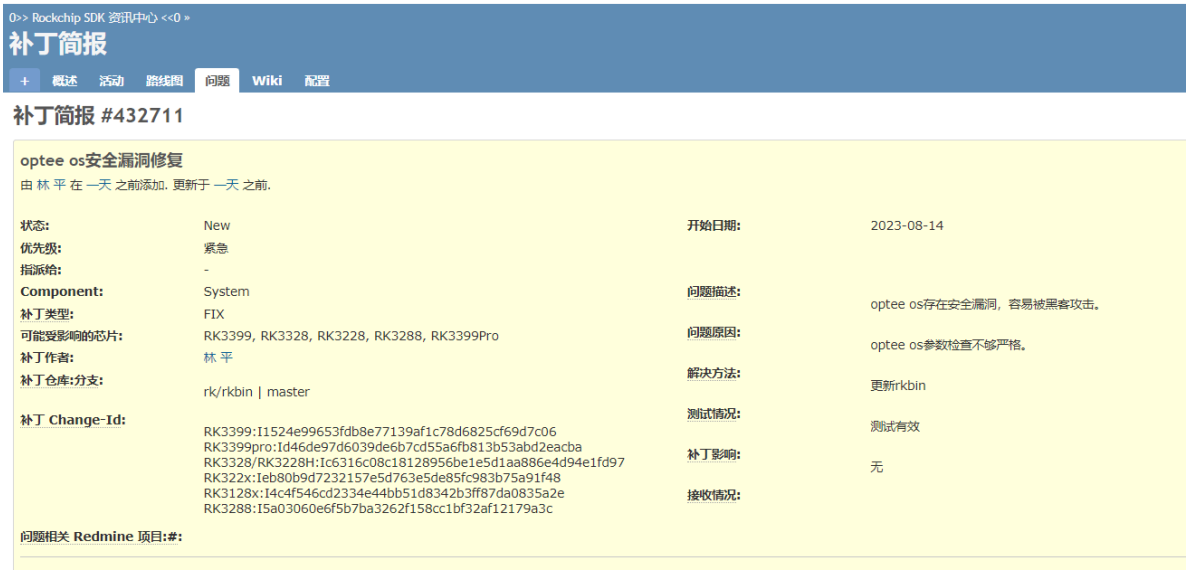
For the current mainstream License list, please refer to the following link

license-list

12. Chapter-14 SDK Security Update

Rockchip SDK security update strategy is divided into several key components:

1. **Core Component Updates**
   - Regularly update mainstream OS, Kernel, and other core components, incorporating relevant security patches.
     Example: Rockchip uses Yocto 5.0 and merges security fixes from Yocto 5.0.0 to 5.0.8 (refer to Yocto Security Fixes).
2. **Chip-Specific Security Advisories**
   - Publish security bulletins and patches for chip-related vulnerabilities.
     Example bulletin:



0>> Rockchip SDK 资讯中心 <<0 »

补丁简报

+ 概述 活动 路线图 问题 Wiki 配置

补丁简报 #432711

optee os安全漏洞修复

由 林 平 在 一天 之前添加. 更新于 一天 之前.

状态:	New	开始日期:	2023-08-14
优先级:	紧急		
指派给:	-		
Component:	System	问题描述:	optee os存在安全漏洞，容易被黑客攻击。
补丁类型:	FIX		
可能受影响的芯片:	RK3399, RK3328, RK3228, RK3288, RK3399Pro	问题原因:	optee os参数检查不够严格。
补丁作者:	林 平		
补丁仓库:分支:	rk/rkbin | master	解决方法:	更新rkbin
补丁 Change-Id:	RK3399:I1524e99653fdb8e77139af1c78d6825cf69d7c06 RK3399pro:Id46de97d6039de6b7cd55a6fb813b53abd2eacba RK3328/RK3228H:Ic6316c08c18128956be1e5d1aa886e4d94e1fd97 RK322x:Ieb80b9d7232157e5d763e5de85fc983b75a91f48 RK3128x:I4c4f546cd2334e44bb51d8342b3ff87da0835a2e RK3288:I5a03060e6f5b7ba3262f158cc1bf32af12179a3c	测试情况:	测试有效
		补丁影响:	无
		接收情况:	
问题相关 Redmine 项目:#:			

Notifications are sent to customers via:

瑞芯微技术信息简报（2020.10.19-2020.10.23）

序号	类别	芯片平台	软件版本	模块	问题严重性	描述（解决问题、信息通知）	软件修改	硬件修改	解决方法（给出详细步骤，包括git提交号，补丁包名称，软硬件修改方案）	备注（风险，是否临时版本等）
1	Security	ALL	Android /Linux	SDK	重要	发布CNVD-2020-58443漏洞修复补丁，基于kernel 3.10 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包：ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-58443/0001-ARM-rockchip-ddr_freq-Avoid-accessing-address-of-use.patch	FTP地址：ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t
2	Security	ALL	Android /Linux	SDK	重要	发布CNVD-2020-58453漏洞修复补丁，基于kernel 4.4 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包：ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-58453/0002-video-rockchip-vcodec-fix-risk-for-session-type-over.patch	FTP地址：ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t

瑞芯微技术信息简报（2020.11.23-2020.11.27）									
序号	类别	芯片平台	软件版本	模块	问题严重性	描述 （解决问题、信息通知）	软件修改	硬件修改	解决方法 （给出详细步骤，包括git提交号，补丁包名称，软硬件修改方案）	备注 （风险，是否临时版本等）
1	Security	ALL	Android/Linux	SDK	重要	发布CNVD-2020-64556漏洞修复补丁，基于kernel 3.10/4.4/4.19 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包： ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-64556 & CNVD-2020-64557/	FTP地址： ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t
2	Security	ALL	Android/Linux	SDK	重要	发布CNVD-2020-64557漏洞修复补丁，基于kernel 3.10/4.4/4.19 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包： ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-64556 & CNVD-2020-64557/	FTP地址： ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t

瑞芯微技术信息简报（2020.12.14-2020.12.18）									
序号	类别	芯片平台	软件版本	模块	问题严重性	描述 （解决问题、信息通知）	软件修改	硬件修改	解决方法 （给出详细步骤，包括git提交号，补丁包名称，软硬件修改方案）	备注 （风险，是否临时版本等）
1	Security	ALL	Android/Linux	SDK	重要	发布CNVD-2020-68355漏洞修复补丁，基于kernel 3.10/4.4/4.19 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包： ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-68355/	FTP地址： ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t
2	Security	ALL	Android/Linux	SDK	重要	发布CNVD-2020-67105漏洞修复补丁，基于kernel 3.10/4.4/4.19 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包： ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-67105 & CNVD-2020-67106/	FTP地址： ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t
3	Security	ALL	Android/Linux	SDK	重要	发布CNVD-2020-67106漏洞修复补丁，基于kernel 3.10/4.4/4.19 SDK需要更新此补丁	是	否	RK FTP可以下载对应补丁包： ftp://www.rockchip.com.cn/10-Security_Patch/CNVD-2020-67105 & CNVD-2020-67106/	FTP地址： ftp://www.rockchip.com.cn 账号名：rkwifi 密码：Cng9280H8t

12.1 Kernel Security Updates

The kernel team conducts ongoing security maintenance:

- **Security Scanning & Upgrading**: Continuously scan the kernel for vulnerabilities, merge stable minor releases, and patch publicly disclosed CVEs.
  Reference: Linux Kernel CVE Announcements.
- **Patch Notifications**: Monthly updates on Rockchip kernel security patches are available at Rockchip Security Patch Tracker.

12.2 Buildroot Security Updates

Refer to the official documentation: Buildroot CVE Management.

Key Example: Buildroot 2024.02 Security Update

Buildroot 2024.02 Changelog

12.2.1 `pkg-stats` Command

Generate detailed reports on software packages, including:

- **CVE Detection**: Scans for known vulnerabilities using the NVD Database.

- **Version Updates**: Identifies available upstream upgrades.
- **Report Generation**: Outputs `pkg-stats.html` (visual) and `pkg-stats.json` (data).

Usage:

```
make pkg-stats
```

Reports are saved in the output/ directory.

- Remediation Steps:

Upgrade vulnerable packages in `package/<package>/<package.mk>`.
Apply patches manually (create .patch files in `package/<package>/`).

12.3 Yocto Security Updates

For example, Yocto 4.0 Security Updates:

Yocto4.0-security

12.4 Debian Security Updates

Debian security updates can be referenced as follows:

Debian-security

Additionally, you can use APT to easily obtain the latest security updates. To get the latest security updates for your Debian operating system, add the following line to your /etc/apt/sources.list file:

```
deb http://security.debian.org/debian-security bookworm-security main contrib non-free non-free-firmware
```

After saving the changes, execute the following two commands to download and install security updates:

```
apt-get update && apt-get upgrade
```

If you discover security issues in your packages or others' packages, be sure to contact the security team via the email address team@security.debian.org. They track existing security issues, assist maintainers in resolving security issues or fix them themselves, and are responsible for sending security advisories and maintaining security.debian.org.

The following page is a brief archive of security advisories published on the debian-security-announce mailing list.

```
[2023/08/07] DSA-5471-1 libhtmlcleaner-java security update
[2023/08/06] DSA-5470-1 python-werkzeug security update
[2023/08/05] DSA-5469-1 thunderbird security update
[2023/08/05] DSA-5468-1 webkit2gtk security update
[2023/08/04] DSA-5467-1 chromium security update
```

```
[2023/08/04] DSA-5466-1 ntpsec security update
[2023/08/03] DSA-5465-1 python-django security update
[2023/08/03] DSA-5464-1 firefox-esr security update
[2023/07/30] DSA-5463-1 thunderbird security update
[2023/07/30] DSA-5462-1 linux security update
[2023/07/30] DSA-5461-1 linux security update
[2023/07/26] DSA-5460-1 curl security update
[2023/07/25] DSA-5459-1 amd64-microcode security update
[2023/07/25] DSA-5458-1 openjdk-17 security update
[2023/07/22] DSA-5457-1 webkit2gtk security update
[2023/07/20] DSA-5456-1 chromium security update
[2023/07/17] DSA-5455-1 iperf3 security update
[2023/07/16] DSA-5454-1 kanboard security update
[2023/07/16] DSA-5453-1 linux security update
[2023/07/14] DSA-5452-1 gpac security update
[2023/07/09] DSA-5451-1 thunderbird security update
```

13. Chapter-15 Rockchip open source information

13.1 Github

Rockchip Code Open Source repositories rockchip-github.

13.2 Wiki

Rockchip open source materials rockchip-wiki

13.3 Upstream

- Rockchip upstream uboot:

```
git clone https://gitlab.denx.de/u-boot/custodians/u-boot-rockchip.git
```

Upstream U-Boot support the following Rockchip SoCs:

`RK3036, RK3188, RK3288, RK3328, RK3399, RK3566, RK3568`

- Rockchip upstream kernel

```
git clone git://git.kernel.org/pub/scm/linux/kernel/git/mmind/linux-rockchip.git
```

Mainline kernel supports:

`RV1108, RK3036, RK3066, RK3188, RK3228, RK3288, RK3368, RK3399, RK3566, RK3568`

14. Chapter-16 SDK FAQ

The following offers answers and clarifications to common questions about various aspects of the SDK. Here are some potential questions that might be included in an SDK FAQ.

14.1 How to confirm the current SDK version and system/kernel version

You can obtain related information through the `/info/` directory or `'/etc/os-release'`, for example:

```
root@rk3572:/# cat /etc/os-release
NAME=Buildroot
VERSION=linux-6.12-stan-rkr1
ID=buildroot
VERSION_ID=2026.02
PRETTY_NAME="Buildroot 2026.02"
BUILD_INFO="wxt@ubuntu Wed Jun 22 10:00:00 CST 2026 - /home/wxt/linux-
develop/linux6.12/buildroot/configs/rockchip_rk3572_defconfig"
KERNEL="6.12 - rockchip_linux_defconfig"
```

14.2 DMA-BUF Related Issues

14.2.1 Introduction to dma_heaps related nodes

`/dev/dma_heaps/` has nodes such as cma, cma-uncached, system, system-dma32, system-uncached, system-uncached-dma32. For example, which memory should RGA use, and which memory should NPU use? The time difference for different memories is quite large. Is there a related explanation?

- sys_heap -> No restrictions on physical memory addresses, and mmap + vmap adopts cached. Users need to call begin/end to maintain data consistency, and CPU access is fast.
- sys_dma32_heap -> Physical memory addresses are limited to 32-bit, others are the same as above.
- sys_uncached_heap -> No restrictions on physical memory addresses, but mmap + vmap adopts write-combine mode, that is, normal-noncached mapping, users do not need to maintain data consistency.
- sys_uncached_dma32_heap -> Physical memory addresses are limited to 32-bit, others are the same as above.
- cma -> Allocates continuous physical memory from the CMA area, mmap + vmap adopts cached mapping, users need to maintain data consistency.
- cma-uncached -> The same as above, but mmap + vmap adopts write-combine mode, that is, normal-noncached mapping, users do not need to maintain data consistency.

Compared with the upstream code, there are additional uncached and dma32 related nodes, which are added for special needs.

- Some peripherals can only access 32-bit addresses, so dma32 nodes are added.
- Some users do not want to maintain data consistency, so uncached nodes are added.

14.3 System Issues

14.3.1 Dual Systems

The current SDK's file system (rootfs) partition is identified based on the partition's UUID. In the kernel, a specific UUID (for example: 614e0000-0000-4b53-8000-1d28000054a9) is specified in a hard-coded manner for mounting. Therefore, to support multiple rootfs partitions, simply create a new partition and burn it into the firmware. When switching systems is required, the UUID can be changed using partition tools such as sgdisk (which depends on the BR2_PACKAGE_GPTFDISK or BR2_PACKAGE_GPTFDISK_SGDISK package).

It should be noted that this dual-system solution requires restarting the device to switch.

In addition to restarting the device, container technologies such as chroot, docker, or kubernetes (k8s) can also be considered to achieve system switching, providing more flexible solutions. The following link provides an implementation method for switching dual systems based on a hardware button, for reference: Implementation of Rockchip RK3588 Android+Linux Dual System.

14.3.2 Year 2038 Problem

32-bit systems may encounter the Year 2038 problem when processing time representations. This is because they use 32-bit signed integers to store timestamps (seconds since January 1, 1970, 00:00:00 UTC), and this representation will not accurately represent time after January 19, 2038, 03:14:07 UTC. The maximum value of a 32-bit integer is 2,147,483,647, which corresponds exactly to the time point of January 19, 2038, 03:14:07 UTC.

To avoid this issue, the following measures can be taken:

- Use 64-bit timestamps: Many modern operating systems and programming languages have adopted 64-bit integers to store time, which greatly extends the range of representable time, far beyond the limitations of 2038. For example, in the Buildroot system, the Year 2038 problem can be resolved by setting the macro `BR2_TIME_BITS_64=y` and recompiling. For more details, you can refer to the discussion in the Buildroot community: Buildroot's Year 2038 Problem
- Update software libraries: Ensure that the software libraries used support 64-bit time representation. The GNU software library provides guidance on how to avoid the Year 2038 problem, which can be found in the manual of the GNU software library: Avoiding the Year 2038 Problem.

Verification method:

`date -s "2038-3-19 3:14:00"` If the system time rolls back to 1970 or 1901 or 2106 after setting the time, the problem may not be resolved.

14.3.3 How to call hardware codec resources and NPU modules on Debian system

Hardware codec demos can be tested through `/rockchip-test/video/*.sh`, basically implemented through `APP+Gstreamer(rockit)+mpp+VPU` for hardware codec calls. For details, refer to the relevant documentation.

AI demos can be tested through `/rockchip-test/npu2/*.sh`, and more references can be found in the `external/rknpu2` documentation for development.

14.4 Partition Size Settings

Partitions are modified through files such as SDK/device/rockchip//parameter*.txt.

If it is a newer SDK, you can also automatically modify and adjust through commands:
`make mod-parts`
`resize-part rootfs 10G`

Or
`make edit-parts`
Change the size following rootfs to 0x07800000

The above is for modifying partition sizes. If modifying partition table names, you can import config.cfg.

14.5 Issues about SDK Building

14.5.1 Sync Issues Caused by Repo

During the update with repo sync -c, the prompt No module named formatter occurs because your host machine is using a newer version of Python, such as Python 3.8+, which has completely removed the formatter module. This issue is caused by the outdated version of the repo tool in the external SDK. To resolve this problem, updating the version of repo is necessary. An example of how to do this is as follows:

```
$ cd .repo/repo/
$ git checkout origin/stable

This branch repo is the 2022 version, and the default master branch is the 2020 version. Or
add `--repo-rev=stable` during the repo init project to switch to the new version repo.

repo init --repo-url ssh://git@www.rockchip.com.cn/repo/rk/tools/repo --repo-rev=stable \
-u ssh://git@www.rockchip.com.cn/linux/rockchip/platform/manifests -b linux -m \
rk3588_linux_release.xml
```

14.5.2 Abnormal Issues Caused by ./build.sh Building

The following error will be prompted:

```
Adding information to /etc/os-release...
Traceback (most recent call last):
file "rk356x/.repo/repo/main.py", line 56, in <module>
from subcmds.version import Version
File "rk356x/.repo/repo/subcmds/__init__.py", line 35, in <module>
ModuleNotFoundError: No module named 'formatter'
```

Due to the matching problem between python3 and project repo versions in buildroot, `device/rockchip` can be modified as follows:

```
--- a/common/post-build.sh
+++ b/common/post-build.sh
@@ -171,8 +171,6 @@ function add_build_info()

     mkdir -p "$INFO_DIR"

-    yes | python3 .repo/repo/repo manifest -r -o "$INFO_DIR/manifest.xml"
```

14.5.3 Compilation issue in the Docker environment with ./build.sh

During compilation, a permission issue is prompted:
rm: cannot remove '/home/zora/myjob/linux/rk3568/output/sessions/latest': Permission denied

Sometimes due to issues with copying SDK source code or the PC environment, it is necessary to change the default permissions to 'zora' (a regular username):

```
chown zora:zora -R .
```

Then proceed with other operations.

14.5.4 Buildroot Building Issues

If some network reasons cause buildroot compilation to fail. This can be solved by using the following ways:

Preset dl directory, dl is a pre-compiled package that can be integrated into buildroot in advance. Reduce download time and improve compilation efficiency.

For example, rk3588 Linux SDK can be modified to add the dl directory as follows

```
$ cd .repo/manifests
$ Change the rk3588_linux_release.xml file as follows:

diff --git a/rk3588_linux_release.xml
b/rk3588_linux_release.xml
```

```
index 5082dea..626f094 100644
--- a/rk3588_linux/rk3588_linux_release.xml
+++ b/rk3588_linux/rk3588_linux_release.xml
@@ -14,6 +14,7 @@

+  <project name="linux/buildroot/dl" path="buildroot/dl" revision="next"/>

$ cd -
$.repo/repo sync -c
```

The modification methods for other chip SDKs are similar.

14.6 Recovery FAQ

14.6.1 How to prevent recovery from being compiled into the update.img

Newer SDKs support disabling recovery; to do this, navigate to make config and deselect the recovery option.

The packaging of update.img is based on the package-file description. Early SDKs used a predefined package-file located in the tools directory. For specific instructions, refer to the relevant documentation and remove the recovery entry.

In newer SDKs, the package-file is automatically generated before packaging the update.img. It is based on the partition descriptions in the parameter partition and the partition images with the same name in the output/firmware directory during the packaging process. If they match, they are automatically added to the package-file and then packaged into the update.img. Therefore, you can either delete the recovery partition in the used parameter.txt or manually delete the output/firmware/recovery.img before running make updateimg to package it.

Newer SDKs also allow you to use make edit-package-file to directly edit the package-file and adjust which images need to be packaged.

14.7 Buildroot FAQ

14.7.1 Dual Screen Asynchronous Clock Times

The clock format can be configured in `/etc/xdg/weston.ini` to compare in seconds:

clock-format=seconds

The default configuration updates the time every minute separately, initiating the update when a specific screen is enabled. The first update aligns with the current time to the nearest whole minute.

From the previous description, it seems likely that the screens were not enabled together, and then the system time changed (either through network synchronization or manual setting), causing the two to be out of sync.

If minute-level accuracy is essential and this scenario needs to be considered, you could modify Weston. Otherwise, using the second-level format should suffice.

```
diff --git a/clients/desktop-shell.c b/clients/desktop-shell.c
index bf75927..67289e2 100644
--- a/clients/desktop-shell.c
+++ b/clients/desktop-shell.c
@@ -531,12 +531,16 @@ panel_launcher_tablet_tool_button_handler(struct widget *widget,
        launcher_activate(launcher->argp, launcher->envp);
 }

+static int clock_timer_reset(struct panel_clock *clock);
+
 static void
 clock_func(struct toytimer *tt)
 {
    struct panel_clock *clock = container_of(tt, struct panel_clock, timer);

    widget_schedule_redraw(clock->widget);
+
+   clock_timer_reset(clock);
 }

 static void
@@ -589,7 +593,7 @@ clock_timer_reset(struct panel_clock *clock)
    clock_gettime(CLOCK_REALTIME, &ts);
    tm = localtime(&ts.tv_sec);

-   its.it_interval.tv_sec = clock->refresh_timer;
+   its.it_interval.tv_sec = 0;
    its.it_interval.tv_nsec = 0;
    its.it_value.tv_sec = clock->refresh_timer - tm->tm_sec % clock->refresh_timer;
    its.it_value.tv_nsec = 10000000; /* 10 ms late to ensure the clock digit has actually
changed */
```

14.7.2 How to Set Buildroot System to Boot into Command Line Mode

To disable Weston in buildroot (remove the include of weston.config in the corresponding board defconfig), then:

- Enable CONFIG_VT, CONFIG_VT_CONSOLE, CONFIG_FRAMEBUFFER_CONSOLE, CONFIG_DRM_FBDEV_EMULATION in kernel defconfig.
- Enable BR2_PACKAGE_FRECON, BR2_PACKAGE_FRECON_VT in buildroot.

The difference is that frecon supports rotation, scaling, and Chinese characters, but it can only display on a single screen.

14.8 Debian FAQ

This chapter is going to answer some frequently asked questions about Debian GNU/Linux based on the Rockchip platform. For other questions, please refer to the official website Debian FAQ.

14.8.1 "noexec or nodev" Issue

```
noexec or nodev issue /usr/share/debootstrap/functions: line 1450:
..../rootfs/ubuntu-build-service/buster-desktop-arm64/chroot/test-dev-null: Permission
denied E: Cannot install into target '/rootfs/ubuntu-build-service/buster-desktop-
arm64/chroot' mounted with noexec or nodev
```

Solution:

```
mount -o remount,exec,dev xxx
(The xxx is the path of project's directory, and then rebuild)
```

In addition, if other compilation exceptions are encountered, first check that the compilation system used is not the system type of ext2/ext4.

14.8.2 Failed to Download "Base Debian"

- Since building Base Debian needs to visit foreign websites, and when using domestic networks to visit foreign websites, download failures often occur:

To uses live build in Debian, configured like followings to change the image source to domestic:

```
32-bit system:

+++ b/ubuntu-build-service/{buster/bullseye}-desktop-armhf/configure
@@ -11,6 +11,11 @@ set -e
 echo "I: create configuration"
 export LB_BOOTSTRAP_INCLUDE="apt-transport-https gnupg"
 lb config \
+ --mirror-bootstrap "http://mirrors.ustc.edu.cn/debian" \
+ --mirror-chroot "http://mirrors.ustc.edu.cn/debian" \
+ --mirror-chroot-security "http://mirrors.ustc.edu.cn/debian-security" \
+ --mirror-binary "http://mirrors.ustc.edu.cn/debian" \
+ --mirror-binary-security "http://mirrors.ustc.edu.cn/debian-security" \
   --apt-indices false \
   --apt-recommends false \
   --apt-secure false \

64-bit system:
--- a/ubuntu-build-service/{buster/bullseye}-desktop-arm64/configure
+++ b/ubuntu-build-service/{buster/bullseye}-desktop-arm64/configure
@@ -11,6 +11,11 @@ set -e
```

```
 echo "I: create configuration"
 export LB_BOOTSTRAP_INCLUDE="apt-transport-https gnupg"
 lb config \
+ --mirror-bootstrap "http://mirrors.ustc.edu.cn/debian" \
+ --mirror-chroot "http://mirrors.ustc.edu.cn/debian" \
+ --mirror-chroot-security "http://mirrors.ustc.edu.cn/debian-security" \
+ --mirror-binary "http://mirrors.ustc.edu.cn/debian" \
+ --mirror-binary-security "http://mirrors.ustc.edu.cn/debian-security" \
  --apt-indices false \
  --apt-recommends false \
  --apt-secure false \
```

If the package cannot be downloaded due to other network reasons, there is a pre-built package to share in Baidu Cloud Network Disk, place it in the current directory and execute the next step directly .

14.8.3 Abnormal Operation Causes an error to Mount /dev

For example, like "askpass command or cannot use one" appears

It may be frequent abnormal operations (CTRL+C) during the compilation process, and the above errors can be fixed by the following way:

```
sudo -S  umount  /dev
```

14.8.4 Multiple Mounts lead to /dev error

For example: sudo: unable to allocate pty: No such device appears

The reason may be that the compilation process has been mounted multiple times, resulting in the above error, which can be fixed by the following way:

```
ssh <username>@<IP address> -T sudo -S umount /dev -l
```

14.8.5 How to Check System Related Information

14.8.5.1 How to Check the Debian Version of Your System

```
root@linaro-alip:~# cat /etc/debian_version
11.1
```

14.8.5.2 How to Check Whether the Debian Display Uses X11 or Wayland

On X11 systems:

```
$ echo $XDG_SESSION_TYPE
x11
```

On X11 systems:

```
$ echo $XDG_SESSION_TYPE
wayland
```

14.8.5.3 How to Check System Partition Status

```
root@linaro-alip:~# parted -l

Model: MMC BJTD4R (sd/mmc)
Disk /dev/mmcblk0: 31.3GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name      Flags
 1      8389kB  12.6MB  4194kB               uboot
 2      12.6MB  16.8MB  4194kB               misc
 3      16.8MB  83.9MB  67.1MB               boot
 4      83.9MB  218MB   134MB                recovery
 5      218MB   252MB   33.6MB               backup
 6      252MB   15.3GB  15.0GB  ext4         rootfs
 7      15.3GB  15.4GB  134MB   ext2         oem
 8      15.6GB  31.3GB  15.6GB  ext2         userdata
```

14.8.5.4 The ssh.service Is Abnormal in System

This is a problem of Debian10 or earlier version, please add the following code in the /etc/rc.local:

```
#!/bin/sh -e
#
## Chapter-16 rc.local
#
## Chapter-16 This script is executed at the end of each multiuser runlevel.
## Chapter-16 Make sure that the script will "exit 0" on success or any other
## Chapter-16 value on error.
#
## Chapter-16 In order to enable or disable this script just change the execution
## Chapter-16 bits.
#
## Chapter-16 By default this script does nothing.
```

```
## Chapter-16 Generate the SSH keys if non-existent
if [ ! -f /etc/ssh/ssh_host_rsa_key ]
then
    # else ssh service start in dpkg-reconfigure will fail
    systemctl stop ssh.socket||true
    dpkg-reconfigure openssh-server
fi

exit 0
```

14.8.6 Debian11 Base Package Fails to Build

An error similar to the following will be encountered:

```
W: Failure trying to run: /sbin/ldconfig
W: See //debootstrap/debootstrap.log for details
```

It is mainly required that the kernel version of the PC should be 5.10+, which is a bug in the old QEMU. There are two solutions:

- The kernel version that comes with the PC must meet the requirements of 5.10+.

The way to check PC's Kernel Version:

```
cat /proc/version
Linux version 5.13.0-39-generic
```

- Update system's qemu

Please refer to qemu.

14.8.7 How to Decompress, Modify and Repackage Debian deb Package

If you want to modify and repackage on the original deb, please refer to the follow way:

```
#Decompress the files in the package to the extract directory
dpkg -X xxx.deb extract/

#Decompress the control information of the package under extract/DEBIAN/:
dpkg -e xxx.deb extract /DEBIAN/

#Modify the file XXX

## Chapter-16 Repackage the modified content to generate a deb package
dpkg-deb -b extract/ .
```

14.8.8 How to Add the Swap Partition in Debian

When the physical memory of the system is not enough, you can add Debian's swap virtual memory partition for the current running program. For example, create a 2G virtual memory

- Create a swap file

```
cd /opt
mkdir swap
dd if=/dev/zero of=swapfile bs=1024 count=2000000
## Chapter-16 count represents the size, here is 2G.
```

- Convert files to swap files

```
sudo mkswap swapfile
```

- Activate the swap file

```
swapon /opt/swapfile

Uninstall:
swapoff /opt/swapfile
```

If it is automatically mounted after booting, you can add it to the /etc/fstab file
eg : /opt/swapfile swap swap defaults 0 0

- Verify whether it is in effect

```
root@linaro-alip:/opt# free -h
               total        used        free      shared  buff/cache   available
memory:      1.9Gi       390Mi        91Mi        75Mi       1.5Gi       1.4Gi
swap:      1.9Gi          0B       1.9Gi
e =h
```

14.8.9 Update Debian System for the First Time Will Restart the Display Service

In general, in order to be compatible with different chips, when Debian starts for the first time, /etc/init.d/rockchip.sh will install various differential packages according to the chip, such as libmali isp and other packages. After installation, the display service will be restarted. If it is an independent project, it can be placed in the image to process this difference.

14.8.10 Errors Occurring in Debian When Calling libGL related dri.so

Introduction as follows:

- EGL is an extension of OpenGL on the ARM platform for the x window system, and its function is equivalent to the glx library under x86.

- Since the driver modesettings used by Xorg will load libglx.so by default (disabling glx will cause some applications which detecting through glx environment fail to start), libglx.so will search for the dri library in the system. However, Xorg 2D acceleration is implemented directly based on DRM and does not implement the dri library, so libglx.so will report the following error during booting.

```
AIGLX error: dlopen of /usr/lib/aarch64-linux-gnu/dri/rockchip_dri.so failed`
```

It has no influence on system operation, please ignore it.

Similarly, the following errors will also be reported during the booting process of some applications, please ignore it for it has not influence on applications operation.

```
libGL error: unable to load driver: rockchip_dri.so
libGL error: driver pointer missing
libGL error: failed to load driver: rockchip
```

14.8.11 How to Confirm that the Hardware Mouse Layer is Useful in Debian

- Configure kernel dts

Similar to the following log:

```
root@linaro-alip:~# dmesg |grep cursor
[ 2.062561] rockchip-vop2 fe040000.vop: [drm:vop2_bind] Cluster1-win0 as cursor plane for
vp0
[ 2.062669] rockchip-vop2 fe040000.vop: [drm:vop2_bind] Cluster0-win0 as cursor plane for
vp1
```

- modetest test whether the layer has been reported

```
102     0       0       0,0             0,0     0               0x00000002
  formats: XR30 AR30 XB30 AB30 XR24 AR24 XB24 AB24 RG24 BG24 RG16 BG16 YU08 YU10 YUYV Y210
  props:
        8 type:
                flags: immutable enum
                enums: Overlay=0 Primary=1 Cursor=2
                value: 2
```

- Check if the summary has called the hardware mouse layer

```
root@linaro-alip:~# cat /sys/kernel/debug/dri/0/summary |grep 64x64
```

If there are steps 1/2, and there are still problems, then check whether /var/log/drm-cursor.log has abnormalities.

14.8.12 The log is Too Large in Debian

Debian provides **logrotate** to manage log files. Logrotate is intended to simplify log file management for systems that will generate many log files. Logrotate supports automatic rotation compression, deletion and sending log related emails. Logrotate can be run daily, weekly, monthly or when the log file size reaches a certain value. Typically, logrotate is run as a daily cron job.

```
apt install -fy logrotate cron
```

14.8.13 Debian Setting Multi User Mode Issue

When the system starts, the 'systemd' process will try to start '/lib/systemd/system/default.target' (usually the graphical interface system is a symbolic link to "graphical.target"). The status can be obtained through the following command:

```
systemctl get-default
graphical.target
```

Setting multi user mode (command line system):

```
systemctl set-default multi-user.target
```

After restarting, it was found that the interface was suspended at the logo and could not access the system. The normal startup sequence of the system is' sysinit ->multi user ->graphic '. If it is set to' multi user. target ', it means that the graphical interface has not been started. In this case, VT2 (terminal interaction needs to be enabled) is required, which means that the kernel needs to open the following two macros

```
CONFIG_FRAMEBUFFER_CONSOLE=y
CONFIG_VT=y
```

14.8.14 Debian Username and Password

The default username and password for the system are 'linaro' and 'linaro',
`Root` can log in without a password, and can be accessed through the command line to' sudo su '`

14.8.15 Debian XFCE Desktop icon Double-click Abnormal

XFCE desktop native bug can be resolved by checking 'Settings ->Desktop ->Icon' and clicking on 'Activate Project'.

14.8.16 Chromium browsers will have a command line flag: --no-sandbox

Unless using non-hardware accelerated, official native browser version. Otherwise, the customized browser version needs to be started with the `-no-sandbox` parameter, because the sandbox is a permission management and controls file access, and only without a sandbox can access to the hardware node is allowed to achieve hardware acceleration.

14.8.17 Setting Up DRI2 Extension in Debian System's X11

In the SDK, `glmark2-es2` uses the `dri2` interface for display with the Mali GPU library.

For specific implementation, refer to the relevant code in `xserver`:

- File Path: `./hw/xfree86/drivers/modesetting/dri2.c`
- Functions: `ms_dri2_get_msc` and `ms_dri2_schedule_wait_msc`

To confirm if Xserver supports DRI2, check the following log:

```
root@linaro-alip:/# grep -i dri2 /var/log/Xorg.0.log
[    47.696] (II) modeset(0): [DRI2] Setup complete
[    47.699] (II) modeset(0): [DRI2]   DRI driver: rockchip
[    47.712] (II) modeset(0): [DRI2]   VDPAU driver: rockchip
[    48.502] (II) Initializing extension DRI2
```

Below is a segment of the test code named `dri2-test.c` for DRI2.

```c
#include <stdlib.h>
#include <stdio.h>
#include <xcb/xcb.h>
#include <xcb/dri2.h>
#include <X11/Xlib.h>
#include <X11/Xlib-xcb.h>

int main(void)
{
    xcb_connection_t *c;
    xcb_dri2_connect_cookie_t cookie;
    xcb_dri2_connect_reply_t *reply;
    Display *display = XOpenDisplay(NULL);
    Window window = DefaultRootWindow(display);
    c = XGetXCBConnection(display);
    cookie = xcb_dri2_connect(c, window, XCB_DRI2_DRIVER_TYPE_DRI);
    reply = xcb_dri2_connect_reply(c, cookie, 0);
    printf("%s[%d] device(%s)\n", __func__, __LINE__, xcb_dri2_connect_device_name (reply));
    c = xcb_connect(NULL, NULL);
    xcb_screen_t *screen = xcb_setup_roots_iterator(xcb_get_setup(c)).data;
    cookie = xcb_dri2_connect(c, screen->root, XCB_DRI2_DRIVER_TYPE_DRI);
    reply = xcb_dri2_connect_reply(c, cookie, 0);
    printf("%s[%d] device(%s)\n", __func__, __LINE__, xcb_dri2_connect_device_name (reply));
    return 0;
```

```
}

build and test:

## Chapter-16 gcc dri2-test.c -lxcb -lxcb-dri2 -lX11 -lX11-xcb -o dri2-test
## Chapter-16 ./dri2-test
main[21] device(/dev/dri/card0)
main[27] device(/dev/dri/card0)
```

14.8.18 Installing GCC Toolchain on Debian

To install, run the following command:

```
apt update && apt install -y build-essential manpages-dev
```

After installation, confirm the GCC version:

```
root@linaro-alip:/# gcc -v
Using built-in specs.
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/lib/gcc/aarch64-linux-gnu/10/lto-wrapper
Target: aarch64-linux-gnu
Configured with: ../src/configure -v --with-pkgversion='Debian 10.2.1-6' --with-
bugurl=file:///usr/share/doc/gcc-10/README.Bugs --enable-languages=c,ada,c++,go,d,fortran
Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 10.2.1 20210110 (Debian 10.2.1-6)
```

14.8.19 Auto-Completion for Installing Packages on Debian

This is a common operation in Linux systems. Normally, you need to install the `bash-completion` package.

For specific instructions, search online, but it generally involves:

```
sudo apt-get install bash-completion
source /etc/bash_completion
```

14.8.20 Supporting DRI3 Extension in Debian X11 System

DRI3 extension, part of X11, provides improved support for direct rendering. It's usually enabled by default in Debian. As a low-level protocol, it's not directly executed without external calls.

To use the DRI3 extension in applications, refer to DRI3 documentation and write code to call the respective interfaces. For example, use DRI3 interfaces provided by the XCB (X protocol C-language Binding) library. Here's how to install and view these interfaces in Debian:

- **Installing XCB DRI3 Development Libraries**

Install the `libxcb-dri3-dev` library with:

```
sudo apt-get install libxcb-dri3-dev
```

This installs the necessary libraries and headers for using DRI3 functions in your applications.

- **Viewing DRI3 Interfaces**

  To view the definitions and functions of DRI3 interfaces, check the `dri3.h` header file:

  ```
  vi /usr/include/xcb/dri3.h
  ```

  This file contains specific implementations of the DRI3 interface, helping you understand how to use the DRI3 extension in your applications.

Follow these steps to start using the DRI3 extension in Debian for developing or improving your graphics applications. Remember to refer to official XCB and DRI3 documentation for correct and efficient use of these interfaces.

14.8.21 Setting System to Boot into Command Line Mode

The general method involves:

- Uninstalling xserver
- Enabling `CONFIG_FRAMEBUFFER_CONSOLE`, `CONFIG_DRM_FBDEV_EMULATION`, `CONFIG_VT`, `CONFIG_VT_CONSOLE` in the kernel

14.8.22 Configuring Screen Rotation

Refer to `/etc/X11/xorg.conf.d/20-modesetting.conf`

```
...
Section "Screen"
    Identifier  "Default Screen"
    Device      "Rockchip Graphics"
    Monitor     "Default Monitor"
    DefaultDepth        24
    SubSection "Display"
            Depth       24
            Modes       "1024x600"
    EndSubSection
EndSection

#### Valid values for rotation are "normal", "left", "right"
Section "Monitor"
    Identifier  "HDMI-A-1"
    Option      "Rotate" "inverted"
    Option      "Position" "0x0"
EndSection
Section "Monitor"
```

```
    Identifier  "DSI-1"
    Option      "Rotate" "left"
    Option      "Position" "0x0"
EndSection
```

14.8.23 Implementing No Black Screen Function in Debian

Black screen refers to the time from X service startup to desktop application display (dependent on the desktop application itself).

To maintain a logo during this time, add the following before executing `Xorg.wrap` in `/usr/bin/X`:

```
export XSERVER_FREEZE_DISPLAY=/.freeze_xserver
touch $XSERVER_FREEZE_DISPLAY
$(sleep 6; rm $XSERVER_FREEZE_DISPLAY)&
```

Freeze for 6 seconds, then display the desktop. Adjust the freeze duration as needed.

Alternatively:

```
{
        export XSERVER_FREEZE_DISPLAY=/.freeze_xserver
        touch $XSERVER_FREEZE_DISPLAY
        while sleep .5; do
                pgrep panel && break # Waiting for status bar service
        done
        sleep 2 # Waiting for status bar rendering
        rm $XSERVER_FREEZE_DISPLAY
}&
```

14.8.24 Removing Desktop Mouse Pointer Display in Debian System

The native mechanism is designed for compatibility with some non-touch applications, where Xserver converts the first touch event into a mouse event.

To bypass, try:

- Using a transparent mouse theme (search online for specifics)
- Modify `hw/xfree86/drivers/modesetting/drmmode_display.c` source code, removing `drmModeSetCursor`, `drmModeMoveCursor` calls
- If the SDK includes a drm-cursor library, modify `/etc/drm-cursor.conf` and add `hide=1`

14.8.25 Steps for Compiling and Porting the rkaiq/rkisp Repository in Debian

To port the `rkaiq/rkisp` functionality from Buildroot to Debian:

14.8.25.1 Overview of Steps

For chips like RK3588, switch to a lower version of GCC and GLIBC for porting to a third-party system.

Step 1: Modify the Buildroot configuration to support GCC 8.

```
[Various changes to the configuration files, including switching from BR2_cortex_a76_a55 to
BR2_cortex_a72_a53 and setting the GCC version to 8.x]
```

Step 2: Set the Buildroot configuration to use GCC 8 and GLIBC 2.28 by default.

```
buildroot# cat configs/rockchip_rk3588_glibc2.28_defconfig

[Configuration for rockchip_rk3588 with GCC version 8.x and GLIBC 2.28]
```

Step 3: Compile with Buildroot using the configured settings.

Set up the Buildroot environment and compile the `camera-engine-rkaiq` module.

```
<SDK># source buildroot/envsetup.sh rockchip_rk3588_glibc2.28
<SDK>## cd buildroot
buildroot# make camera-engine-rkaiq
```

After compilation, port the generated files (e.g., `output/rockchip_rk3588_glibc2.28/build/camera-engine-rkaiq-1.0/camera-engine-rkaiq-1.0.tar`) to the Debian system.

14.8.26 How to Download Offline Deb Packages in Debian

```
root@linaro-alip:/# apt-get download <package name>
```

14.8.27 How to Check glibc Version in Debian

```
root@linaro-alip:/# ldd --version
ldd (Debian GLIBC 2.31-13+deb11u7) 2.31
or
## Chapter-16 /lib/libc.so.6
GNU C Library (GNU libc) stable release version 2.35.
```

14.8.28 Support for Screen Splitting in Debian Systems

For a physically split screen with a resolution of 3840x2160 cut into 3840x720, the output resolution must be 3840x2160 to light up the screen, but only the 720 part is displayed.

The latest xserver added padding support.

Setting method:

```
+++ b/overlay/etc/X11/xorg.conf.d/20-modesetting.conf
[Configuration changes to set VirtualSize and Padding for the display]
```

Modify `/etc/X11/xorg.conf.d/20-modesetting.conf` as above. `VirtualSize` is configured as the desired resolution, and `Padding` is configured for the pixels cut off in the order top, bottom, left, right.

For unclear text on the screen, modify the padding order.

```
Option "VirtualSize" "DSI-1:1920x316"
Option "Padding" "DSI-1:0,0,0,764"
```

Mouse movement is relative and usually doesn't require special modifications. If there are ratio or position issues, consider using a software cursor (remove the mouse layer in kernel dts configuration or configure SWcursor in the modesetting conf).

To remove the cursor in kernel dts, search and delete, like:

```
kernel/arch/arm64/boot/dts/rockchip# ag cursor
rk3566-evb2-lp4x-v10-linux.dts
12:     cursor-win-id = <ROCKCHIP_VOP2_CLUSTER0>;
```

Or configure SWcursor in modesetting conf:

```
--- a/overlay/etc/X11/xorg.conf.d/20-modesetting.conf
+++ b/overlay/etc/X11/xorg.conf.d/20-modesetting.conf
@@ -2,6 +2,8 @@ Section "Device"
     Identifier  "Rockchip Graphics"
     Driver      "modesetting"

+    Option      "SWcursor"       "TRUE"
+
```

14.8.29 Debian XFCE4 Desktop Lock Screen Not Working

The lock screen function in the top-right corner of the XFCE desktop requires the installation of the `light-locker` package:

```
sudo apt install -y light-locker
```

14.8.30 Debian Partition Repair Issue

In some cases of unexpected power loss and reboot, the file system may become corrupted. To address this issue, the following methods can be considered:

1. The `fsck` used by default in `systemd` only performs basic repairs rather than a full repair. It is recommended to modify the `bootargs` (cmdline) in the kernel device tree (DTS) by replacing the `rw` parameter with `fsck.repair=yes`. This allows the system to automatically check and attempt a full file system repair during boot.
2. Implementing an A/B system scheme can further improve system safety and reliability in scenarios involving unexpected power loss.

14.8.31 Debian Old Version Mirrors Unavailable

Older Debian LTS versions have stopped maintenance. The available mirrors that can still be used are as follows:

```
https://debiansupport.com/mirrors/
```

14.9 Linux Video Related Issues

14.9.1 When playing videos, there is stuttering, and the logs show frame dropping errors

Using fpsdisplaysink to confirm the maximum frame rate. If the maximum frame rate is close to the expected frame rate, you can specify `sync=false` to solve the problem, some platforms can enable AFBC. About video playback freezes, you can also check the hardware running time by `echo 0x100 > /sys/module/rk_vcodec/parameters/mpp_dev_debug`

14.9.2 gst-launch-1.0 Camera Video Preview Command

You can use v4l2src plugin, such as `gst-launch-1.0 v4l2src ! autovideosink`

14.9.3 The Playback Screen Jitters after Turning on AFBC

Screen jitter is generally caused by insufficient DDR bandwidth. You can try the fixed performance mode. In addition, due to the display hardware implementation, If there is scaling in the vertical direction, the required performance and bandwidth will be relatively high, which is easily insufficient, and other scaling methods should to be used (such as rga/gpu).

14.9.4 How does GStreamer implement zero-copy

Use dmabuf related interfaces for data processing to achieve zero copy

14.9.5 How to Test the Highest Decoding Performance of gst-launch-1.0

Set the performance mode, use the official fpsdisplaysink to check the frame rate, and the driver's debugging interface to check the driver frame rate.

14.9.6 The Screen Jitters or Ripples Appear during Playback

Jitter is generally caused by insufficient performance of the display hardware, mostly due to insufficient DDR bandwidth. DDR performance mode can be used to fix the highest frequency

14.9.7 How to Quickly Connect GStreamer to Opengles

Generally, the synthesis is supported by the gl plug-in, and the display is supported by a third-party display service. Display services that can be synthesized internally through opengles (such as Weston)

14.10 Third-party OS Porting Issues

14.10.1 Is There Any Introduction to Kylin System Porting, Which is to Download the Standard iso Image and Extract rootfs.squashfs for Porting

You can refer to the introduction of the porting document in the SDK, which is also applicable to Kirin os, but Kirin os is relatively closed. If you want better performance, you can ask Kirin for images of RK platform.

14.10.2 Which Domestic OS Have Been Adapted

Tongxin and Kirin have adapted. In addition, the HarmonyOS community is currently working on the adaptation of RK3588, RK356X, and RK3399. For detailed progress, you can also consult our company's business, or follow the HarmonyOS open source community.

14.10.3 Whether to Support UEFI Booting

RK3588 will give priority to supporting UEFI

14.11 Display Related Issues

14.11.1 Mouse Blinking Issue

"Refer to the SDK documentation on VOP configuration to set the primary and mouse layers to non-cluster layers."

Some chips support types of layers such as smart, esmart, and cluster, with the cluster layer being the default setting for the mouse layer, relying on some special processing in the SDK.

Some customer applications use methods that do not support cluster layers and require the use of other types of layers (such as smart, esmart) as the mouse layer, for example:

```
&vp0 {
    cursor-win-id = <ROCKCHIP_VOP2_SMART0>;
};
```

14.11.2 How to Send Video to the Video Layer

There are interface of drm to query the type of plane. For details, please refer to kmssink method of gstreamer.

14.11.3 How to Configure the Position of Each Screen in Wayland Multi-screen with Differential Display Mode, Such As Left and Right or Up and Down Positions

Weston's differential display only supports left and right arrangement, according to the screen loading order. please refer to `<SDK>/docs/en/Linux/*/Rockchip_Developer_Guide_Buildroot_Weston_EN.pdf` 2.9 Multi-screen configuration for details.

14.11.4 What is the Debian Xserver Version

Debian10 uses xserver1.20.4, Debian11 uses xserver1.20.11

14.12 Browser Related Issues

14.12.1 Chromium R114 Scaling Issue

Using the parameter --force-device-scale-factor=1.5 will make the entire screen display smaller, resulting in an inability to display in full screen.

For the Chromium Wayland interface, this feature has only been adapted for Google's customized Aura Wayland protocol in newer versions, and there are bugs present under the standard Weston.

15. Chapter-17 SSH Public Key Operation Introduction

Please follow the introduction in the “Rockchip_User_Guide_SDK_Application_And_Synchronization_EN” to generate an SSH public key and send the email to sw.fae@rock-chips.com, to get the SDK code.
This document will be released to customers during the process of applying for permission.

15.1 Key Authority Management

Server can monitor download times and IP information of a key in real time. If an abnormality is found, download permission of the corresponding key will be disabled.

Keep the private key file properly. Do not grant second authorization to third parties.

15.2 Reference Documents

For more details, please refer to document
“/docs/en/Others/Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf"